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MAGNETICALLY INDUCED FORCES ON
EPICARDIAL LEADS
EXPERIMENTAL DEVICE DEVELOPMENT AND CLINICAL TESTING

by

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A THESIS

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Abstract

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For Dr. Nichols

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Chapter 1

Introduction

Medical diagnostics heavily rely on imaging tools such as MRI, a powerful and evolving application capable of showing high contrast sensitivity to soft tissue while not depending on ionizing radiation. Instead MRI uses strong magnetic fields and radio waves to excite hydrogen protons in the body, then detects the signals these excited protons emit as they return to their original state to create detailed images of internal tissues [1].

The magnetic nature of MRI makes it a hazardous environment where surgical equipment is limited. For example, during open-heart surgery cardiac implantable electronic device (CIED) systems are frequently placed to regulate cardiac rhythm [2] which are composed by several conductive materials such as temporary epicardial pacing leads. Furthermore, there is a recognized practice, if significant resistance is encountered, to cut the epicardial lead wire, intentionally leaving a small segment inside the patient to prevent immediate complications [3, 4, 5]. These cases introduce long-term repercussions, most notably the loss of access to MRI scanners.

In the United States alone, more than 300,000 individuals are receiving new CIED implants every year, as the Heart Rhythm Society (HRS) reported in 2024 [6], and a growing interest has focused on the interactions between the MRI environment

and various metallic or conductive objects and medical devices. Key standards from the ASTM International F04 committee address these concerns through standardized test methods for evaluating safety parameters such as magnetically induced displacement force (ASTM F2052 [7]), torque (ASTM F2213 [8]), and radiofrequency (RF)-induced heating (ASTM F2182 [9]), as well as uniform labeling practices for MRI safety (ASTM F2503 [10]).

According to ASTM F2503, a device labeled as MRI safe is expected to be non-conductive, non-electrical, non-magnetic, and to present no known hazards in the MRI environment [10]. However, this classification does not apply to all epicardial leads, as they are rarely physically characterized after removal, nor properly labeled by vendors. By the time they are cut, these leads have fulfilled their clinical purpose, and systematic safety evaluations are seldom performed.

For instance, conductive items such as electrocardiogram (ECG) leads, even if non-functional and relatively short, are generally considered unsafe or require extreme caution in the MR environment [11, 12, 13]. Main concerns include magnetically induced translational forces and torques, radiofrequency (RF) heating, or inadvertent cardiac stimulation [14, 15, 16]. First, the strong magnetic field and its spatial gradient can exert translational forces and torques on the lead, resulting in mechanical displacement along the direction of the magnetic field. Second, the lead can absorb energy from the MRI's (RF) field, potentially causing localized heating that poses a thermal risk to the patient. Third, the RF field may induce alternating currents within the lead, which can electrically stimulate nearby cardiac tissue, potentially leading to dangerous physiological effects [17]. Effectively preventing patients with CIED or cardiac surgery history, who are most likely to need a MRI scan during their lifetime, to not be able to enter MRI room [11, 18].

There is still a dearth of focused studies evaluating the precise dangers associated with retained epicardial leads, despite the frequent use of such leads and their reper-

cussions [11, 19, 20]. Existing clinical recommendations tend to be too cautious and frequently disqualify these individuals from MRI eligibility even when there is little evidence to support it. Recent studies regarding this problem have shown both no image quality degradation and, most importantly, no serious safety events such as, device malfunction, myocardial injury or arrhythmias, associated with MRI examinations in patients with CIEDs abandoned leads, including epicardial leads [19, 21]. While these studies have evaluated, to some extent, the resulting consequences of RF-induced heating, mechanical forces and conductivity, additional study is needed.

Specifically, focused quantitative investigations are required to address key unanswered questions regarding the physical forces experienced by lightweight devices such as retained epicardial pacing leads in the MR environment. While substantial research has quantified the projectile hazards posed by ferromagnetic or heavier hospital objects in the MR environment [18, 22, 23](e.g., oxygen tanks, scissors, or monitors), far less attention has been given to the much smaller forces experienced by thin, low-mass conductive leads. The importance of accurately assessing such low-magnitude forces is underscored by limitations in standard test methods [7, 8], which often require adaptations for ultra-lightweight or needle-like devices where conventional measurements struggle to capture micronewton-level effects [24]. To properly situate this work, the following section reviews pertinent literature on MRI QA devices, MRI safety principles, and previous studies to establish current state of the art regarding epicardial leads on MRI environment.

1.1 Literature Review

From a safety standpoint, the static magnetic field (B_0) represents the main source of hazard and must be shielded properly. A huge number of diagnostic clinical scans are daily completed without incidents indicating that magnetic interactions with nor-

mal tissues are within the bounds of safety up to the highest fields now in use for MRI [17]. However, there had been incidents associated with ferromagnetic objects either implanted in the patient or located in the fringing field of the magnets [23, 25].

MRI safety protocols are grounded in a system of absolute and relative contraindications. Absolute contraindications include non-MR-conditional cardiac implantable electronic devices (CIEDs), metallic intraocular, and foreign bodies, among others. Relative contraindications consider implants such as stents, surgical clips, and programmable shunts that may be scanned under controlled conditions [26]. The safety classification of each device is guided by testing standards and certification data, with CIEDs representing a critical category due to their susceptibility to electromagnetic interactions [17, 27].

The American College of Radiology (ACR) Manual on MR Safety (2024) [28] outlines the current regulatory and procedural framework for MRI safety. Recommendations include structured zoned access control and interprofessional collaboration through clearly defined positions (MR Medical Director, MR Safety Officer, and MR Safety Expert) to avoid unscreened individuals or ferromagnetic objects from inadvertently entering the magnet room. Pre-scan screening, implant model and status verification, and continuous patient monitoring are still necessary for this safety infrastructure [26].

In addition to current safety guidelines, studies have centered on qualitative findings. For example, a large observational study by Schaller *et al.* (2021)[29] evaluated 200 MRI examinations in 139 patients with a total of 243 abandoned leads. Their objective was to assess the safety of 1.5-T MRI in patients with at least one abandoned CIED lead and to evaluate any potentially harmful effects on the active CIED leads.

Before the scans, safety was a priority. Each case was discussed by radiologists and the ordering healthcare professionals in order to balance the advantages of alternative tests against the dangers associated with MRI. Next, baseline battery voltage, sensing,

pacing thresholds, and lead impedances were recorded. During the MRI, patients were monitored continuously via electrocardiography telemetry and pulse oximetry and the scans were performed in a normal specific absorption rate (SAR) mode, ensuring they did not exceed 2.0 Watts per kg for the whole body. In some cases involving the brain, a transmit-receive radiofrequency coil was used to avoid the central location of the leads relative to the coil.

Immediately after the MRI, the CIED evaluation was repeated to check for changes in capture threshold, sensing, and lead impedance. The research found a low rate of adverse events (3%), with no serious complications such as sustained arrhythmias or device resets. The observed events were primarily transient, non-life-threatening changes in lead sensing parameters that resolved spontaneously. Notably, this safety profile held true even for thoracic and cardiac MRIs, where the theoretical risk is highest.

Complementing this, a retrospective study by Vuorinen *et al.* (2021) [21] focused on 26 MRI scans in 17 patients with epicardial leads. With a similar objective, on a 1.5-T MRI, researchers distinguished between “old” epicardial leads (implanted before 2000) and “modern” leads (implanted in 2000 or later) to compare safety outcomes.

They followed a dedicated institutional protocol, previously implemented, which included measuring the length of the leads using chest x-rays prior to MRI scan to determine the lead reached lengths, approximately 24cm, that might increase heating risks. Similarly, researchers documented CIED generator and lead models and parameters with implantation dates. Finally, data analysis utilized the Mann-Whitney U test for group comparisons of patient characteristics using SPSS Statistics.

The study’s key finding was that MRIs performed on patients with “modern” epicardial leads had no detectable adverse events. However, one definite and one possible adverse event were noted in patients with “old” leads, suggesting that lead age and technology may be a factor in the risk profile.

These observations rely on clinical monitoring and empirical outcomes rather than direct quantification of physical interactions with the static magnetic field. To bridge this gap and enable more precise risk evaluation, standardized quantitative protocols are essential for measuring the magnetically induced forces on such objects.

1.1.1 Quantification of Magnetically Induced Displacement Force and Torque

ASTM F2052-21 [7] offers a standardized technique for assessing magnetically induced displacement force on medical equipment in order to set safety standards for such situations. The methodology includes a non-magnetic pendulum apparatus with a protractor that allows for the a suspended subject to be pulled by the magnetic field. The characterization of the static magnetic field and its spacial gradient is required. For this purpose, Ferreira *et al.* (2017)[30], developed a custom-designed fixture with an AlphaLab Gaussmeter probe to map the spatial gradients inside a Siemens Magnetom Trio™ 3T MRI scanner. Measurements of the magnetic field vector components (X, Y, and Z) were collected along the Z-axis in 5cm increments, correctly capturing the static magnetic field and ensuring repeatability and showing a steady spatial gradient.

To quantify the magnetically induced displacement force, Woods *et al.* (2021)[31], conducted a similar experiment following ASTM F2052-21 [7] on eleven metal alloys commonly used in medical implants. For deflection force testing, nine specimens (in three nominal mass categories: 2, 20, and 200 g) of each alloy were tested in a Siemens MAGNETOM Trio™ 3T MR system with the objective to establish evidence for MR conditional device labeling.

The main results regarding magnetically induced deflection force showed that for seven of the eleven tested alloys, the magnetic force was found to be less than the device weight in any current clinical 1.5 T, 3.0 T, and 7.0 T MR system. These safe

alloys included various grades of CP titanium, Ti-6Al-7Nb, Ti-6Al-4V ELI, Ti-15Mo, Biodur 108, and Co-28Cr-6M. However, for the other alloys, including 316L stainless steel and 22Cr-13Ni-5Mn, the magnetically induced force was calculated to be greater than the device weight in certain high-field systems, such as 3.0 T and 7.0 T clinical scanners.

In terms of torque measurements, Woods *et al.* (2021) [31] claimed that for medical instruments made from their list of alloys the magnetically induced torque will always be less than the worst case torque produced by gravity in a 3.0 T or less static magnetic field. And emphasized that *“No movement was observed for any of the magnetically induced torque specimens at 3.0 T.”*

Despite the fact that this particular study reported no movement, and therefore no measurable torque, it set an important precedent. The authors established an upper bound on magnetically induced torque by measuring the coefficient of static friction between the specimens and the test surface. They showed that the magnetic torque was less than roughly 30% of the gravity torque for all tested alloys using the ASTM conservative safety criterion that magnetically induced torque should not exceed gravitational equivalent torque $\tau < Lmg$. This result suggests that under typical clinical MRI circumstances, torque is not a limiting safety concern for implant-grade versions of these materials. This finding is important because it demonstrates that torque dangers are significantly less serious than displacement forces for conventional implant materials. Consequently, device designers can prioritize addressing displacement force effects while being confident that torque is generally well-controlled under typical clinical field strengths.

However, the study was probably done using ASTM F2213-17 [8] “The Low Friction Surface Method” which estimates magnetically induced torque by placing the device on a low-friction, non-metallic, non-conductive surface near the MRI isocenter. The device is rotated step-wise, and any spontaneous alignment or rotation toward

the static magnetic field is observed. If no movement occurs, the maximum possible torque is estimated from the device’s weight and the surface’s friction coefficient outside the MR environment. ASTM F2213-17 states that, if rotation is observed, more precise tests should be used instead which means this method relies on macroscopic motion and may not be able to detect mN forces.

In this regard ASTM F2213-17 define other methodologies, mainly the Torsional Spring or Pulley Method. A study by Seo *et al.* (2021) [32], utilized “the Pulley Method” for the safety assessment of magnetically induced torques on active implantable medical devices (AIMD) in 1.5 T and 3.0 T magnetic resonance imaging (MRI) environments. They constructed a cylindrical shaped apparatus entirely from acrylic, including moving parts like the bearings, to minimize magnetic field resistance.

This methodology determines the maximum magnetically induced torque by fixing the AIMD onto a rotating plate placed at the MRI isocenter. A thread attached to the plate is connected to a force gauge located far enough to avoid interference with the magnetic field. As the plate rotates to align with field, the thread tension is measured.

Prior to MRI deployment, the apparatus was validated using calibrated masses applied at known radii to generate controlled torques. Across a range of applied loads, the measured torque agreed with theoretical values to within an average error of approximately 1.5%. The preliminary experiment was configured to replace the magnetic field’s effect with known gravitational forces (weights). Weights ranging from 100–500 g were placed on the rotating plate at the point where the AIMD would be located, which was 2.9 cm from the center pillar. A force gauge, connected via a thread, was placed 4.9 cm from the center and used to pull the plate in the opposite direction of the weights. The force gauge used was load-cell-based with a resolution of 0.01 N. Each test was performed ten times for each weight (100 g, 200 g,

300 g, 400 g, and 500 g), and measurement was performed only when the equipment was stable.

Additionally, [32] addressed the friction issue not by eliminating it entirely, but by quantifying and compensating for it mathematically within their final torque calculation. The friction of the rotating plate (F_f) was determined beforehand in a prior experiment. Then, the friction value was explicitly used on the formula for magnetically induced torque (τ) calculations, which follows the principles of their Pulley Method. They explained that the torque (τ) is obtained by multiplying the distance from the center of rotation (r) to the AIMD's position by the difference between the measured force (F_m) and the friction of the rotating plate (F_f), $\tau = r(F_m - F_f)$.

In the clinic, they tested two types AIMD. First a bone conduction implant which assists patients with hearing loss by magnifying sound and transmitting it directly through the skull to the cochlea. These devices utilize magnets to collect external sound energy via a microphone and transmit it wirelessly to the internal stimulation device [32]. Second, the nonmagnetic devices, such as pacemakers and implantable cardioverter-defibrillators (ICD) which are generally not significantly affected by magnetic fields because their enclosures are typically made of MRI-compatible metallic materials like titanium or zirconium.

The main results showed that the bone conduction implant consistently exhibited the highest average torque (9.460 Ncm at 1.5 T and 14.618 Ncm at 3.0 T), while the pacemaker showed the lowest (0.255 Ncm at 1.5 T and 0.668 Ncm at 3.0 T). AIMD containing magnets, such as the cochlear and bone conduction implants, generally showed significantly higher torques than devices without magnets like ICD and pacemakers.

Similarly, Stoianovici *et al.* (2024) [24] showcases a variation of the ASTM F2213 Torsional Spring measurement technique, referred as the 2-string method. This method address the difficulty of measuring very low magnetically induced torques

on lightweight, slender medical devices, such as needles. This approach utilizes two strings to suspend the device by its ends and creates a torsional spring mechanism that is frictionless.

The needle rotates due to magnetic torque, the torque is calculated analytically based on the measured rotation angle and compared to the weight of the object. The main results showed that the method successfully measured torques between 0.0001 Nm and 0.0018 Nm in the MRI environment. The maximum magnetically induced deflection angle remained below the 25° limit required by the ASTM F2213.

1.1.2 Quantification of Magnetically Induced Heating

Similarly, the ASTM F2182 guideline [9] offers a standardized test procedure for detecting RF-induced heating on or near passive implants in tissue-mimicking phantoms in order to quantify the heating effects of MRI scans on AIMD. Similar methods were used in earlier research by Aboyewa *et al.* (2021) [33] to examine the interaction of leads with the RF field, assessing the energy absorbed and the ensuing heat generation.

The experiment was performed on a 3T Siemens Tim Trio™ MRI scanner using a turbo spin echo pulse sequence. The pulse sequence was set to produce a whole-body Specific Absorption Rate (SAR) of approximately 2 W/kg and fluoroptic probes were used to measure the temperature rise. Several factors on induced heating were tested, including lead length (such as 6.33 cm, 12.65 cm, and 18.98 cm), lead type (bipolar vs. unipolar), orientation, and shape.

It showed that lead length, type, and orientation significantly influence heating behavior. This study found that short leads (< 13 cm) posed no significant thermal hazard during MRI at a specific absorption rate (SAR) of 2 W/kg but the maximum temperature rise observed at the lead tips. For both lead types investigated, the amount of heat induced increased with increasing lead length. Notably, bipolar leads

showed slightly higher heating than unipolar leads and the straight lead configuration produced the highest maximum temperature rise.

The overall significance of these studies is in establishing reliable experimental and computational frameworks for quantifying MRI-induced mechanical effects, particularly where conventional ASTM approaches fall short. Stoianovici *et al.* (2024)[24] solves the technical challenge of measuring very low magnetically induced torques on slender, lightweight medical devices, such as needles, where conventional ASTM F2213 methods often become unreliable due to friction bias with their new methodology variant. Similarly, Seo *et al.* (2021)[32] validates the pulley methodology for AIMD. The modeling of clinically relevant alloys done by Woods *et al.* (2021)[31] shows experimental basis to rely on the listed alloys but requires further testing in terms of torque. While the modeling of the static magnetic field done by Ferreira *et al.* (2017)[30] provided a practical way to accurately map the magnetic field and calculate its spacial gradient. Finally, measurements by Aboyewa *et al.* (2021)[33] focused on the RF-heating rather than mechanical effects. They employed the same MRI system and testing framework we utilize here, reinforcing the feasibility of our experimental setup and situating our work within established practices.

1.2 Scope of Research

This document focuses primarily on the first of the three, previously mentioned, mechanisms of interaction, magnetically induced translational displacement forces and torques acting on retained epicardial pacing leads near and inside the MRI scanner bore, and to map how those forces evolve with distance from the isocenter. It follows that, this study investigates whether retained epicardial pacing leads are subject to torque or translational magnetic forces during MRI that reach clinically sig-

nificant levels, potentially justifying existing safety restrictions. We aim to provide experimental evidence and addressing a gap in the literature.

The core scientific aim is the development and implementation of adapted methodologies capable of reliably measuring forces at the micro-Newton scale. First, in a controlled laboratory environment and subsequently in the clinical 3T MRI setting. The methodology is capable of testing micro-Newton forces with precision for clinical application pushing towards these tools and clarifying whether the static magnetic field in MRI poses a real risk to patients with retained leads.

Additionally, radiologists frequently request guidance on challenging or uncommon cases involving patients with retained cardiac leads during MRI safety evaluations. One key objective of this work is therefore to create practical educational resources that help clinicians confidently address these difficult clinical scenarios. Specifically, we aim to develop a straightforward set of decision-support tools and teaching materials that enable radiologists, particularly radiology residents, to systematically evaluate the risks of static magnetic field-induced translational forces and torques on retained leads, improving safety decision-making and resident training in real-world practice.

To achieve these aims, the work proceeds in two main phases. In the laboratory phase, investigation used a Polyvinyl Chloride (PVC) pendulum-based measurement setup following ASTM F2052 [7] for the displacement angle and a custom-made solenoid simulating a magnetic field similar to that found on a MRI scanner bore entrance at scale. Displacement in millimeters of a metallic rod exposed to the field were recorded to ultimately estimate the translational force.

A second setup was specifically developed to quantify magnetically induced torque, which is particularly challenging for slender, lightweight structures where conventional methods may suffer from friction, low sensitivity, or inadequate resolution for micro-scale effects. While our torque apparatus does not strictly adhere to any single

ASTM F2213 method, it draws conceptual similarities from established low-friction techniques. For example, to further reduce potential errors from string-based systems, our design elevates the test sample to be positioned at a isocenter using non-magnetic supports for near-frictionless suspension. Torque is opposed and measured via a side-ways lever arm and a thin, non-magnetic rod attached to the fixture, which bends slightly under the counter-force from induced rotation. This deflection induces strain, detected by bonded strain gauges integrated into a custom circuit for direct, quantitative readout of torque magnitude under controlled orientations.

These methods allow for high sensitivity in detecting small magnetic forces in a controlled environment, without the complications of patient variability or direct MRI scanner use. Measurements are interpreted in the context of magnetic susceptibility theory and compared to their gravitational equivalents, which is a common normalization approach in MRI safety literature to assess clinical significance [7, 34].

Building on this validated foundation, a limited clinical phase in a 3T MRI environment follows. Given the challenges associated with clinical access for experimental testing, such as MRI scanner weekly availability and short measurement periods, the clinical experiments were therefore designed with two complementary goals. First, to verify device performance under real conditions. And second, to identify workflow bottlenecks, operator errors, or opportunities for optimization in the measurement chain.

The techniques developed in this document will help with more thorough evaluations of MRI safety in patients with retained temporary epicardial leads and set the stage for a future clinical adaptation of the protocol employing a 3T MRI scanner.

1.3 Summary

Epicardial leads have the problem of being broadly labeled MRI-unsafe under ASTM F2503. This classification originates from theoretical concerns over static magnetic field interactions (translational force and torque), RF-induced heating, and inadvertent electrical stimulation, yet it restricts MRI access for cardiac patients most in need of diagnostic imaging.

Hundreds of MRI tests using abandoned leads, including contemporary epicardial types, have shown no significant adverse events, indicating good safety profiles in large observational studies. There is still a significant gap in the accurate, micro-Newton-scale characterization of mechanical forces on thin, lightweight epicardial leads, despite the fact that standardized testing (ASTM F2052, F2213 [7],[8]) and earlier research on implant alloys confirm low magnetic interactions in many materials. The methodology on this project follows that of the literature reviewed in this chapter. Featuring a static field mapping fixture, a pendulum for deflection angle determination, and a newly developed torque device that resembles a suspension method but uses a low friction bearing to enable rotation and a highly sensitive sensor to measure mN forces, which is further described in Chapter 3.

Building on this foundation, Chapter 2 will provide essential background on the underlying physics of MRI scanners. It includes the generation and spatial distribution of static field, along with a broad introduction to gradient, and RF fields. Alongside physics, a detailed description of epicardial lead design, material composition, and implantation configurations is provided. Key concepts from electromagnetism (magnetic susceptibility, force, and torque on conductive elements) and mechanics (deflection, rotation, and tissue anchoring) will form the thematic backbone, enabling a rigorous analysis of lead-MRI interactions and informing the interpretation of experimental results in subsequent chapters.

Chapter 2

MRI Fundamentals and Safety Concerns regarding Epicardial Leads

This chapter provides a review of the most important topics to fully understand the findings of this current work. It includes an introduction to MRI scanners, the physics behind them and, more importantly, the possible interactions of the system with matter. Moreover, an emphasis on the static magnetic field of a MRI scanner and mechanical interactions with epicardial leads is included. A brief discussion on epicardial leads design is included followed by implications for the clinic and the lab setting.

Most of this chapter content is based on physics and imaging from well known textbooks, including Griffith's Introduction to Electrodynamics [35], and Bushberg's Physics of Medical Imaging [1].

2.1 Magnetic Resonance Imaging

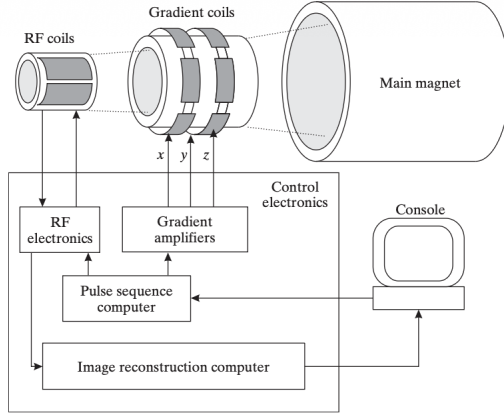
MRI or Magnetic Resonance Imaging is a medical imaging modality that uses magnetic fields and radiofrequency (RF) energy to produce images, rather than ionizing radiation like X-rays. In fact, MRI modality has exceptional utility in clinical settings, particularly for neurological and musculoskeletal imaging, because it produces images with high sensitivity to anatomical variations and superior soft tissue contrast [1].

This contrast comes from the magnetic properties of different tissue types such as fat, gray matter, white matter, and cerebral spinal fluid when exposed to a large magnetic field. The strength of a magnetic field, also known as magnetic flux density (B), is measured in Tesla (T). An older, alternative unit is the gauss (G), where $1 \text{ T} = 10,000 \text{ G}$. For perspective, a typical clinic MRI scanner is 1.5 to 3 Tesla, orders of magnitude stronger than Earth’s magnetic field, in the range of 30,000 to 60,000 nT[36]. Therefore, the exceptionally strong static magnetic fields required for clinical imaging are sufficient to interact with materials normally considered non-magnetic in everyday contexts.

This makes AIMDs particularly prone to safety risks, as even trace amounts of ferromagnetic materials in device components can experience mechanical forces and torques within the scanner environment. Understanding these risks requires examination of two fundamental aspects, the static magnetic field that the MRI scanner produces continuously, and the effects of this field on different materials. Primarily biological tissue in typical clinical use, but critically, the specialized alloys used in medical device construction, most importantly in epicardial leads.

To fully characterize the risk environment for epicardial leads, we must first understand the parts that make the MRI system and their potential risks. We highlight one of the five main components in the image formation process, the magnet strong static field. Figure 2.1a shows all these components and Figure 2.1b provides an ex-

ample of a actual, clinically relevant MRI scanner like the one used at CHI Health Creighton University Medical Center - Bergan Mercy, where experiments took place. Each of these pieces are accompanied by interaction complications if a foreign object lays around the test space. For example, as explained in section 1.1 heat and electrical interactions are a concern of RF fields or Gradient Coils [29, 33]. For mechanical interactions, the interest falls under the mechanisms in which the static magnetic field could possibly interact with wire or thin cylinder like devices. The next section offers a full explanation of the main magnet, enabling a clear understanding of both the source and the spatial distribution of the static magnetic field itself.



(a) Diagram illustrating the main five components of an MRI scanner.

(b) Siemens 3T Magnetom Trio™ clinical MRI system.

Figure 2.1: MRI scanner schematic and a clinical MRI system. (a) Simplified MRI scanner design highlighting the main magnet, gradient coils, and RF system, Control electronics, and Console. Adapted from Prince [37]. (b) Siemens Magnetom Trio 3T system, like at CUMC Bergan Mercy Hospital. As a modern whole-body MRI scanners, adapted from [38].

2.1.1 The Static Field

Magnets are at the core of the MR system, their performance is judged by field strength, temporal stability, and homogeneity. Magnetic fields can be induced by moving charges in a wire, where the direction depends on the charge's sign and direc-

tion, described by the "right-hand rule". Coiling a current-carrying wire, creating a solenoid or electromagnet, superimposes these magnetic fields, significantly augmenting the field strength inside the coil and causing a rapid falloff outside. The amplitude of the current I and the number of coil turns n determine the overall magnitude of the magnetic field strength for an ideal solenoid; $B = \mu_0 nI$, the produced magnetic field is parallel to the cylinder's long axis, usually horizontal. Common MRI are typically made of superconducting wire-wrapped cylinders. These wires are made from niobium-titanium alloys and show no resistance to electric current when maintained at extremely low temperatures and align the field along the patient's cranial-caudal axis in clinical systems.

A good depiction of the magnetic field can be obtained by modeling a finite solenoid of length L and radius R using a formula derived from Biot-Savart law with $B = 3\text{T}$ as follows

$$B(z) = \frac{\mu_0 n I}{2} \left[\frac{z + \frac{L}{2}}{\sqrt{R^2 + \left(z + \frac{L}{2}\right)^2}} - \frac{z - \frac{L}{2}}{\sqrt{R^2 + \left(z - \frac{L}{2}\right)^2}} \right]. \quad (2.1)$$

where μ_0 is the permeability of free space. The first term in the brackets represents the contribution from the far end of the solenoid, and the second term represents the contribution from the near end. This formula provides the axial field along the solenoid, with the maximum occurring at the center ($z = 0$) and decreasing smoothly toward the ends. A simulation of this model for a 1.4m length by 0.6m diameter ideal solenoid, following a real MRI, such as the Siemens Magnetom™ system, [38] provides the information need to recognize key points in space in the MRI room.

Figure 2.2 showcases this model in two parts. First, the magnetic field (B_0) along the z-axis, where $z = 0$ corresponds to isocenter and a grey dotted line shows the edges of the solenoid at half the length or 0.704 m away from isocenter. The field is normalized to a maximum of $B_0 = 3\text{ T}$, with no shielding or distortion applied.

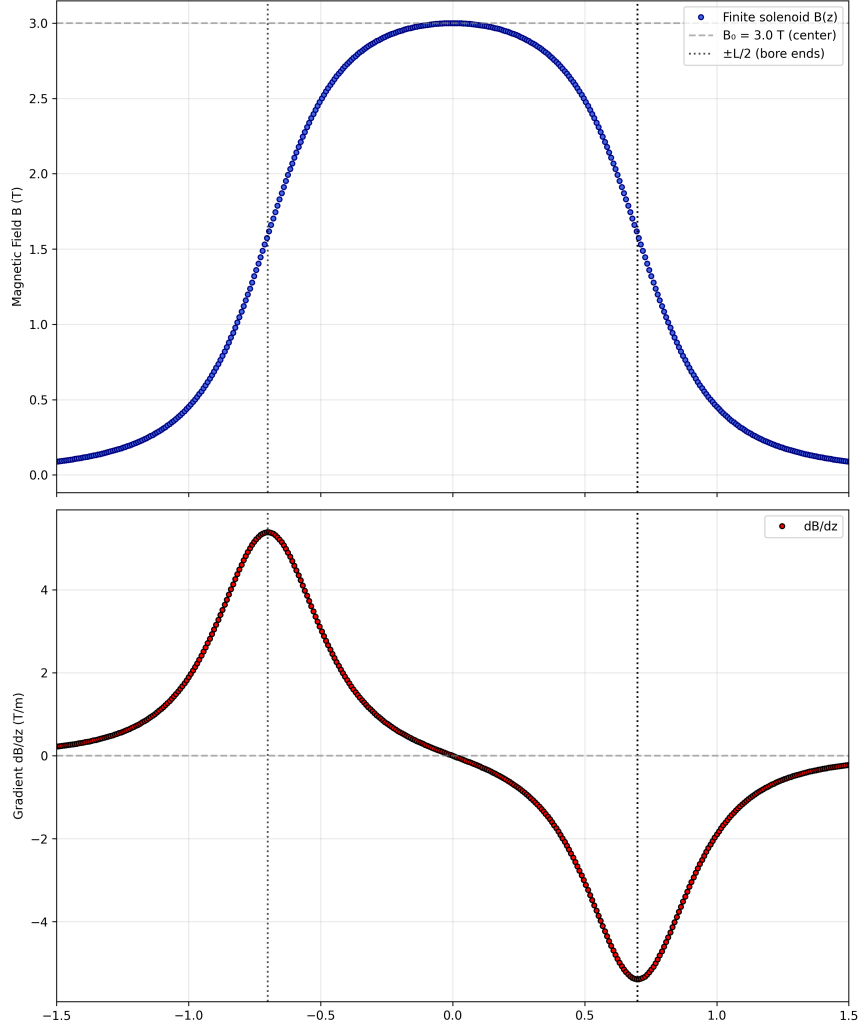


Figure 2.2: Axial magnetic flux density B_z and spatial gradient for an idealized 3T MRI solenoid ($L = 1.4\text{m}$, $R = 0.3\text{m}$). Top: The magnetic field peaks at 3T at the isocenter ($z = 0\text{m}$) and decreases symmetrically toward both ends. Grey dotted lines indicate the solenoid edges at half length (bore entrances). Bottom: The spatial gradient dB_z/dz reaches its maximum magnitude at the bore entrances ($z = \pm 0.704\text{m}$).

Second, the axial field $B(z)$ from the finite solenoid model yields a spatial gradient dB_z/dz , its maximum value is confirmed to be at the bore entrance position [7, 30] ($z \approx \pm 0.704\text{ m}$ for this 1.4 m length example), as shown in the bottom panel of Figure 2.2. These strong gradients (often several T/m near entrances) exert attractive forces on paramagnetic/ferromagnetic objects (pulling toward higher $|B|$) and repulsive on diamagnetic ones (pushing away), posing projectile hazards [1, 7, 30]. This gradient-

driven behavior supports ASTM displacement force testing and motivates shielding designs in modern systems. High-susceptibility materials experience the strongest pull toward the isocenter, with maximum risk at bore entrances.

Having established the spatial distribution of the static field, we now turn to how this field interacts with matter. These interactions are governed by fundamental electromagnetic theory and provide the foundation for classifying materials by magnetic susceptibility which quantifies MRI safety risks for implanted devices according to ASTM standards [7, 8].

2.1.2 The Static Field interaction with matter

Any material placed in a magnetic field develops a magnetization, $\mathbf{M}$, defined as the net magnetic dipole moment per unit volume, $\mathbf{M} \equiv \frac{\mathbf{m}}{V}$ [35]. In a way this $\mathbf{M}$ represents a “smoothed” magnetization of an average field effect over many atoms. At the atom level, the magnetic field inside a material fluctuates rapidly due to particles in motion. This phenomena makes the basis for MRI image generation. However, for the purpose of this thesis, we focus on the macroscopic behavior mostly. To simplify the macroscopic analysis of materials in magnetic fields, physicists define the auxiliary magnetic field $\mathbf{H} \equiv \frac{1}{\mu_0} \mathbf{B} - \mathbf{M}$, where $\mathbf{B}$ is the macroscopic magnetic flux density and μ_0 is the permeability of free space. $\mathbf{H}$ accounts for the part of the magnetic field produced by free currents, separating it from the bound currents induced by magnetization [35]. This distinction becomes useful when analyzing mechanical effects on magnetic dipoles.

Different materials respond differently to an applied field. As magnetization is also directly proportional to the field strength $\mathbf{M} = \chi_m \mathbf{H}$, the degree to which a material becomes magnetized is quantified by its magnetic susceptibility, χ_m , which is typically small in biological tissues. Diamagnetic materials, such as water, calcium, and most organic compounds, develop a magnetization opposite to the applied B-field due to

paired electrons. Paramagnetic substances, like molecular oxygen, deoxyhemoglobin, or gadolinium-based MRI contrast agents, align weakly with the field, enhancing the local magnetic flux. Ferromagnetic materials, by contrast, exhibit strong self-sustained magnetization and are avoided in MRI due to safety hazards [39]. Contrast agents, such as gadolinium or iron-oxide, represent a separate class of regulated pharmaceuticals administered only in carefully controlled doses[40]. Figure 2.3 shows a spectrum of magnetic susceptibilities highlighting the region MRI where water and most human tissue is found.

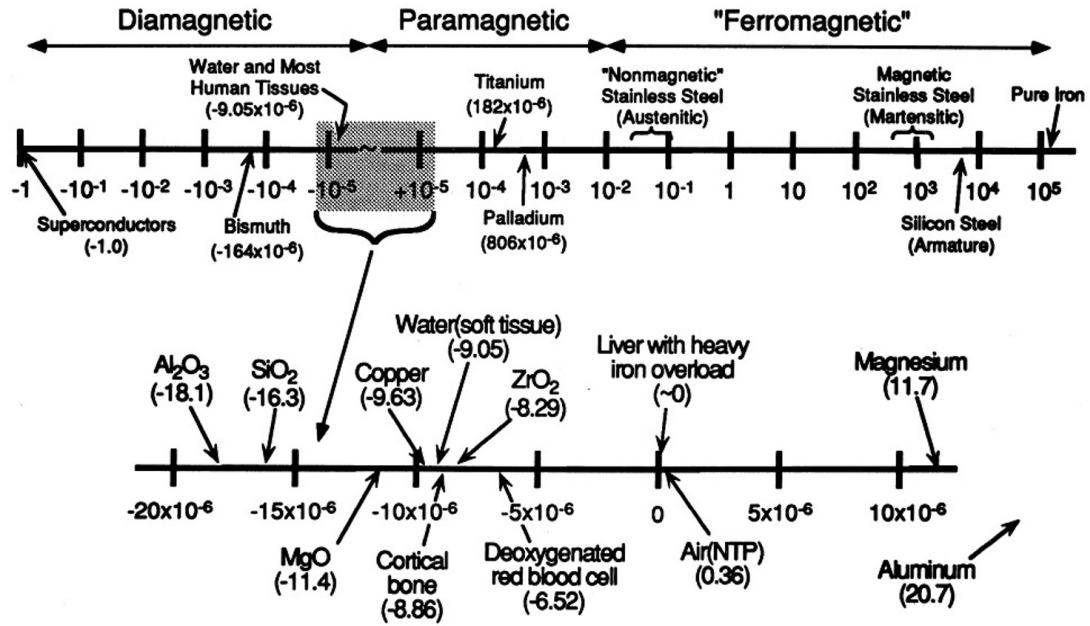


Figure 2.3: Spectrum of magnetic susceptibilities. The upper diagram uses a logarithmic scale to indicate the full range of observed magnetic susceptibility values, extending from -1.0 for superconductors to 100,000 for soft ferromagnetic materials. The bottom diagram uses a linear scale (in ppm) to indicate the properties of some materials with 20 ppm. The susceptibilities of most human tissues are in the range from -7.0 to -11.0 ppm. Adapted from Schenck [17].

For materials that obey magnetization - field strength proportionality, called *linear media*, the magnetic flux density can be expressed as

$$\mathbf{B} = \mu_0(1 + \chi_m)\mathbf{H}, \quad (2.2a)$$

$$\mathbf{B} \approx \mu\mathbf{H}, \quad (2.2b)$$

here, can define the *permeability* of the material as $\mu \equiv \mu_0(1 + \chi_m)$. Moreover for paramagnetic materials which susceptibility is tiny ($|\chi_m| \ll 1$) $\mathbf{B}$ can approximate to Eq 2.2b.

2.1.3 Image generation

While macroscopic magnetization (M) and susceptibility (χ_m) govern bulk mechanical interactions and safety risks for implants, the diagnostic power of MRI arises from the microscopic magnetic properties of atomic nuclei. Hydrogen (specifically its proton, 1H) is the most utilized element in clinical MRI because it has the largest magnetic moment and is highly abundant in biological tissues, primarily in water and fat, which constitute the majority of soft tissue composition. In fact, MRI was originally named Nuclear Magnetic Resonance (NMR), because it involves the spectroscopic study of the magnetic properties of the atomic nucleus as it resonates and aligns with applied magnetic field changes. On the scale of elementary particles, these interactions involve the nuclear spin and charge distribution of protons and neutrons within the atomic nucleus.

Normally, these hydrogen atoms have a random orientation of their nuclear magnetic moments (protons) due to thermal energy. When placed in a strong static magnetic field (B_0), these nuclear magnetic moments align in parallel (same direction of B_0 , low-energy state) and antiparallel (opposite direction of B_0 , high-energy state) directions.

In addition to aligning with B_0 , the protons also experience precession, much like the Earth's rotation around its tilted axis or a spinning top under gravity. This precession occurs at an angular frequency (ω_0) known as the Larmor frequency, which is directly proportional to the magnetic field strength (B_0), as described by the Larmor equation $\omega_0 = \gamma B_0$.

An RF energy pulse (B_1) is then applied, synchronized to the Larmor frequency of protons (about 42.58 MHz per tesla), magnetizing the protons and tipping the net magnetization (M) away from the longitudinal direction (M_z). This pulse provides just enough energy for some protons to move into a higher energy state, also called RF excitation. In doing so, it also causes the magnetic moments of many protons to become synchronized, or phase-coherent. They start to precess together in the transverse plane, creating what is called transverse magnetization (M_{xy}).

The instant the RF pulse is turned off, the relaxation process starts, these precessing protons start to lose that synchronization due to slight variations in their local magnetic environments. This causes the transverse magnetization signal to gradually fade over time. The oscillations of this signal are detected by the same RF coils that sent them.

First, the signal in the transverse plane, (M_{xy}), decreases as the protons desynchronize, this is described by T_2 (Transverse Relaxation). And the process by which the system regains equilibrium is described by T_1 (Longitudinal Relaxation). The longitudinal magnetization (M_z) returns, and the particles realign with the main magnetic field (B_0). The final MRI image's depiction of a tissue's brightness or darkness is determined by the RF signal produced during relaxation processes taken together.

However, this covers one a single signal from a particular spot. To make an image rather, the MRI system must know where each signal is coming from. This is achieved using small, precisely controlled variations in the magnetic field known as gradient fields. These gradient coils, superimposed on B_0 , make the magnetic field

slightly stronger or weaker depending on position, causing the Larmor frequency of the protons to vary with location.

Three gradient systems work together to map space. First, Slice Select Gradient (SSG) is applied during excitation so that only protons in a particular “slice” of the body resonate at the RF pulse frequency. This defines the plane of imaging. Then, Phase Encoding Gradient (PEG), briefly changes the precession speed of protons in one direction, giving each row of tissue a unique phase “signature”. Finally, Frequency Encoding Gradient (FEG) is applied during signal readout so that protons along another axis precess at slightly different frequencies.

These gradients must switch rapidly and with large current flows, typical slew rate ranges from 5 to 250 mT/m/ms, with currents on the order of 100–200 A [1, 37], to accomplish encoding. The resulting signals are stored in a matrix called k-space. Each line in k-space represents one measurement with a specific gradient configuration. When all necessary lines are filled, the Fourier Transform reconstructs the data into an image, where each pixel corresponds to a small volume element (voxel) of tissue.

The control and computing subsystems coordinate every stage of image acquisition and reconstruction. The host computer and operator console provide the interface through which the technologist selects and supervises scanning protocols, while the pulse programming and measurement control systems synchronize gradient and RF activity [1]. Supporting units such as the digitizer, image processor, and compute engine translate the raw analog signals into interpretable digital images through Fourier-based reconstruction algorithms [37]; Finally, the dedicated power supplies, patient table mechanics, and electromagnetic shielding ensure stable operation, patient positioning, and noise minimization.

These same magnetic field components can pose hazards to implanted conductive objects. Although most modern MRI systems are designed for patient safety, implanted devices and retained surgical materials introduce additional

considerations. One clinically relevant example is epicardial leads, which are commonly placed during cardiac surgery to provide temporary postoperative pacing for arrhythmias. These leads consist of conductive wires (often stainless steel or similar alloys) sutured directly to the epicardial surface of the heart and may be removed postoperatively or, in some cases, cut short and abandoned in situ to avoid extraction risks.

2.2 Epicardial Leads

CIED systems refer to cardiac implantable electronic devices, including pacemakers, ICDs, cardiac resynchronization therapy (CRT) devices, and implanted rhythm monitors. These are advanced medical devices used to monitor and regulate heart rhythm disorders. Out of these components, cardiac leads are a type of very thin, ~ 4 to 5 F (French gauge, meaning 1.3 to 1.7 mm), electrical wire that connects a pulse generator to the heart enabling two-way communication between them. Their main functions include, monitoring, pacing, and defibrillation.

There are cardiac lead types differentiated by three main things, placement in the heart, polarity, and their fixation mechanism, which is the method used to secure the lead tip to the heart tissue to prevent movement. In practice, Figure 2.4 shows the two types of leads that are used, epicardial leads which are implanted directly on the epicardium (the outer surface of the heart) and transvenous leads, inserted through the veins and positioned inside the heart's chambers, making contact with the endocardium (the inner lining of the heart). Both are prone to complications such as fracture, infection, and venous obstruction (transvenous only) requiring complex extraction procedures [41, 42].

The position of leads is an important distinction. Clinically, these leads are indicated for different populations and diagnostics therefore carry a variety of man-

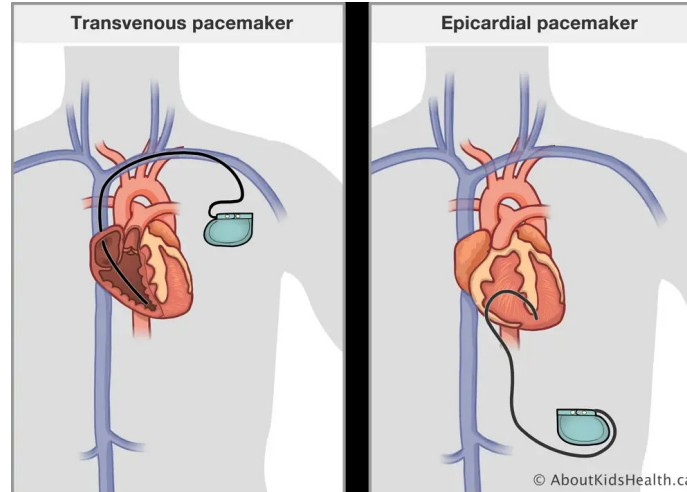
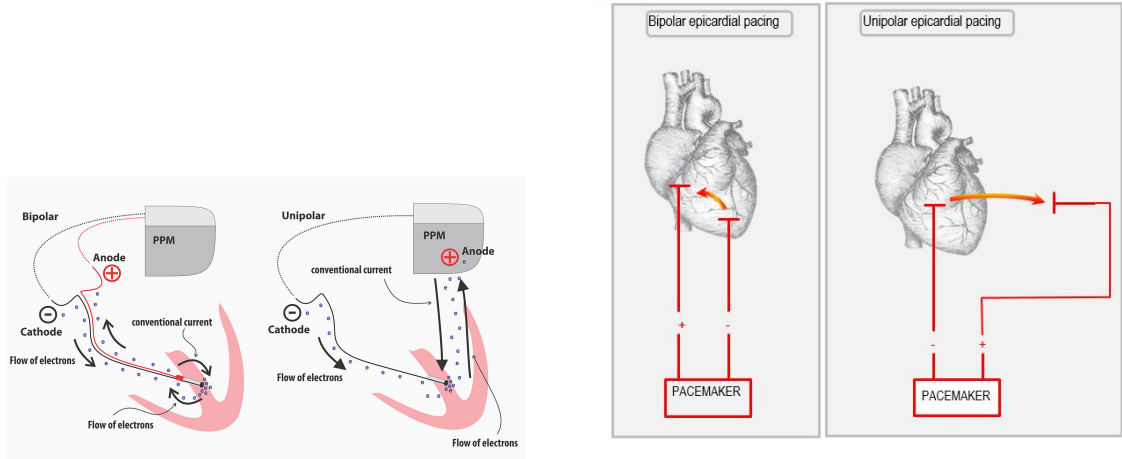


Figure 2.4: Comparison of transvenous vs epicardial lead placement in pacemakers (adapted from [43]).

agement decisions (leaving vs. extracting) and different complications (percutaneous lead extraction or transvenous lead extraction). [19, 44] Physically, different contact with tissue means mechanical interaction, an example is that transvenous leads may be cooled by blood affecting its overall heating capacity. Beyond thermal effects, any MRI-induced displacement force or vibration on transvenous leads may be somewhat dampened by surrounding blood and vessel walls, while for epicardial leads by being sutured directly to the epicardial tissue, it translates directly to mechanical stress.

Within epicardial leads, design may vary, including lead polarity and body structure. Unipolar leads contain a single conductor and one electrode (the cathode) are in contact with the heart. Bipolar leads contain two conductors that connect to two electrodes on the lead itself, a tip electrode (cathode) and a ring electrode (anode). In figure 2.5 a comparison of these types is shown. On the left side 2.5a depicts the current flow for each case, while 2.5b shows the position of the anode relative to the heart.

Bipolar leads exhibit further structural variation in their body design. As shown in Fig. 2.6b, coaxial design features an inner coil conductor nested within an outer coil conductor, each wrapped in a separate insulation. Coradial designs consist of a

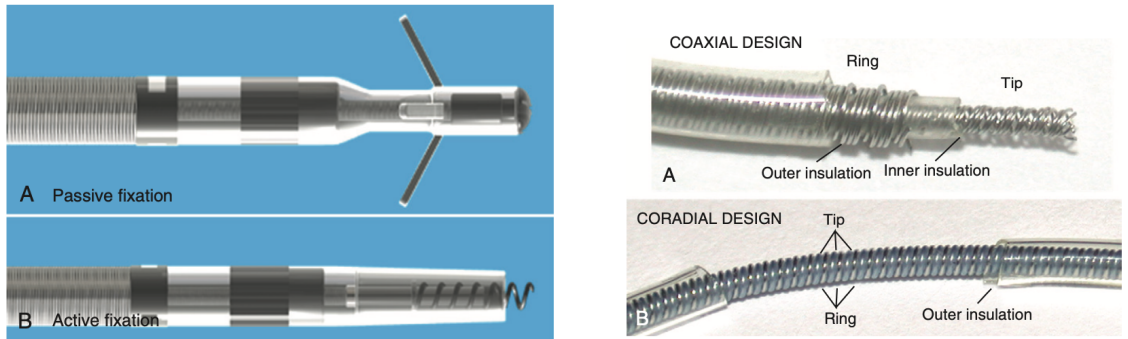


(a) Unipolar vs. bipolar epicardial lead configuration.

(b) Relative electrode positioning of unipolar and bipolar epicardial leads.

Figure 2.5: Comparison of unipolar and bipolar epicardial pacing leads. (a) Illustrates the distinction in current flow pathways using a schematic of the heart, adapted from [45, 2]. (b) Depicts the relative anode and cathode positions in unipolar and bipolar epicardial leads, adapted from [46, 2].

single coil containing two parallel, intertwined conductor strands, each individually insulated [2].



(a) Lead fixation mechanisms.

(b) Coaxial and coradial lead body design.

Figure 2.6: Lead fixation and body design variations in pacing leads. (a) Passive-fixation leads rely on flexible tines that anchor within myocardial trabeculae, while active-fixation leads employ a retractable helix that screws into the myocardium. (b) Coaxial designs use two nested coil conductors with separate insulation layers, whereas coradial designs incorporate parallel, intertwined conductors within a shared lumen. Adapted from [2].

The several components of an epicardial lead are made of different materials chosen for their biocompatibility, durability, and electrical properties. Electrodes are

made from inert materials to minimize corrosion and inflammatory reactions over the long term. These include titanium, platinum-iridium alloys, Elgiloy (an alloy of cobalt, iron, chromium, and other metals), and vitreous carbon. Similarly, conductors which carry the electrical signals between the pulse generator and the electrode are commonly made out of MP-35N (alloy of nickel, cobalt, chromium, and molybdenum.) Insulation, a protective outer coating of the lead that prevents short-circuiting is made of different polymers like Silicone Rubber (Polysiloxane), Polyurethane, Fluoropolymers.

Finally, epicardial leads are affixed directly to the heart's outer surface during a surgical procedure in mainly two ways, as shown in figure 2.6a. Suturing and Screwing, meaning the electrode is designed to be sutured directly onto the epicardium. While screwing refers to an active-fixation helix (screw), typically made of a platinum-iridium alloy or a similar metal, screwed directly into the myocardium [2]

From figure 2.6 one can see some of the challenges when retracting the epicardial lead, for this practical reason they might be abandoned after its purpose has been fulfilled [3]. An abandoned epicardial lead could measure full-length or fragments after being left behind. Some fragments are described as long as 5cm, being 2cm an average length. Further, longer leads are reported up to 30 cm or couple fragments adding to about 21 cm. In diameter size may be compared to standard transvenous pacing leads, that of about half a millimeter to 1.5 mm. And are typically narrower than standard transvenous leads ($\sim 5F$, $\simeq 1.7$ mm outer diameter), often measuring between 0.5 mm and 1.5 mm.”

The physical length of the wire can impact the resulting interactions with the magnetic field, as more of the material in the human body more of it can interact. Table 2.1 summarizes the materials found on a epicardial lead and relates to its magnetic susceptibility.

Table 2.1: Qualitative magnetic susceptibility of common epicardial lead materials and their relevance for MRI safety. [10, 47, 48, 9]

Component	Material	Magnetic susceptibility	Relevance for MRI safety
Electrode	Titanium	Paramagnetic	Low; minimal torque/force in MRI
Electrode	Platinum-Iridium alloy	Paramagnetic	Low; generally MRI safe
Electrode	Elgiloy (Co-Cr-Fe-Ni alloy)	Paramagnetic	Moderate; may experience slight translational force
Electrode	Vitreous carbon	Diamagnetic	Minimal; very low MRI interaction
Conductor	MP-35N	Weakly paramagnetic	Low; minimal contribution to MRI forces
Insulation	Silicone rubber (polysiloxane)	Diamagnetic	Negligible; non-conductive and non-magnetic
Insulation	Polyurethane	Diamagnetic	Negligible; non-magnetic
Insulation	Fluoropolymer (PTFE)	Diamagnetic	Negligible; MRI inert

Current institutional recommendations regarding MRI in patients with retained epicardial leads remain conservative despite evolving evidence. Major guideline bodies, including the Heart Rhythm Society (HRS), the European Society of Cardiology (ESC), and the Centers for Medicare & Medicaid Services (CMS), allow MRI in patients with MRI-conditional or certain legacy cardiac implantable electronic devices (CIEDs), but continue to restrict or discourage imaging in patients with abandoned or epicardial leads [21].

Specifically, at the time of the HRS 2017 consensus, the 2021 ESC pacing guidelines noted that there was insufficient data to comment on the safety of MRI with an abandoned cardiac lead. And if a patient had an MRI-conditional system along with an abandoned lead, lead extender, lead remnants, fractured lead(s), or surgically implanted epicardial leads, the system was considered non-MRI-conditional. Leading to a Case-by-Case consideration for whether to image or not and in which modality. CMS similarly excludes abandoned or epicardial leads from coverage due to perceived insufficient safety data, even though large observational studies indicate that modern epicardial leads (implanted after 2000) may undergo MRI safely under carefully monitored conditions [21, 29].

This cautious institutional stance demonstrates the need for experimental characterization of mechanical interactions between retained epicardial leads and MRI fields. This supports the rationale for the present study. By quantifying these forces under controlled laboratory conditions, the work shown in this document directly addresses

whether current institutional restrictions are justified and lays the groundwork for evidence-based updates to clinical safety policies.

2.3 Safety Concerns

After briefly reviewing the image generation sequence, observe that it involves the primary components illustrated in Figure 2.1a. These components are associated with various risks and safety concerns. Disregarding control systems and peripherals, the three main sources of hazard are the gradient coils, the RF coils, and the main magnet, the latter presenting the most critical safety issues due to its powerful static magnetic field.

2.3.1 Hazards attributed to the encoding Gradient Fields

First, the quickly shifting magnetic gradients introduce hazard mechanisms beyond magnetic field strength. For example, when the gradient currents are repeatedly turned on and off, large forces called Lorentz forces, are applied to the coils in the strong magnetic field, leading to significant vibration. That, is the source of the loud "knocking" sounds during MR scans. Noise levels in modern MRI systems can easily exceed 130 dB, which is roughly a million times more intense in sound pressure than normal conversation or similar to an airplane taking off about 100 meters away, leading to temporary hearing loss as reported in early studies [18]. Consequently, hearing protection is now considered mandatory during MRI procedures.

Moreover, the rapid temporal variation of MRI gradient fields induces electric fields within conductive tissues via Faraday's law. The effect scales with both the rate of change of the magnetic field and the spatial extent of the conductive path through tissue, so even though the absolute magnetic field perturbation from gradients (10–60 mT/m, or less than 1% of B_0) is small, the rapid temporal ramping

can induce voltages large enough to reach neural activation thresholds [49]. Typical clinical systems, limited by FDA guidelines to ~ 40 T/s, remain below the threshold for cardiac excitation, but research systems capable of 300 mT/m gradients and higher slew rates approach that regime, posing theoretical cardiac stimulation risks [1, 18, 37]. Although these fields are much weaker than those associated with RF transmission in terms of tissue heating the time-varying electric potentials can depolarize excitable membranes, producing peripheral nerve stimulation (PNS) or, in extreme cases, involuntary muscle twitching [18, 22].

2.3.2 Heating as consequence of Radiofrequency pulses

Regarding the heating concern, the RF field produced by an MRI scanner consists of both magnetic and electric components. Where, following Faraday’s law, the time-varying magnetic field induces electrical currents within conducting loops or materials. Because there is a significant difference in electrical conductivity between the metallic implant and the surrounding human tissue, these currents experience Joule losses [18, 33]. This energy is dissipated as heat directly into the adjacent tissue, causing a localized rise in temperature.

For straight metallic implants like epicardial leads, the antenna effect is a dominant mechanism for heating. The tangential component of the RF electric field that runs parallel to the lead causes free charges on the conductor’s surface to migrate towards the tip of the lead. Then they accumulate creating a scattered electric field resulting in a much higher total electric field at the lead tips [33, 50]. Heating is further amplified if the physical length of the lead is comparable to the wavelength of the RF pulse in the body. Here, the lead can act as a resonator, creating a standing wave of current along its length with maximum heating typically occurring when the lead length is approximately half the wavelength of the RF frequency in tissue. For a 3 T MRI system, this resonant length is approximately 12–13 cm [33].

Finally, the extent of the heating action is determined by several variables, Specific Absorption Rate (SAR), lead configuration, and lead structure. SAR is used to quantify the power deposited per unit mass in the tissue; higher SAR levels generally lead to higher temperature rises [18, 33]. The shape, orientation, and termination significantly modify how the RF field couples with the lead [18, 33].

2.3.3 Risks associated with the Static Magnetic Field

In terms of the static magnetic field, the safety concerns are visible and much straight forward. The main magnet is the most distinguishing feature and is simultaneously the greatest source of risk associated with the device. This immense strength can cause everyday objects to behave in non-intuitive and dangerous ways, most notably turning ferromagnetic items into high-velocity projectiles [18].

A ferromagnetic object experiences a translational magnetic force that pulls it toward the scanner [17, 18]. This force is not strongest at the center of the magnet. Rather, it is determined by the spatial gradient of the B_0 field. This aligns with theoretical insight of ASTM F2052-15 [7]. Similarly, objects like those with asymmetric shapes, such as cylinders or needles, experience torque. Torque is the force that tries to twist an object so that its long axis aligns with the magnetic field lines and is dependent on this field with ASTM F2213-17 explaining some methodologies [8]. In the following section a derivation of the equations needed for the quantification and comparison of these forces is given.

In linear media, the magnetic flux density is directly proportional to the applied field strength. For magnetic dipoles $\mathbf{m}$ (or objects well-approximated as dipoles, such as cylindrical shapes), the mechanical interactions with the magnetic field, are governed by the following fundamental expressions from classical electromagnetism,

$$\mathbf{F} = (\mathbf{m} \cdot \nabla)\mathbf{B}. \quad (2.3)$$

$$\boldsymbol{\tau} = \boldsymbol{m} \times \mathbf{B}. \quad (2.4)$$

where $\boldsymbol{m}$ is the magnetic dipole moment and B is the magnetic field.

These equations represent the general physics of dipole-field interactions and hold universally for induced or permanent dipoles in the low-susceptibility regime relevant to MRI safety testing. The torque expression (2.4), in particular, is independent of any specific measurement protocol; however, its practical evaluation in the laboratory, such as the determination of $\boldsymbol{m}$ or the orientation relative to the field, depends on the experimental setup and will be detailed further in the context of our measurements. For weakly paramagnetic materials, the induced magnetic moment magnitude can be estimated as

$$m_0 = \frac{1}{\mu_0} \chi_m B_0 V. \quad (2.5)$$

where χ is the magnetic susceptibility, V is the volume of the material, and μ_0 is the permeability of free space. With these foundational expressions laid out, we can now develop the specific physical framework for the translational force relevant to our experimental setup.

2.3.4 Translational Forces induced by the Static Field

In the MRI environment, the dominant safety concern for non-ferromagnetic objects arises not at isocenter, where the static field $\mathbf{B}_0$ is uniform and $\nabla \mathbf{B} \approx 0$, but in the fringe field region outside the bore, where strong spatial gradients exist.

Consider a simple pendulum with a metallic object at its end. This object is free to swing along the z-axis of an MRI. A static magnetic field along that same axis is then applied. The pendulum would be pulled towards the magnet with a force F_m until it reaches equilibrium with its weight mg and tension T_s .

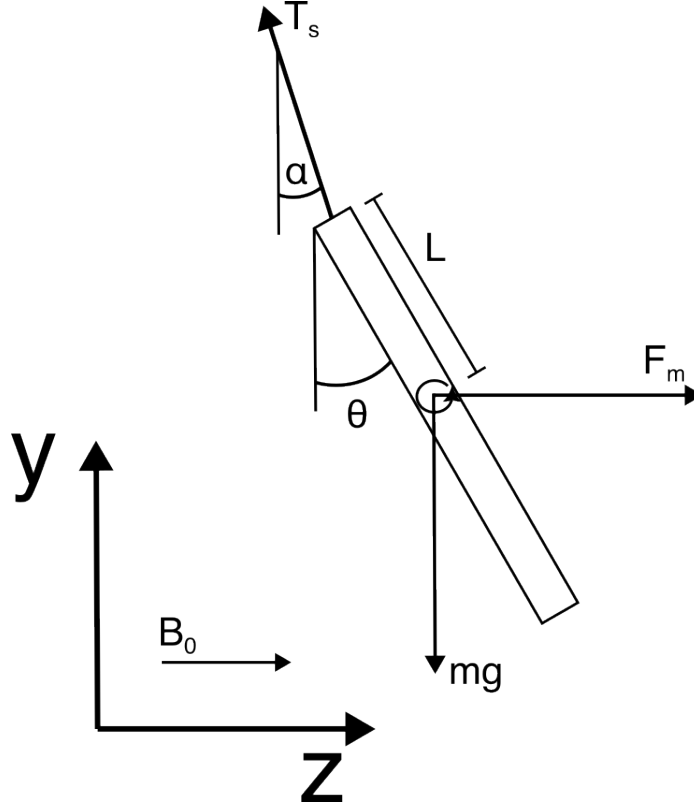


Figure 2.7: Free-body diagram of device in magnetic field (adapted from ASTM F2052-15 [7], Appendix X2).

Figure 2.7 shows a complete free body diagram. After applying Newton's laws at equilibrium the equation $F_m = mg \tan(\alpha)$. Noting that this solution is independent of the point of attachment of the string and holds if the center of magnetic force does not coincide with the center of mass [7]. The magnetic force on a paramagnetic object F_m is also determined from equation (2.3). By using the magnitude of the dipole moment equation (2.5) we can write the magnetic force in a single dimension as

$$F_m = \frac{\chi_m V}{\mu_0} B \nabla B \quad \Rightarrow \quad F_m = \frac{\chi_m V}{\mu_0} B_0 \frac{dB_0}{dz} \quad (2.6)$$

therefore following ASTM F2052-15, the translational force exerted on a device due to a static magnetic field spatial gradient can be evaluated through angular deflection.

In the pendulum, test subject is suspended in equilibrium, and the balance of forces leads to

$$\frac{F_m}{F_g} = \frac{F_m}{mg} = \tan(\alpha), \quad (2.7)$$

where α is the deflection angle from vertical, F_m is the magnetically induced translational force, m is the mass of the object, and g is the gravitational acceleration.

This is valid even in cases when the magnetic force does not operate exactly at the center of mass. As a result, the approach is reliable for evaluating a variety of devices. The formulation of equation (2.7) explains why the measured deflection angle α (and thus F_m) depends not on χ or B_0 independently, but on their product. Practically, this insight justifies the lab methodology. Although magnets in our lab, a solenoid or neodymium magnets, generate small magnetic fields (only about 20 to 300 Gauss), we can use ferromagnetic samples, with proportionally larger χ_m to replicate clinically relevant force-to-weight ratios (F_m/F_g). An estimation for the expected χ_m can be found on appendix B.1 and summarized in table 2.2 below.

The measurable magnetic deflection depends on the ratio between the mechanical term and the magnetic force term. To reproduce the same deflection, 1 cm, at laboratory field strengths much smaller than those in MRI systems, the product must be preserved. Because $B_0 \nabla B_0 \propto B_0^2$ for a geometrically similar field, scaling the field down by a factor of 10^2 requires increasing the sample's effective susceptibility by roughly 10^4 to maintain the same mechanical response.

Table 2.2 summarizes the scaling. For typical solenoids producing 30–300 G the factor of χ_m/ρ spans from approximately one hundred-thousandth to one-thousandth of that found in ferromagnetic materials. Consequently, to appropriately mimic translational forces under these laboratory conditions, a realistic value of χ_m for a ferromagnetic substitute should lie between one and ten.

Table 2.2: Order-of-magnitude summary of laboratory scaling for required susceptibility. Values shown are representative from a scaled solenoid model in figure ??.

B_0 (T)	$B \cdot \nabla B$ (T ² /m)	χ/ρ	χ_m
3.0×10^{-3}	1.707×10^{-5}	7.21×10^{-3}	5.05×10^1
3.0×10^{-2}	1.707×10^{-3}	7.21×10^{-5}	5.05×10^{-1}
3.0×10^0	1.707×10^1	7.21×10^{-9}	5.05×10^{-5}

According to ASTM F2052 [7], a deflection below 45° occurs when the magnetic force is less than the object’s weight. Though not a strict acceptance criterion [31]. In contrast, a nonmagnetic stainless steel wire ($\chi_m \approx 10^{-4}$) might show a magnetic force of 30% of the objects weight, whereas ferromagnetic metals like nickel can exceed two million percent. That is more than enough to become dangerous projectiles unless restrained [33, 31, 18].

In the other hand, the effect of the static field on torque are expected to be less significant than the pull as it only dependent on the field strength and not its gradient. As the curl is decomposed into the torque magnitude we observe that only components parallel to the field have an effect on the it, with that in mind lets address the problem.

2.3.5 Torque induced by the Static Field

Consider a metallic rigid cylinder shaped subject (a wire or rod) free to rotate, in a low friction surface, and align itself to a static magnetic field (B_0). The cylinder experiences a torque and has magnetic dipole moment according to eq. (2.4). For a thin, linear paramagnetic material, the induced magnetic moment depends on its orientation relative to the static magnetic field. It can be expressed as

$$\mathbf{m} = m_0 \cos(\theta), \quad (2.8)$$

where m_0 is the maximum induced dipole moment, which occurs when the material's axis is aligned parallel to the B_0 field ($\theta = 0^\circ$). When the material is oriented perpendicular to B_0 ($\theta = 90^\circ$), the effective component of the dipole moment along the field direction vanishes. This follows from the projection of the dipole moment vector onto the magnetic field axis.

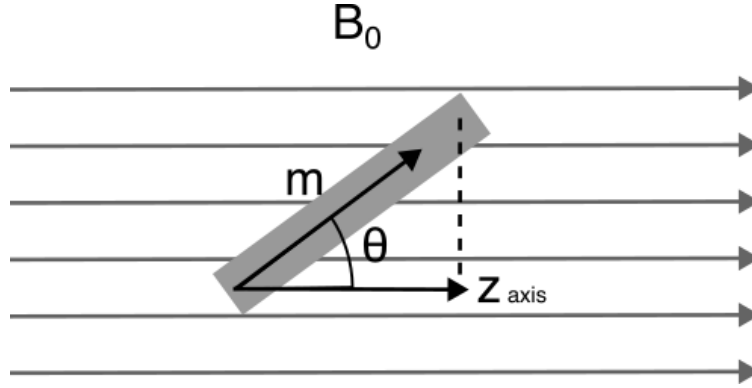


Figure 2.8: Vector diagram of a magnetic dipole $\mathbf{m}$ in a uniform magnetic field $\mathbf{B}$ directed along the z -axis.

In Figure 2.8, the dipole of length a , experiences a torque magnitude $|\boldsymbol{\tau}| = |\mathbf{m} \times \mathbf{B}| = mB_0 \sin \theta$. From the geometric projection of the dipole moment onto the magnetic field direction (eq. 2.8), and the torque experienced by the dipole we can write

$$|\boldsymbol{\tau}| = m_0 B_0 \sin(\theta) \cos(\theta) = m_0 B_0 \frac{\sin(2\theta)}{2}.$$

Using equation 2.5 for the dipole moment, we then write

$$\tau(\theta) = \tau_{max} \sin(2\theta), \text{ or} \quad (2.9a)$$

$$\tau_{max} = \frac{\chi_m B_0^2 V}{2\mu_0}. \quad (2.9b)$$

Once the torque is found, a direct comparison to the body’s weight can be made. According to ASTM F2213 [8, 24], the criterion (c_τ) for evaluating magnetically induced torque on medical devices in a magnetic resonance environment is

$$c_\tau = \frac{\tau_{max}}{mgl}, \quad (2.10)$$

where τ_{max} is the maximum torque measured or obtained by equation 2.9b, meaning that the maximum magnetically induced torque is comparable than the product of the device’s longest dimension (l) and its weight (mg). Under the ASTM gravity-based acceptability criterion, a device is considered acceptable when $c_\tau < 1$.

With the tools developed in this chapter, such as comparison against weight equations (2.7, 2.10), we aim to estimate the magnetic susceptibility and quantify the force and torque of ferromagnetic objects as they interact with lab-size magnets. We also project these forces and torques onto materials with much lower susceptibility in much higher magnetic fields for clinical applications, keeping in mind the proportionality shown in Table 2.2.

2.4 Concluding Remarks

This chapter has reviewed the essential theoretical foundations for evaluating MRI safety in patients with retained epicardial leads. MRI fundamentals, static field (B_0), RF excitation, gradient encoding, and relaxation (T_1 , T_2), were explained to provide enough information about the scanner function. To assess the interaction with implants, the emphasis was on the static field. It is capable of inducing translational forces and torques on conductive implants, and the physics that rule the interaction seeking to characterize such forces. Magnetic susceptibility (χ_m) was used to classify materials as diamagnetic, paramagnetic, or ferromagnetic, with most human tissues and the primary constituents of modern epicardial leads (MP-35N conduc-

tors, platinum-iridium or titanium electrodes, and polymer insulation) occupying the low-to-moderate paramagnetic range.

Epicardial leads, as parts of CIEDs, differ from transvenous leads in placement, design (unipolar vs. bipolar, coaxial vs. coradial), fixation (suturing vs. screwing), and material composition (paramagnetic electrodes and conductors; diamagnetic insulation). Their direct attachment to the epicardium and frequent abandonment (often as 5–30 cm fragments, 0.5–1.5 mm diameter) raise concern for mechanical interactions with MRI scanner fields, despite low-to-moderate magnetic susceptibilities that suggest minimal risk [17].

Safety hazards associated with the three primary field components were systematically examined. Gradient fields produce acoustic noise and possible peripheral nerve stimulation, radiofrequency pulses can induce localized heating via the antenna effect, and the static field generates both translational forces, maximized at bore entrances where spatial gradients are steepest, and torques on conductive implants [17, 18].

Current guidelines (HRS 2017, ESC 2021, CMS) remain conservative, considering systems with abandoned or epicardial leads non-MRI-conditional due to insufficient data, despite evidence that post-2000 leads may be safe under monitoring [21, 29].

The material presented here serves as a physical framework needed to move from theoretical prediction to empirical validation. Quantifying static-field forces and torques on epicardial lead under controlled conditions in the next experimental chapter addresses the evidence gap that currently restricts MRI access for a growing patient population.

Chapter 3

Materials and Methods

This chapter details the experimental setups and measurement procedures developed to quantify MRI-induced translational forces and torque acting on post-surgical epicardial pacing lead when exposed to its static magnetic field.

Three non-magnetic test fixtures were constructed, specifically designed to measure these forces with the precision required for clinical applications, in accordance to ASTM guidelines [8, 7, 9, 10]. First, a spatial field mapping platform to characterize B_0 and ∇B_0 . Alongside but independently, a pendulum-based deflection apparatus to assess translational force through angular displacement. Finally, a strain sensor with a low-friction pole and lever arm measurement system for rotational force (torque) assessments.

All three fixtures were developed to be tested in a laboratory framework first. As such, this chapter is divided into the three experiments we described. Each, constituted of three parts, a technical description of the fixtures developed, the in-lab testing, and a first clinical methodology. As we explain our setup, rationale, and proven modifications to the instruments, their sensitivity, and precision, in order to provide a reproducible map of action to test our instruments.

In the laboratory, the complete suite of instruments and associated control/acquisition software was constructed and tested in a controlled low-field environment (≈ 30 G). Although this field strength is two orders of magnitude below the 3 T clinical scanner, it was deliberately chosen to enable precise evaluation of the fixtures' accuracy, sensitivity, resolution, and repeatability without requiring scanner time. The laboratory setup also allowed direct observation of the fundamental physical phenomena described in Sections 2.3.4 and 2.3.5 on representative ferromagnetic test samples.

Importantly, both translational force and torque scale predictably with the magnetic field strength, $F \propto B_0 \frac{dB_0}{dz}$ and $\tau \propto B_0^2$. Consequently, the dimensionless relationships in the behavior remain identical between the laboratory (30 G) and clinical (3 T) regimes. We reproduce the same normalized force and torque signatures that will be encountered in the MRI suite by scaling the effective susceptibility χ_m and density ρ of the test objects (see Appendix B.1), that is using ferromagnetic objects under small magnetic field. The laboratory results are therefore directly valuable, they confirm that the instruments function as intended and that the physics is captured correctly before any clinical measurements are attempted.

As outlined in section 1.2, the several constraints of clinical MRI access necessitated thorough laboratory validation of the instruments and protocols beforehand. Through institutional collaboration, CHI CUMC Bergan Mercy Hospital, accommodated a dedicated measurement session on their Siemens Magnetom Trio™ 3T MRI scanner, in which measurements were conducted under radiologist supervision to ensure safety protocol around bringing custom fixtures into the MRI suite. Accordingly, the clinical session pursued the two complementary goals previously defined, namely to verify device performance and identify workflow issues. Moreover, a clinical session provided the first dataset on the MRI-induced translational forces and torques acting on post-surgical epicardial pacing leads, thereby closing the loop between bench validation and in-vivo applicability.

The first challenge presented when estimating the induced magnetic force in an object is to know the spatial behavior of the magnetic field around the object that experience the force given by equation 2.6. The next section, or first experiment, explores this behavior, explains how to measure it both in the lab and at the clinic.

3.1 Mapping the Magnetic Field

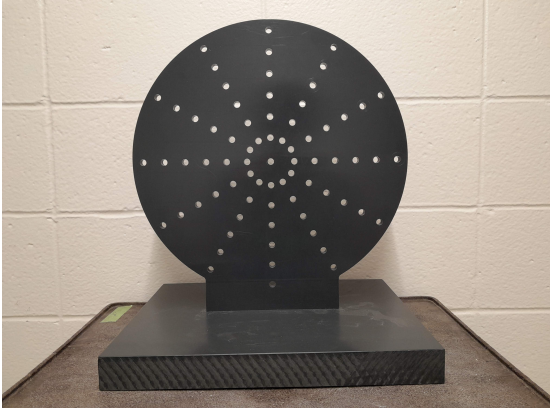
Magnetic field strength (B_0) and gradient (∇B_0) measurements were performed using a commercial Gaussmeter (Model GM2, AlphaLab Inc., Salt Lake City, UT). The GM2 operates on the Hall effect principle, wherein a voltage is generated across a standard axial probe when exposed to a perpendicular magnetic field. This voltage is proportional to the local magnetic flux density, enabling precise spot measurements of field magnitude. The gaussmeter used has an accuracy of 1% of the DC reading in the 16°C to 29°C range [51].

The first step in accurately mapping the static magnetic field of an MRI scanner is to know the precise position of the measurement probe at all times relative to the scanner bore entrance or isocenter. To achieve this spatial accuracy and ensure reproducible field measurements in a clinical environment, a rigid non-magnetic fixture constructed from PVC was developed [34].

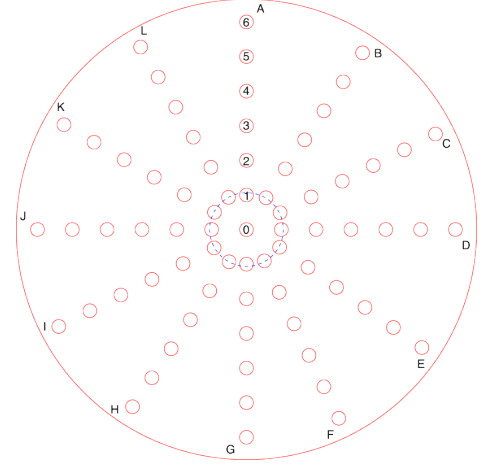
3.1.1 Mapping Platform

The fixture or platform consists of a 300(W) $\times$ 420(L) $\times$ 40(H) mm rectangular prism as base and a 20 mm diameter vertical circular plate, and has a total weight of about 9kg and height of 35 mm with its center at 21 mm for aligning with the MRI scanner isocenter. Its plate is perforated with a polar grid of probe-access holes, with 12 radial lines spaced at 30°. Each line contains 5 holes at uniform radial spacing of 20mm, sharing a single central hole as shown in Figure 3.1 for 49 different positions.

Each access point was numerically indexed to facilitate 2D field mapping in the X-Y plane.



(a) PVC-based mapping platform.



(b) Example field map output from MRI fringe region.

Figure 3.1: PVC-based magnetic field mapping platform. Used for spatial B_0 and ∇B_0 measurement. based on design from Ferreira, et al. (2017) [34, 30].

These holes in the fixture’s plate are used to insert the gaussmeter probe for data capture using LabVIEW, chosen for its flexibility and precision. Our expectation is to move the couch along the z axis with minimal disturbance, thus data acquisition protocol customization is needed. The manufacturer’s default software, AlphaApp, provides basic data-logging, recording, live screen streaming, and plotting of measurements from USB-enabled AlphaLab meters[52].

However, AlphaApp’s control over sampling frequency is limited by design, therefore limiting fine temporal resolution when the probe is moved into steep field gradients. In contrast, using LabVIEW Virtual Instrument (VI) interface allows for configurable sampling intervals, buffer control, and precise timing protocols, enabling higher accuracy as the probe approaches and crosses high gradient regions within the MRI fringe. We developed that VI by leveraging the AlphaLab Data Acquisition Communication Protocol [53] to command the device directly via a USB series port, providing control over timing and sample rate.

The entire fixture is shifted axially along the Z-direction of a horizontal bore MRI scanner using controlled couch increments (e.g., 10 cm or continuously). This translation allows for volumetric sampling of B_0 over several planes. The spatial magnetic field gradient ∇B_0 can then be estimated numerically using finite difference calculations between adjacent measurement points on a field strength map along the z-axis

$$\frac{dB}{dz} \approx \frac{B_2 - B_1}{z_2 - z_1} \quad (3.1)$$

where B_1 and B_2 represent the measured magnetic flux densities at positions displaced symmetrically from the location of interest. This fixture has previously been applied to MRI mapping in the work by Ferreira [30].

3.1.2 Laboratory size: Solenoid and Neodymium Magnets

Seeking to achieve the most reliable approximation of clinical MRI conditions, magnetic fields in the lab were generated using two approaches. The methods involved in our experiments used either neodymium permanent magnets or a copper-wound solenoid. Each approach presents advantages and limitations.

For example, a neodymium magnet can produce a very strong magnetic field (B_0 up to 1–1.4 T near its surface), but its strength at the sample/probe position depends heavily on distance from the magnet (falling off approximately as $1/r^3$) and results in a steep spatial gradient. These properties make neodymium magnets ideal for generating strong localized torque on small samples, where the sample can be positioned very close to the magnet to exploit the peak intensity without requiring power supplies or cooling. The fixed nature of the field, however, limits fine adjustments without physical repositioning.

In contrast, the magnetic field of a solenoid depends directly on the current flowing through its wire, allowing precise, continuous tuning of the field without mechanical repositioning. With appropriate coil design, solenoids can also provide better field uniformity over a defined region, making solenoids better suited for controlled translational force assessments, where variable field strength and gradient are needed to measure forces consistently across distances.

The approach used in this section was the latter. A custom-built air-core solenoid of approximately 15 cm in length and 10 cm in outer diameter was densely wound with copper wire (~ 1 mm diameter) meaning a winding density of 50,000 turns per meter, forming an inner core of approximately 5 cm. A Pasco Scientific SF-9584 DC power supply was used to deliver currents in the range of 0–4 A, corresponding to magnetic field strengths from 1 to 18.9 Gauss at about 8cm from the aperture along its long axis (z-axis) for this solenoid.

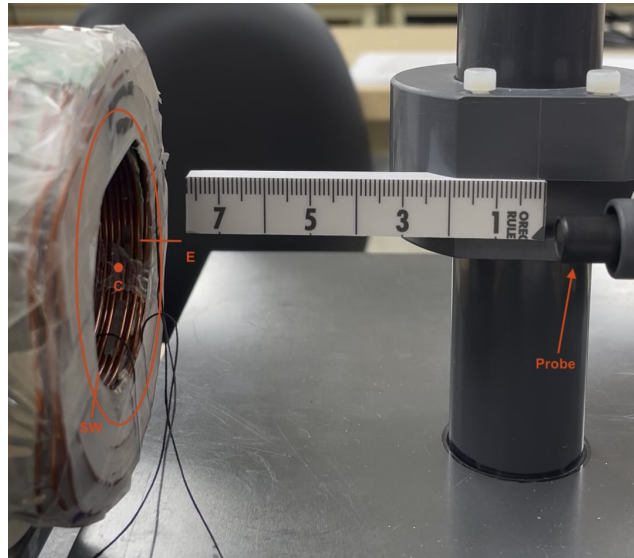


Figure 3.2: Probe mapping magnetic field generated from solenoid.

Figure 3.2 shows the experimental setup used to test the probe and obtain a reference map of the magnetic field of our solenoid. At the solenoid entrance, three positions were selected to place the gaussmeter probe to check for homogeneity, at the center ($r=0$ cm, $\theta=0$), southwest ($r=10$ cm, $\theta=225$), and east ($r=10$ cm, $\theta=0$).

This was achieved by using a translational stage positioning the probe 10 away from the solenoid’s entrance [54]. The stage uses 3 servomotors, one for each cartesian axis, to accurately position the probe with changes less than a tenth of a millimeter. However, the distance across the z-axis was limited to 11 mm by the servo’s maximum extension. This setup was used for experiment in section 3.2. To make a quick check across a wider range, a ruler was fixed at the same three positions described and the probe was steadily translated about 20 cm to the solenoid entrance and back. Data was recorded using the manufacturer’s default software, AlphaApp [52] with a sampling time of 1 reading per second, for about 2 min. Video recording was used to map time, and distance enabling a rough map of the B field and its gradient.

3.1.3 Clinical Work

To characterize the magnetic field gradient along the scanner’s longitudinal (z) axis, the circular grid fixture was mounted on the patient table. Couch motion controls were first calibrated, allowing precise positioning in millimeters via the four-digit display. Fixture-to-bore clearance was verified, and the fixture proved marginally over-sized for full insertion, constraining the maximum travel depth.

The gaussmeter probe was routed through the fixture plate into the MR scanner room with a dedicated acquisition computer, located in the control room. Measurement range was defined from 1 cm away of the bore entrance to approximately 1 m away.

Prior to each acquisition series, the fixture was aligned to scanner isocenter using the built-in laser crosshair and a calibrated ruler. Its position was permanently marked on the table surface for reproducibility. Grid coordinates examined were the central position (00) and off-axis locations A3, A6, G6, and G3 (vertical plane), with the physical heights of G6 and G3 recorded to ensure correspondence with trans-

lational couch motion. Uniformity across the grid was additionally confirmed by measurements at right-side coordinate D3.

At each grid coordinate, the couch was translated continuously while the gauss-meter logged data in paired passes, one advancing into the bore and one retracting, producing independent datasets for hysteresis assessment. In total, 15 raw data tables were acquired, four at the central position (00), two at each off-axis lettered coordinate (A3/A6, G3/G6, D3), and one repeated file following a LabVIEW crash. All files were saved as Comma Separated Values (CSV) with the structure illustrated at appendix D.2.

Post-acquisition, the CSV files were imported into Python using the pandas library. Column headers were standardized, and data integrity checks (duplicate removal, timestamp validation) were performed. The flux column was subsequently converted to physical units, and the longitudinal gradient was extracted by numerical differentiation and polynomial fitting routines to determine the optimal representation of the field profile for later modeling.

With the spatial behavior of B and ∇B explored, we move towards the test object itself and how it physically responds to that field. In the next section, experiment two, we present the fixture used to measure the actual displacement a small test object experiences, record the resulting deflection angle, and use that angle, together with Eq. 2.7, to compare the magnetic force directly against the object’s weight. At the same time, we take the measured force and work backwards using the field values to determine (or confirm) the material properties, such as effective susceptibility. This cross-check confirms that the underlying physics holds as expected. In the clinic, we assess deflection angle independently from magnetic field behavior providing first insight onto positive control materials and epicardial lead configurations.

3.2 Translational Force Measurement via Angular Deflection

To assess MRI-induced translational forces, our lab developed a pendulum-style apparatus or fixture following the geometry and physics outlined in ASTM F2052-15 [7]. This apparatus enables measurement of small deflection angles to derive displacement forces via gravitational torque balance (see Eq. 2.7). The following subsections describe the fixture design and features, the test objects used (including positive controls and epicardial lead holders), the smallest measurable force and associated limitations, and the measurement protocols implemented both in the laboratory (low-field solenoid) and in the clinical 3T MRI environment.

3.2.1 Measurement Device and test objects

Figure 3.3d presents CAD renders of the pendulum fixture (full assembly and detailed views of the upper and lower rings), created in CATIA to document the design intent and engineering drawings (full set in Appendix A.1). Figure 3.3 shows corresponding photographs of the constructed apparatus, with real-world implementation of these features.

The fixture was constructed entirely from PVC components, including a vertical post of about 1.5 meters and a rigid base of dimensions 30(L)×20(W)×4.5(H) cm for stabilization. The vertical post includes a set of two rings fixed with nylon screws, to adjust heights, a default suspension height of $L = 1$ meter between these two rings was used for consistency across trials, maintaining the geometry and proportions showed in equation (3.2). This distances were measured from the last contact point of the string to the rubber on the top ring to the sample hanging free at ruler’s height. The lower ring, in Figure 3.3c, features a slit where a custom made Kydex™ ruler is held by smaller nylon screws. This ruler is detachable and allows for easier transportation and

ruler size exchange. The upper ring in the CATIA model (Fig. 3.3b) was simplified for demonstration, however the final implementation shows a rubber clamp holding the string.

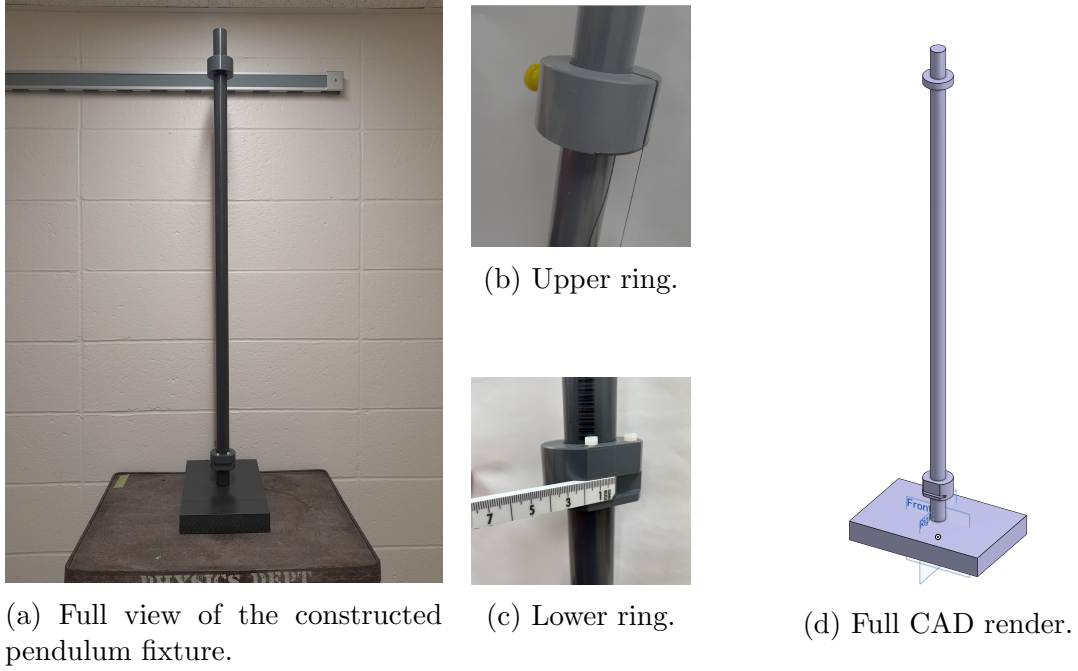


Figure 3.3: Full and detailed views of the pendulum fixture. Panel (b) captures the full ensemble CAD model. Panel (c) shows the upper ring was designed to secure the pendulum string with clamp and rubber attachment. Panel (d) shows the lower ring containing the slit used to align a 1cm-wide by 20 cm ruler perpendicular to the string.

After describing the physical implementation of the pendulum fixture, we now evaluate its measurement capabilities. In particular, we determine the smallest detectable translational force along the z-axis by analyzing the geometry of the deflected pendulum at equilibrium. Whether forces on the order of millinewtons, comparable to the weight of 1 g, can be resolved depends on the finite precision of the ruler-based deflection measurement and the pendulum's length.

3.2.1.1 Smallest measurable translational force

The smallest measurable translational force can be obtained by considering geometry. The pendulum from figure 3.3a forms a right triangle where the vertical line is the pole " L ", the horizontal gives the position on the pendulum " x " and the hypotenuse is the pendulum's string. Both of these measurements have a 1mm precision, yielding a $L=1\text{m}$ and $z=1\text{mm}$ triangle (Figure 3.4). Then, the smallest deflection angle capable of measurement is given by $\alpha = \tan(z/L) \sim z/L$ for small angle which is about of a 10^{-3} rad (~ 0.057 deg).

Since the angular deflection corresponds to the ratio of magnetic to gravitational forces (see eq. 2.7) the smallest measurable magnetic force is

$$F_{m,min} = \tan(\alpha_{min}) = mg\left(\frac{z}{L}\right). \quad (3.2)$$

For a 1g test mass¹, this corresponds to a detection threshold of $F_{m,min} \sim 10^{-6}\text{N}$ ($1\mu\text{N}$). Thus, the pendulum fixture can resolve magnetic force as small as one-thousandth of the object's weight.

This resolution defines the lower limit of measurable magnetic susceptibility using the pendulum method. For nonmagnetic or weakly paramagnetic materials, such sensitivity ensures that even subtle translational forces can be quantified, making the setup suitable for assessing device safety under MRI static field gradients.

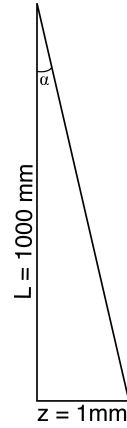


Figure 3.4: Right-angled triangle formed by the pendulum geometry at equilibrium.

3.2.1.2 Test Objects and Baskets

Testing for low field lab measurements a solid ferromagnetic cylinder was used as sample, showed in figure 3.5. Its dimensions were captured with caliper, ruler, and scale, yielding length = 6 mm , diameter = 3 mm , and mass = 0.3648 g therefore density $\rho = 8.6\text{ g/cm}^3$. The sample was suspended by a 955 mm -long sewing thread, tied

¹Sample calculations can be found on appendix B.2

with a simple knot at its center. For clinical applications, a 0.25 mm nylon monofilament is recommended. A monofilament nylon thread has low stretch, complies with MR-safe properties (Non-metallic, Non-conductive, Non-magnetic), and reduces variability introduced by torsional drift or thread elasticity [24, 7]. The bottom of the sample was free to swing, forming a pendulum under gravity effectively aligning with model on figure 2.7 and equation 2.6. A horizontal ruler, fixed along the Z-axis, served as the visual reference to quantify lateral displacements.

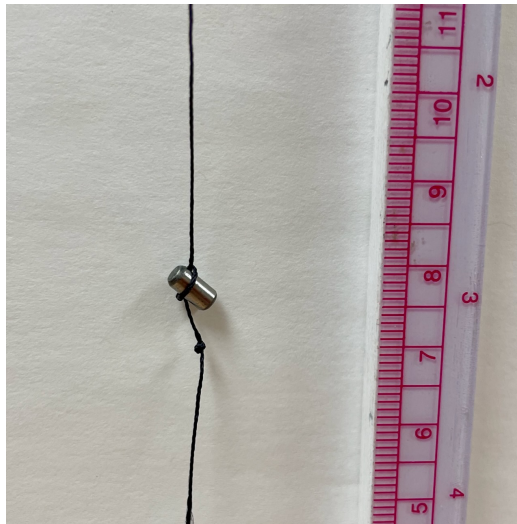


Figure 3.5: Ferromagnetic object and string attached, a ruler in centimeters and inches is showed for size reference

However, this ferromagnetic test object produces a strong response in the small solenoid field and serves mainly to demonstrate the sensitivity of the pendulum setup in the laboratory. To validate the measurement protocol with materials that are expected to exhibit negligible or zero translational force under clinical MRI conditions and to characterize the epicardial lead itself, positive controls were prepared from two common biocompatible, non-ferromagnetic metals, titanium and 316L stainless steel wire, both sourced from TEMCo Industrial (USA) and supplied on their standard spools [55].

The straight titanium wire (Surgical Grade 1) had a diameter of 1.02 mm, a cut length of 2.3 cm, and a measured mass of 0.1549 g with handling tape for adhesion.

The straight 316L stainless steel wire (annealed medical/industrial grade) had a diameter of 0.76 mm, a cut length of 2.2 cm, and a measured mass of 0.1690 g (including handling tape). An additional titanium coil (mass = 1.7964 g) was tested as a further positive control to assess geometry effects.

For the clinical epicardial lead measurements, the lead itself (mass = 0.1386 g, diameter = 0.46 mm, length = 378 mm) was mounted in a custom 3D-printed Polylactic Acid (PLA) basket described below.

To accommodate epicardial lead positioning while maintaining MRI compatibility, a lightweight, non-magnetic hexagonal comb mesh was developed and fabricated from PLA. The mesh dimensions were 16 (L) $\times$ 14 (W) $\times$ 1 (H) mm, for a total volume of less than 100mm^3 equivalent to around 100 mg given the 1.24 grams per cubic centimeter (g/cm^3) PLA density. Figure 3.6 shows a 3D render of the mesh, with the desired proportions. This mesh was 3D printed and has a final weight of 130mg. The mesh was suspended such that the lead axis was aligned approximately parallel to the magnetic field gradient. The mesh's lightweight design aimed to minimize its contribution to the total magnetic force, however its mass was not explicitly subtracted from the translational force calculations.

So far, we have described the materials used in both the lab and the clinic, from the solenoid (in Section 3.1.2) to the translational fixture and test objects for both environments. The following two sections explain the methodology followed in each of the two environments.

3.2.2 In Lab Measurement Protocol

The solenoid magnet was particularly useful for generating a variable magnetic field for the translational forces measurement tests. It allows for a range of magnetic fields strengths (B_0) at a given distance from the magnet without the need of displacing our fixture. The magnetic field was measured in a different way from Section

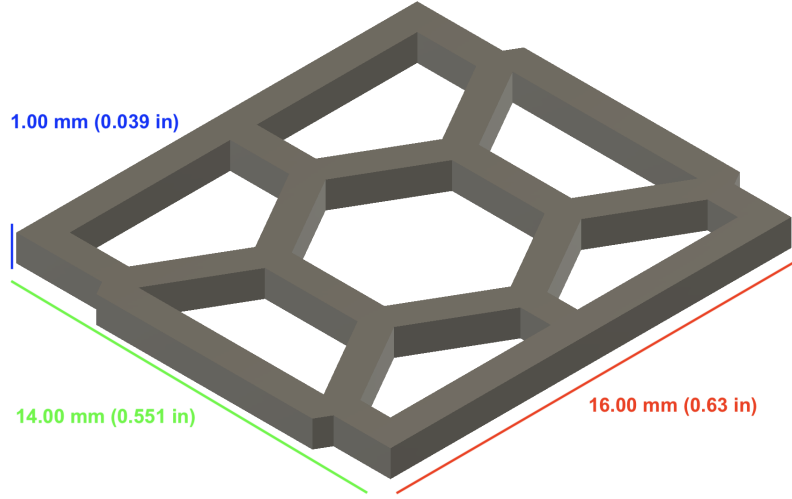


Figure 3.6: The design of the lead holder was made as a rectangular lightweight hexagonal comb mesh to wrap the epicardial lead around. It is held by the pendulum string

3.1.2. This time it was measured at a stable point along the pendulum trajectory for a given current or magnetic field (B_0) and then half a millimeter around that point to find the gradient as per Eq. 2.3. The gaussmeter in the VI was operated in a single mode; with rest periods between trials due to the time required to stabilize the pendulum after each adjustment. This methodology is detailed below.

The setup used our custom-made solenoid connected to a Pasco power supply capable to deliver up 8 A acting as variable magnetic field generator for our experiment. The test object, as described in section 3.2.1, was attached to a string pending from the MRI safe pole and angle of deflection was measured visually with a ruler on the subject horizontal displacement towards the coil. The gaussmeter was positioned to make magnetic field measurements at the same time as angle deflection.

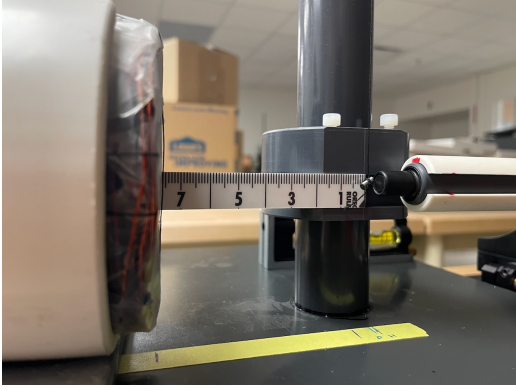
Figure 3.7 shows the experimental lab setup used to assess the methodology accuracy and reliability with all components. The magnetic field generation instruments are on the left side, whereas the gaussmeter with the probe on its translational stage is shown on the right side of the figure.



Figure 3.7: In-lab experimental setup for MRI translational force assessment.

As briefly explained in section 3.1 the translational stage uses three Thorlabs travel motorized actuators driven via the Kinesis software interface [56]. While Kinesis-controlled stages support sub-micrometer incremental motion and repeatability on the order of microns [54], our experiment only required positional accuracy on the order of 0.5 mm. Thus the stage system offered more than sufficient precision and stability for repeatable setup between trials. The translation speed was set up using Kinesis for 0.3 m/s and the maximum displacement of each actuators was 12 mm. Initially, the test object remains at rest (Figure 3.8a) once the current is flowing the magnetic field on the solenoid pulls the subject towards it (Figure 3.8b).

We chose increments of 1 mm from the resting position up to 11 mm. Induced currents remained low because the ferromagnetic body barely moved, the translation



(a) Subject hanging in pendulum at rest.



(b) Translational force pulls the subject towards the magnetic field source.

Figure 3.8: Comparison of the subject behavior with and without translational forces during magnetic field exposure.

stage's full range was used, the subject never entered the solenoid unintentionally, and the subject's position was always precisely known. For each displacement, the current required to produce a magnetic field strong enough to pull the pendulum to that point was measured and recorded alongside the field flux. A magnetic-field map was then acquired at each corresponding current level, ensuring the equilibrium position and its ± 0.5 mm neighbors were carefully sampled. With this method, the value of B_0 at equilibrium was known for every current level. The gradient was calculated for each point with the two neighbors. Appendix D includes sample data from these trials.

The measurements taken for characterizing the local field gradient, at the pendulum's equilibrium position (B_0), 0.5 mm below ($B_{-0.5}$), and 0.5 mm above ($B_{+0.5}$) were sampled at least nine times per current level, and gradients were computed using Python pandas library with the central finite difference method (Eq. 3.1) producing

$$\frac{dB}{dz} \approx \frac{B_{+0.5} - B_{-0.5}}{1 \text{ mm}}, \quad (3.3)$$

Using a 1-mm separation ($z_2 - z_1$) keeps the approximation in the linear regime while remaining resolvable on the field map. The 1mm increments in displacement was translated to angular displacement as

$$\alpha = \tan^{-1} \left(\frac{x}{L} \right) \quad (3.4)$$

Furthermore, to determine the magnetic susceptibility χ_m of the test sample. We use Equations. (2.6) and (2.7),

$$\chi_m = \frac{\mu_0 m g \tan(\alpha)}{V B_0 \frac{dB_0}{dz}}, \quad (3.5a)$$

$$\frac{\chi_m}{\rho} = \mu_0 g \frac{\tan(\alpha)}{B_0 \frac{dB_0}{dz}}. \quad (3.5b)$$

providing a direct estimation of magnetic susceptibility from angular deflection measurements, provided the field and its gradient are known. For experimental purposes, a plot of the deflection angle versus the product of field strength and field gradient allows one to extract χ_m from the slope of a linear fit, following

$$\tan(\alpha) = \frac{\chi_m}{\rho \mu_0 g} \cdot B_0 \frac{dB_0}{dz}. \quad (3.6)$$

In general, these forces are described as a function of the static field strength (B_0), its spatial gradient (∇B_0), and the material properties such as magnetic susceptibility (χ) and density (ρ). This equation later allows for plotting and interpretation.

All measurements were logged in Microsoft Excel, and photos/videos were captured via smartphone to document behavior. A minimum of three trials were performed for each current level; however, only the final (most consistent) dataset was retained for analysis. Data were exported to a Jupyter Notebook for clean up and processing with **pandas**, then plotted to evaluate $\tan \alpha$ vs. field gradient force. Propagated uncertainties in α and dB_0/dz were computed using standard error propagation formulas. All analysis code and data are publicly available [57].

3.2.3 Clinic Measurement Protocol

To quantify magnetically induced translational forces on implantable samples within the MRI fringe field, the custom translational fixture (Figure 3.3a) was positioned on the patient table. The fixture supported the lightweight baskets (Figure 3.6) suspended by a common sewing string, allowing samples to displace freely under magnetic force while measuring equilibrium position shifts relative to the zero-force state.

The translational fixture and its central basket were aligned to scanner isocenter using the room lasers, with the zero position permanently marked using tape on the couch surface. The distance from the couch edge to the fixture base was measured to enable reproducible placement of the equilibrium position. Due to the strong magnetic forces near the bore entrance, achieving no visible displacement of the empty basket was only possible either outside the MR room or with the basket only, positioned far from the 5-gauss line. This meant repeated repositioning of the fixture and multiple attempts to establish a true zero reference, adding considerable time to the protocol.

Measurements were performed using a titanium (Ti) sample as a positive control, given its known paramagnetic response in clinical MRI contexts. Three replicate trials were conducted with the Ti sample, recording both the distance from isocenter and the observed displacement on the fixture scale at each position. The couch was advanced in 50 mm increments toward the bore entrance and 100 mm increments away from it to capture the spatial dependence of the force.

The titanium sample was configured as a loosely coiled spring-like shape, oriented horizontally so that its long axis aligned with the scanner's z-direction (longitudinal field) and its diameter was perpendicular to B_0 . In contrast, the epicardial lead was coiled tightly and oriented vertically (perpendicular to B_0), a configuration dictated by its greater thickness and stiffness, which facilitated tighter packing.

Recognizing that coil geometry and orientation strongly influence induced force and torque, additional trials were performed using straight segments of both the titanium material and 316L stainless steel. These straight configurations followed the identical incremental positioning protocol.

In total, eight raw data tables were acquired, three for the coiled titanium sample, three for the coiled epicardial lead, one for the straight titanium segment, and one for the straight 316L stainless steel segment. All displacement measurements were documented manually and later digitized for analysis. Also publicly available [57].

Once the translational force and material properties are validated through the combined field mapping and deflection measurements, we turn to the second key interaction, the torque induced by the magnetic field on the test object. The following section, experiment three, describes the fixture developed to measure this torque with high sensitivity. This includes sensor fabrication and methodology development. We record the induced rotation (or counter-deflection), convert it to torque magnitude, and compare the result against the expected torque from the known field strength and the material properties already confirmed in the translational experiment.

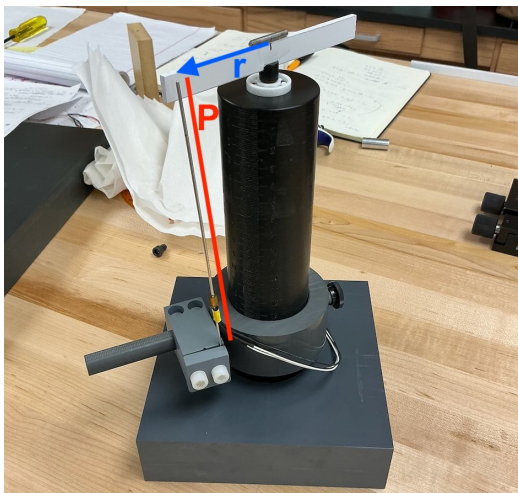
3.3 Torque Measurement via Counter-torque

The final part of this force and torque assessment involves the measurement of the magnetically induced torque on a horizontally oriented object. ASTM 2213-17 [8] lists five different methodologies, calculation based on measured displacement force method, low friction surface method, suspension method, torsional spring method, and pulley method. In section 1.1, the results on three of these methods were revised. Those included low friction, pulley and a variation of torsional spring methods [31, 24, 32], which showed good results but further considerations are needed.

Similar to suspension method or string method, we developed a torque measurement device capable of holding both a custom made sensor and test object. The next section will explain further specific physics and the fixture's components, which include sensor and circuit implementation explained in subsection 3.3.3.

3.3.1 Torque Measurement Apparatus

A non-magnetic fixture was fabricated with clinical considerations in mind. Figure 3.9 shows the main torque fixture, this fixture was designed to include a base and a ring to hold a sensor featured in Figure 3.9a. The main component is a pole made from Delrin[™], 200 mm tall with 51 mm of diameter. Underneath it a PVC base can be attached for support with a nylon screw. On top, it features a space for a non-magnetic ceramic, which is meant to reduce as much friction for the rotation as possible. Finally, around it a ring with a clamp will hold a custom made sensor which will be explained in detail in section 3.3.3.1.



(a) Main torque fixture pole. Key adjustable parameters for maximizing measured torque are the sensor's tip position P (red) and the distance from center to sensor along the transfer arm r (blue).



(b) Close up of the transfer arm on top of fixture (from bottom to top): magnetically susceptible test object, Kydex[™] transfer arm, plastic alignment notch, and ceramic bearing for low-friction rotation.

Figure 3.9: Non-magnetic torque fixture designed for horizontal subject orientation. The pole provides rigid support, the ceramic bearing minimizes rotational friction.

A problem to overcome is that, as explained in section 2.2, epicardial leads are malleable (they can be bent or curved into any shape). The model assumes a rigid dipole that spins uniformly with the magnetic field (B_0), for that reason a rigid torque transfer arm should be considered. Figure 3.9b shows the ceramic bearing and a plastic notch capable of holding this transfer arm. The transfer arm is 10 mm thick and is made from KydexTM similar to the ruler described in section 3.2.1.

The magnetically susceptible test piece experiences a magnetic torque following equation (2.9) and is exerted around the attachment point. The torque transmitted through the joint becomes a force, that can be measured with a sensor located at a known distance (r_{sensor}). The physics of this device follows that of a lever arm, $\tau_{max} = F_{sensor}r_{sensor}$, since the sensor stops the rigid arm making an angle of 90 degrees. Because torque is conserved along the rigid body, it provides a practical means to extend the moment arm beyond the base geometry ensuring sufficient space for accurate force sensing while maintaining constant torque transmission.

In Figure 3.9a the sensor is highlighted by a red line P and with blue line r showing its position relative to bending point and the rotation axis respectively. In this case, a thin rod stops the arm and bends following the cantilever beam deflection equations. These equations indicate that the strain at the fixed end increases linearly with both the applied force and the beam length, suggesting a mechanical advantage for measuring small forces. Then the strain on the sensor caused by the force applied is measured via strain gauges leading to resistance change. More on this is explained on section 3.3.3.1. A half-bridge Wheatstone circuit is configured so the output voltage corresponds to strain on the cantilever (see Section 3.3.3.2 for more details). Furthermore, the sensor and circuit design will determine the signal resolution. The system is linearly calibrated by applying known weights to the free end of the cantilever, yielding the expected relationship

$$\Delta V = k\Delta W \quad (3.7)$$

where k , the slope of fit, is the calibration factor between voltage change (ΔV) into the corresponding change in applied force (ΔW Weight). Implementation of this sensor can be found on section 3.3.4 and a further investigation on how small of a force can these devices measure is given in section 3.3.2.

Once the measuring device is described, we dedicate the next section to assess the smallest measurable torque with the custom made sensor in detail. Followed by the design of a circuit and sensor that aligns with this theory. The description of the circuit and sensor are additional work carried on during this project with the laboratory methodology laid out in section 3.3.5

3.3.2 Smallest measurable torque

For torque measurements, the sensitivity of cantilever-based torque sensors is determined primarily by the relationship between the applied tip load and the strain induced at the fixed end of the beam. Figure 3.10 shows a diagram for a thin cylindrical beam of length L , diameter D , and Young's modulus E , fixed at one end and subjected to a transverse point load F at the free tip. Classical beam theory [60] gives the maximum surface strain as

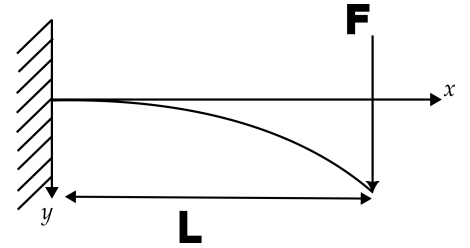


Figure 3.10: Cantilever Beam of length L with concentrated load F at the free end.

$$\epsilon = \frac{32FL}{\pi ED^3}. \quad (3.8)$$

This relationship indicates that the strain is inversely related to the Young's modulus and the cube of the diameter and directly proportional to beam length. As a result,

longer and thinner rods increase the observable signal by producing greater stresses for the same applied force.

Following the geometry used by Chiu *et al.* [58], a wire of $L = 69.85 \text{ mm}$ and $D = 0.64 \text{ mm}$ fabricated from Nitinol ($E = 66.2 \text{ GPa}$) achieved an empirical detection limit of 0.49 mN , corresponding to an output change of approximately 5 mV . Using Equation 3.8², this translates to a geometric sensitivity factor on the order of $L/D^3 \approx 2.66 \times 10^8 \text{ m}^{-2}$. This term isolates the geometry-dependent amplification of strain and therefore serves as a design metric for mechanical sensitivity.

A similar configuration can, in principle, achieve micro-Newton sensitivity when paired with a low-noise amplification circuit, provided the strain gauge and readout maintain linear response and the system noise floor remains below the equivalent strain corresponding to the smallest measurable force.

In the actual implementation, a metallic foil strain gauge bonded to the beam surface, which its position is shown in Figure 3.11 and 3.12, converts the mechanical strain into a measurable change in electrical resistance. For small elastic deformations, the fractional change in resistance is linearly related to strain through the gauge factor G_F [61, 62, 63]

$$\frac{\Delta R}{R} = G_F \epsilon, \quad (3.9)$$

where G_F is typically in the range of 2.0–2.2 for constant foil gauges. Combining Equations 3.8 and 3.9 gives the total resistance change due to an applied tip force.

$$\Delta R = R_0 G_F \frac{32FL}{\pi E D^3}. \quad (3.10)$$

This resistance change is converted to a differential voltage when the gauge is part of a Wheatstone bridge circuit. For a half-bridge configuration, utilized by Chiu, *et*

²Sample calculations on this limit can be found on appendix B.3

al. (2016) [58], where two active strain gauges form adjacent arms of the bridge, the small-signal output voltage under excitation V_{exc} is approximately [61, 64]

$$V_{out} \approx \frac{V_{exc}}{2} G_F \epsilon. \quad (3.11)$$

Assuming perfect linearity, little temperature drift, and perfect bridge balancing. The direct proportionality between applied force and bridge voltage may be obtained by substituting Equation 3.8:

$$V_{out} = 16 G_F V_{exc} \frac{L}{\pi E D^3} F. \quad (3.12)$$

In practice, the smallest detectable force is governed by the combined noise of the gauge, bridge, and amplifier system. If the noise-equivalent output voltage V_{noise} corresponds to a minimum detectable strain $\epsilon_{min} = 2V_{noise}/(V_{exc}G_F)$, then the force detection limit follows from Equation 3.8 as

$$F_{min} = \frac{\pi E D^3}{32 L} \epsilon_{min}. \quad (3.13)$$

Theoretically, the sensor can resolve forces in the micro-Newton range by optimizing the ratio L/D^3 , minimizing circuit noise, and utilizing materials with moderate modulus (such as aluminum or Nitinol). Then, assuming a bridge noise ($\sigma_V = 0.0005$ V) and the calibration slope for a particular sensor the smallest torque (σ_τ) or limits of detection can be found. These restrictions can be decreased by averaging (σ scales as $1/\sqrt{N}$), by increasing mechanical leverage (slope grows almost linearly with cantilever length), or by enhancing the bridge/amplifier noise performance. For quasi-static measurements, the limits of detection may be moved from tens of mg into the single-mg or sub-mg range using a combination technique (longer/thinner cantilever + averaging + low-noise amplifier); dynamic measurements will be constrained by

bandwidth and $1/f$ drift. In our lab we are limited by the materials available, meaning a test on different sensor was done.

3.3.3 Sensor and Circuit Design for Torque Measurements

A sensor paired with an amplifier circuit was made following the ideas described in Chiu *et al.*(2016)[58]. For our use case the sensor was fabricated from different non-magnetic materials to assess which would make the better sensor. While our circuit introduced the ability to zero the reference level at 0 V with two trimming potentiometers.

3.3.3.1 Sensor Design

Figure 3.12 shows a sample set of seven beam force sensors which were produced in the laboratory. The sensor is formed by two main pieces the beam itself and a pair of OMEGA[®] precision strain gauges (SGD-3/350-LY11) [59]. This strain gauges were glued on opposite sides of the rod around

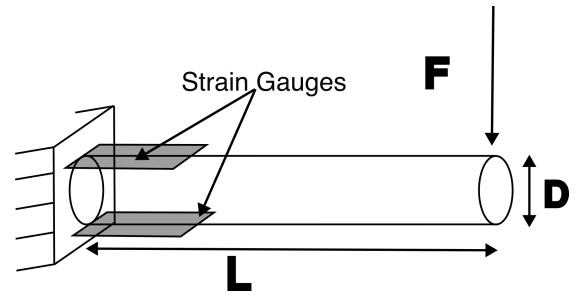


Figure 3.11: Cantilever setup showing the fixed cylinder and strain gauges "sandwiching" the rod at its base.

3 to 5 cm from its end, depicted in Figure 3.11. This two variable resistances or strain gauges are part of a Half-Wheatstone bridge configuration and are connected to the circuit with soldered jumper wires, described in section 3.3.3.2. The beam materials included glass, aluminum, titanium and two types of plastic. Table 3.1 summarizes their characteristics, they were made of similar sizes to test repeatability. Two of the seven available sensors were found to be irreparably damaged, so no further measurements were possible on them.

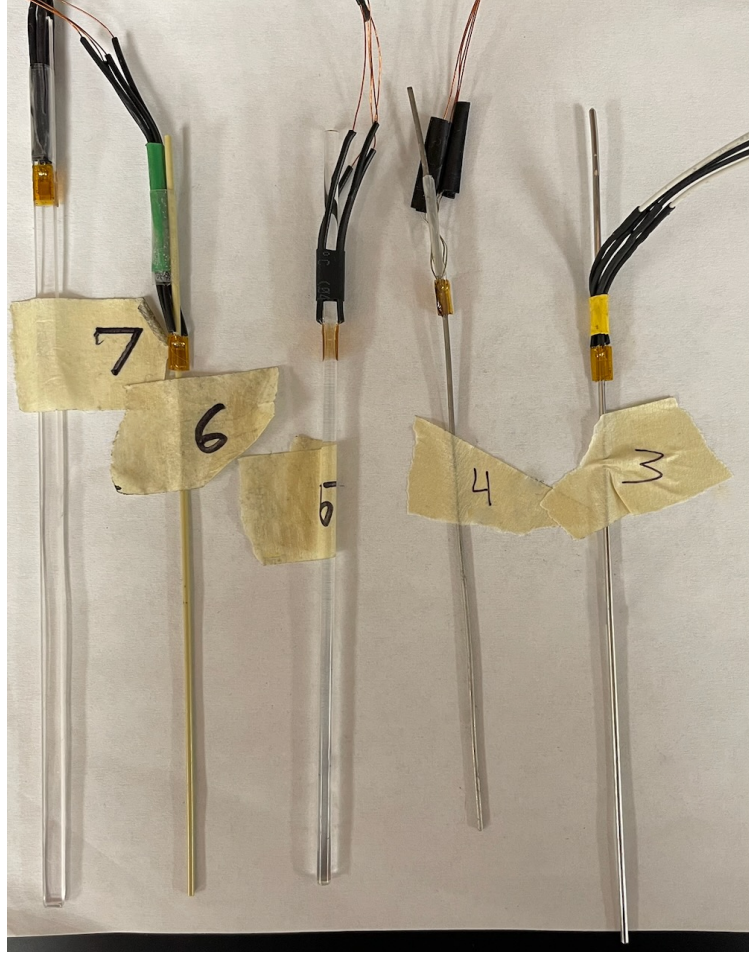


Figure 3.12: Rods fabricated during the torque experiment design. From left to right, (7) Clear Plastic, (6) Yellow plastic², (5) Glass, (4) Titanium ¹, (3) Aluminum

Table 3.1: Characteristics of the rods.

Rod ID	Material	Diameter (mm)	Length (mm)
Rod1	Aluminum (Al)	1.6	145.0
Rod2	Al	1.6	182.00
Rod3	Al	1.6	182.00
Rod4	Titanium	< 1.6	220.00
Rod5	glass	3.2	166.0
Rod6	yellow plastic	1.6	220.00
Rod7	clear plastic	2.9	180.0

3.3.3.2 Circuit Design

Distinctively, the introduction of the ability to zero the reference level at 0 V with two trimming potentiometers was implemented.

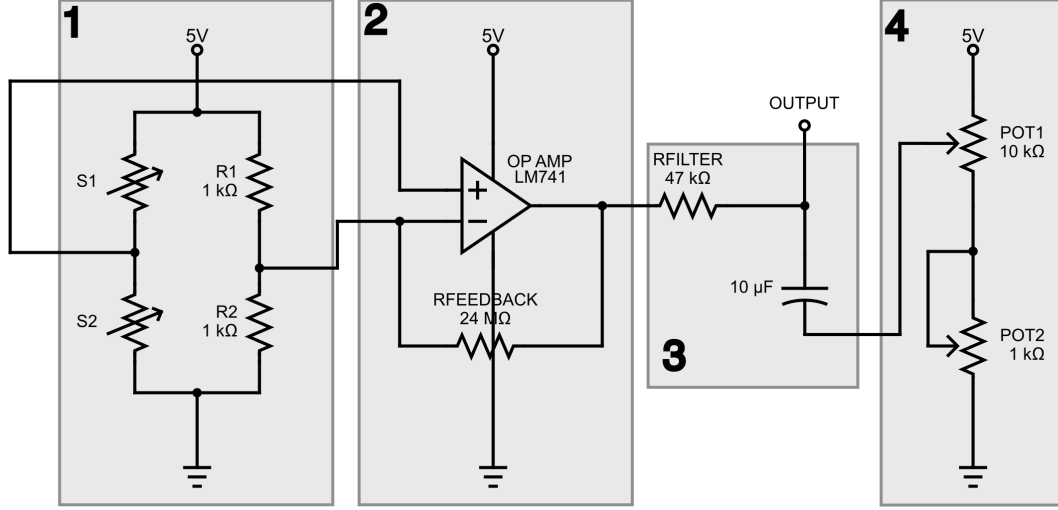


Figure 3.13: Circuit developed for the torque measurement setup, following the design principles of Chiu *et al.* (2016)[58]. Features include, 1. half Wheatstone Bridge Strain Gauge, 2. Amplifier, 3. Low-pass filter, 4. Coarse-fine adjustable reference network.

The measurements for the experiment in section 3.3 were done utilizing the already existing circuit prototype, a description and electronic analysis is given that followed its scrutiny. Figure 3.13 shows the circuit main parts, it includes four main sections, the Wheatstone bridge, the amplifier, a low pass filter and the ground offset potentiometers. From a Kirchhoff analysis for the first two sections, the transmission function was derived as

$$\frac{V_{out}}{V_{in}} = \frac{R_f}{R} \frac{S2 - S1}{S2 + S1} + \frac{S2}{S2 + S1} \quad (3.14)$$

where $R = R1 = R2$, $S1$ and $S2$ are the strain gauge and R_f is the feedback in figure 3.13. The output voltage of the circuit is driven by the relative difference between the two strain gauges compared to their total resistance. The operational

amplifier multiplies this small imbalance by the gain factor $\frac{R_f}{R}$. The bridge sets a baseline through the voltage division $\frac{S_2}{S_2+S_1}$, whereas the amplified differential term $\frac{S_2-S_1}{S_2+S_1}$ determines the sensitivity of the circuit to strain. For the complete derivation see appendix B.4.

After the instrumentation–amplifier stage, the output passes through a first–order RC low–pass filter composed of a 47 k Ω resistor in series with a 10 μ F capacitor to ground, as in Figure 3.13 part 3. The time constant is then $\tau = 0.47$ s meaning a 0.33 Hz cutoff frequency. Finally, a first–order system with this time constant has a 10 $\rightarrow$ 90% rise time of approximately 1.03 s. In other words, whatever the Wheatstone–op–amp transfer function generates, this RC stage only slows its dynamics rather than altering its steady–state gain. It suppresses high–frequency noise from the strain gauge bridge and smooths fast fluctuations that the rod could not physically produce anyway.

In part 4 of Figure 3.13, the DC reference entering the filter is set by the two potentiometers analyzed in Appendix B.5. Both pots sit in series, forming a programmable voltage divider from 0 to 5V, and the capacitor of RC network simply charges toward whichever reference these pots define. From the voltage-divider derivation in Appendix B.5, the node voltage obeys

$$V_{\text{ref}} = V_{\text{in}} \frac{R_2}{R_1 + R_2}, \quad (3.15)$$

so sweeping the fine pot moves R_2 over a narrow range ($\sim 0\text{--}1$ k Ω), which shifts the reference only within a small, controllable window. The coarse pot changes the large R_1 segment (~ 10 k Ω), expanding or collapsing the overall span.

3.3.4 Sensor Calibration

Prior to any torque measurement, each sensor or rod from table 3.1 was tested or calibrated to accomplish two things, ensure use of the most sensible sensor and to acquire the conversion factor or slope of voltage to weight explained in section 3.2.1.1. Additionally orientation and position of the rods was tested to ensure optimal values on these parameters for torque measurements.

For calibration in the lab, sensor was set in a cantilever position as shown in Figure 3.14, so the bar face horizontally and a small cone-shaped basket made out of tape was attached to hold increments of water. Water was added in μL increments (equivalent to milligrams) simulating torque loads from 0 to 800 mg. Initial increments of 10 up to 200 μL covered the expected force range of epicardial leads weight, followed by 50 μL increments to demonstrate linearity and sensor stability.

The change in voltage and water is then plotted and a fit can be found, see Section 4.3.1.

Then, in order to find the best position for sensor we set the following experiment, a set of neodymium magnets were aligned to the ceramic bearing to get a static magnetic field, and a platform angled at 45° carried the sensor as we moved it inwards. From section 2.3.5, it was shown that the maximum torque appears at this angle. Then we tested for the optimal length (L) and position from the center (r) as shown in Figure 3.9a. This process was repeated to find the sweet spot on distance from the center up to a 100 mm radius and the best height was tested to different lengths along the rod sensor.

3.3.5 In Lab Methodology and Setup

During the experiment, 2 measurements were performed. Magnetic field flux in Gauss using the AlphaLab Inc. Gaussmeter, described before, and Voltage from the strain gauge plus circuit sensor.

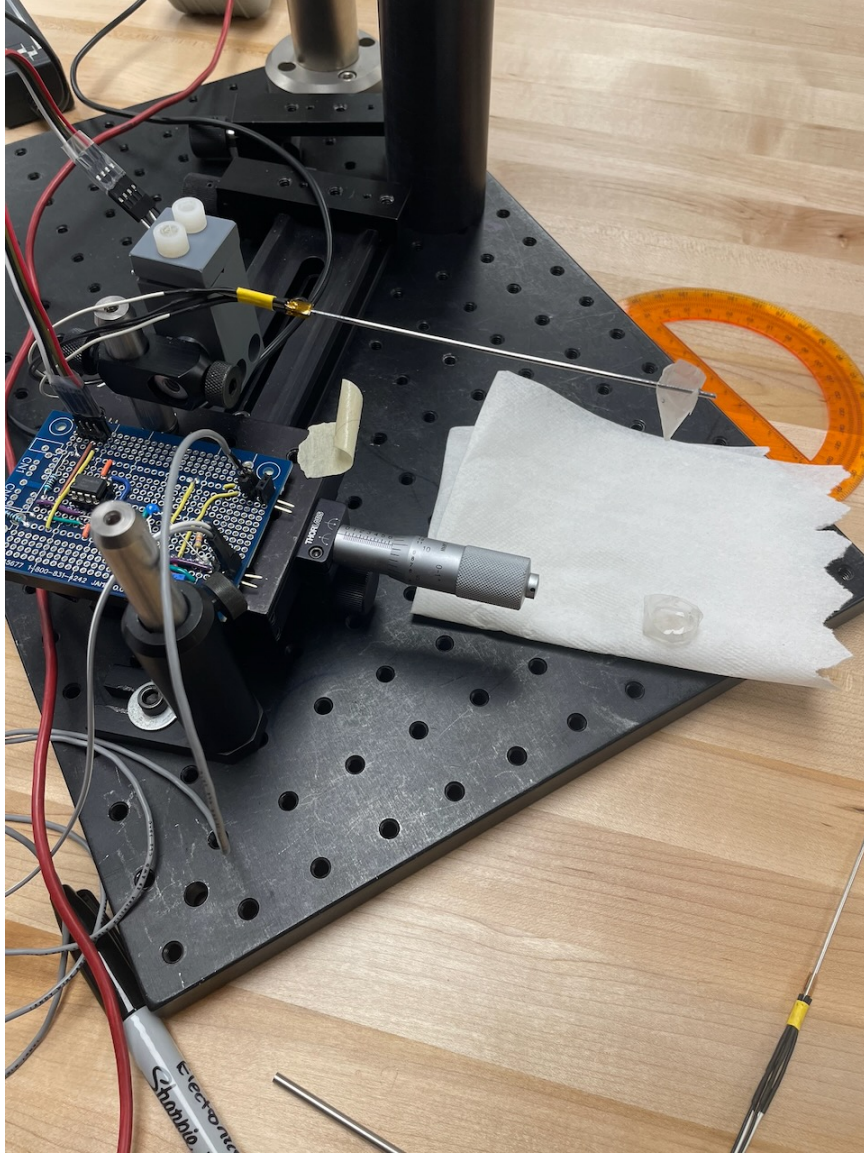


Figure 3.14: Circuit and sensor. The sensor was clipped with a custom made PVC clamp horizontally to act as a cantilever beam, a boat at length L was added, then a pipette was used to add water and see voltage changes with weight.

To show the effects of the magnetic field on the setup, a big ferromagnetic dowel was positioned on top of the ceramic bearing. It was around 20cm long and with it a strong dipole moment could be induced, serving as a visual confirmation that the system is working.

However a smaller dowel is needed, this dowel would experiment much smaller torques based on size only. And following the same logic as in translational this

would be a better test subject at lab scale. With a small magnetic field, high magnetic susceptibility and small volume, following equation 3.7 proportionality is ensured.

The experiment consisted of, with the sensor fixed in the ideal position, vary the magnetic field produced by the Helmholtz coils -mimicking the behavior in distance as a real MRI-increasing as we get closer to isocenter. Then the respective voltage produced by the deformation of the sensor is recorded. Finally, the voltage to weight calibration serves to estimate the induced torque of the system.

experiment done 3 times, 1st with helmholtz coils for 2 different lenght rods
and then with a set of 4 neodymium magnets in a different holder for the smaller rod.

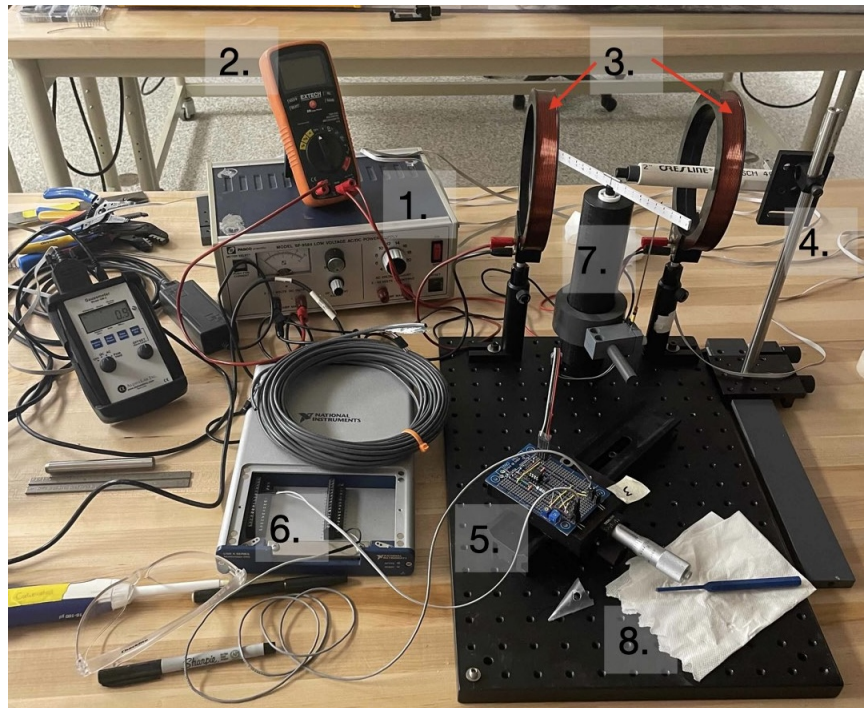


Figure 3.15: Experimental setup for torque. Includes magnetic field generation and measurement devices, 1 through 4. 5 is the circuit explained in section 3.3.3.2, 6. is a Data Acquisition (DAQ) system connected to a laptop and 7. is the main device where the dowel subject sits.

3.3.5.1 Magnetic field conditions

For the initial configuration, a permanent-magnet–based torque setup was implemented as shown in Figure 3.16a. This first arrangement employed four neodymium magnets, each magnetized through their thickness, with nominal dimensions of $1/4$ in thickness and $5/16$ in outer diameter. The magnets were mounted symmetrically on a rigid mechanical frame, with their magnetization axes oriented to produce a static magnetic field across the region occupied by the torque arm. Threaded lateral adjusters and a central support allow controlled positioning and alignment of the magnets relative to the rotating element.



(a) Initial permanent-magnet torque measurement setup. The apparatus employs four neodymium magnets symmetrically mounted on a rigid frame to generate a static magnetic field across the torque arm.



(b) Helmholtz coil configuration used to generate a controllable and approximately uniform static magnetic field at the coil center. The coils are driven with equal currents in the same direction to superpose their magnetic fields.

Figure 3.16: Comparison of the two magnetic field generation setups used for torque measurements: (a) a permanent-magnet–based configuration providing a fixed static field, and (b) a Helmholtz coil configuration enabling controlled variation of magnetic field strength.

A second setup (Figure 3.16b) utilized Helmholtz Coils EM-6722 from PASCO Scientific to create a controlled, nearly uniform static magnetic field that could be varied in strength and observed in real time. These Helmholtz coils consist of two identical coils, each wound with 200 turns, they were wired so current flowed in the same direction. They had an inner radius of about 10.06 cm and outer radius around 10.58 cm, and they sit on a optical base that lets us adjust the center-to-center spacing

close to where the field is most uniform (about one diameter apart). The coils used a maximum DC current of 2 A, which we supply with the PASCO Scientific SF-9584 DC power supply to increase the magnetic field incrementally; as current increases, so too does the magnetic flux density between the coils, allowing us to observe corresponding torque changes on our contraption.

3.3.5.2 In Clinic Measurement Protocol

Torque measurements were conducted ensuring consistency and reproducibility. Prior to setup, both researchers confirmed that all personal belongings and ferromagnetic items had been removed from their pockets. The VI was opened with the necessary parameters configured, and the data acquisition card (DAC) was powered on. A 10-meter jumper wire was connected to the DAC and integrated into the circuit, with the computer system interfaced to the DAC for real-time data collection.

The experimental apparatus was then fitted with the test object—either the magnetically susceptible calibration material or the epicardial lead—mounted on the rotating ruler. Then it was positioned on the MRI room bench at a distance of 1 meter from the bore, measured from the base of the apparatus to the bore face. Finally, the ruler arm was aligned to rest against the strain gauge, oriented as close as possible to 45° relative to the bore axis.

In the control room, torque measurements were acquired according to a predefined flowchart procedure, with results recorded on the measurement worksheet. A section of the worksheet was considered complete only when all data fields were marked with an “X.” Section 1 of the worksheet corresponded to measurements performed with the magnetically susceptible calibration object. Following these measurements, the apparatus was retrieved from the MRI room, and the calibration object was replaced with the epicardial lead affixed to the ruler. Section 2 of the worksheet was then completed by repeating the flowchart process with the lead in place.

Data collection in each section was performed at incremental positions along the MRI bore axis. For both the calibration object (Section 1) and the epicardial lead (Section 2), torque measurements were acquired at eleven translational positions: 100, 90, 80, 70, 60, 50, 40, 30, 20, 10, and 0 cm from the bore face. This systematic sampling allowed for direct comparison of torque response between the calibration object and the epicardial lead across the same magnetic field gradient.

3.4 Experimental Setup Summary

Chapter 4

Results

So far, the foundational layers have been set. Usually epicardial leads are labeled MRI-conditional at best [29, 27]. Leaving uncertainties on whether patients with these leads can undergo a MRI scan, despite the fact that these patient population is more likely to need one.

The different lead types, uni-polar vs bipolar, materials they are made of, size and configuration, and case by case complications pose a real challenge in the identification and testing of these leads under a MRI environment. Moreover, the effects on the leads are split by physical interaction. Being three the major concerns RF induced-heating, magnetically induced forces and torques, and electrical currents produced by gradient encoders.

Fortunately, the physics around such environment has been lay out. In Chapter 2 an explanation on the equations and expected measurement limits for the particular case of magnetically induced force and torques is given. Leaning into our methodology which has a three step approach.

For this chapter, we present the results found on each of the experiments presented in Chapter 3, that is Magnetic field mapping, Translational Force Measurements, and

Torque measurements. In each of these experiments, the focus is on both laboratory work and clinical work, as far as possible.

In all following sections the results are two fold. First, the laboratory work shows the expected behavior in a safe, controllable manner and allows us to predict the results on the clinic. Second, the clinical work will orient us and allow for assumption check and methodology fixes. This changes will be addressed in Chapter 5.

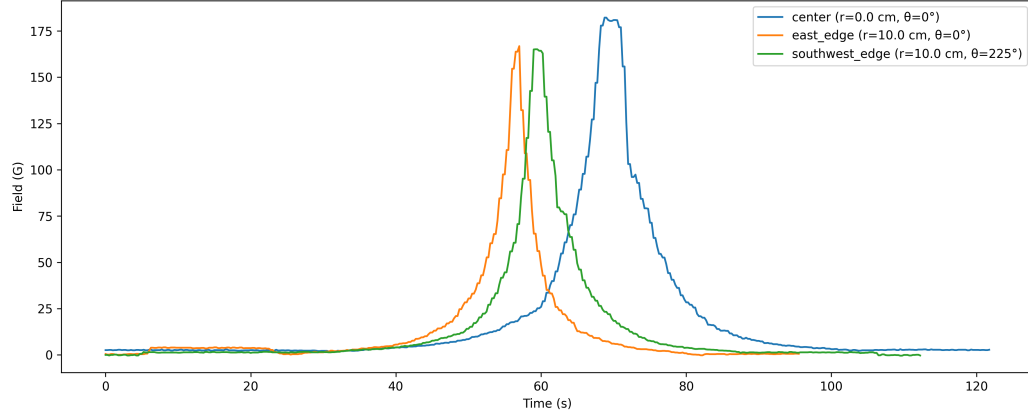
On our first experiment, mapping the magnetic field in both frameworks showed consistency with the expectations from theory at their respective scales. From this experiment, the extraction of both B and ∇B is crucial, as the results for the next two experiments will follow.

4.1 Magnetic Field Mapping

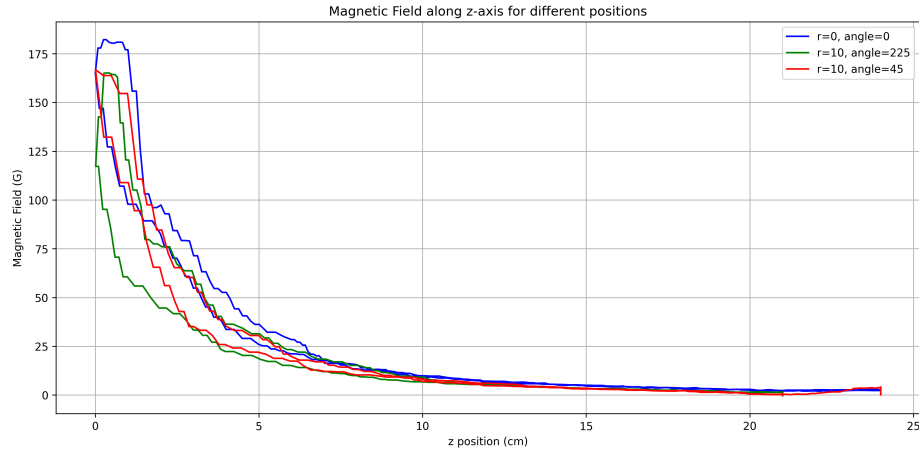
4.1.1 Mapping for a Solenoid in Lab

The magnetic field B generated in section 3.1, yielded to the following figures. In this particular set of measurements the main objective is to capture the spatial gradient behavior of an static magnetic field produced by a solenoid, which has a constant current flow throughout the experiment. Figure 4.1 shows two plots from the field mapped, Fig. 4.1a shows the magnetic field flux B in gauss as a function of time going in and out to the solenoid front face, while Fig. 4.1b shows the estimated respective positions of the probe along the z -axis.

The three colors represent the position of the probe on each trial, this can be seen clearly on Figure 4.2 where the map was plotted as a representative line for their respective side. It is important to notice the similarities between runs on the southwest side and the east side of the front as this expected homogeneity allow us to extrapolate the data around the solenoid similar to Fig ?? confirming the theoretical background and speeding our clinical magnetic field acquisition protocol.



(a) B field as a function of time.



(b) B field as a function of z -position.

Figure 4.1: Magnetic field measurements plotted against time and axial displacement. Used to estimate stability and spatial variation of the mapping setup.

Figure 4.3 shows both the calculated gradient from equation 3.1, and the multiplication by field strength B at the particular current of 2.966 A. The later being useful on experiment 3.2 for finding magnetic force, as per equation 2.6, and for its fit (see Eq.3.6) making one point out of the set in the translational experiment described in section 3.2.2.

In the translational experiment, the current applied to our solenoid was varied, and the magnetic field was measured in specific points within 7 to 8 cm away from the solenoid, this lead to a different results which are the points of equilibrium of the pendulum. These results are meaningful and will be used in section 4.2.1.

3D Magnetic Field Magnitude from Measurements with Solenoid Circle

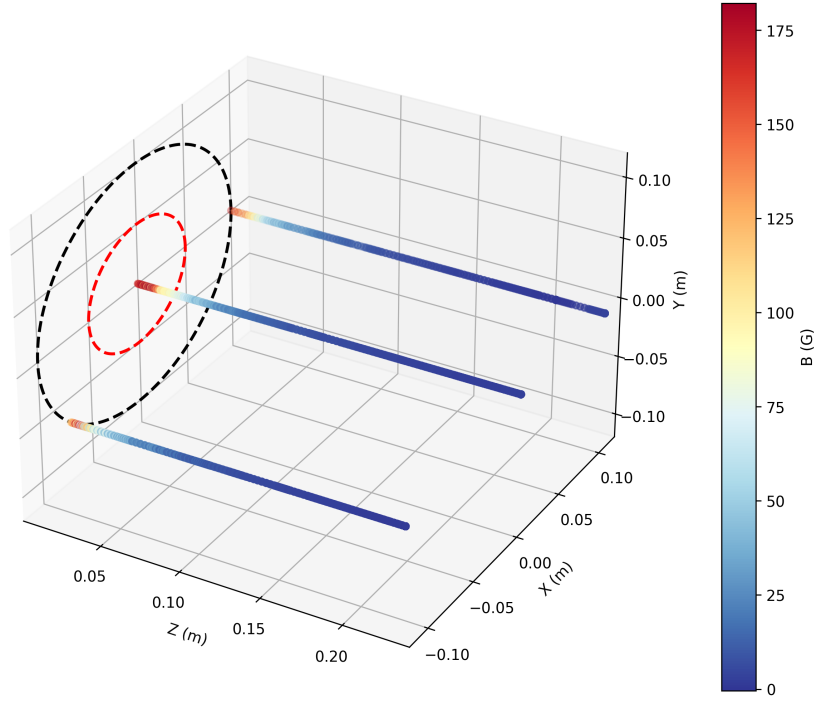
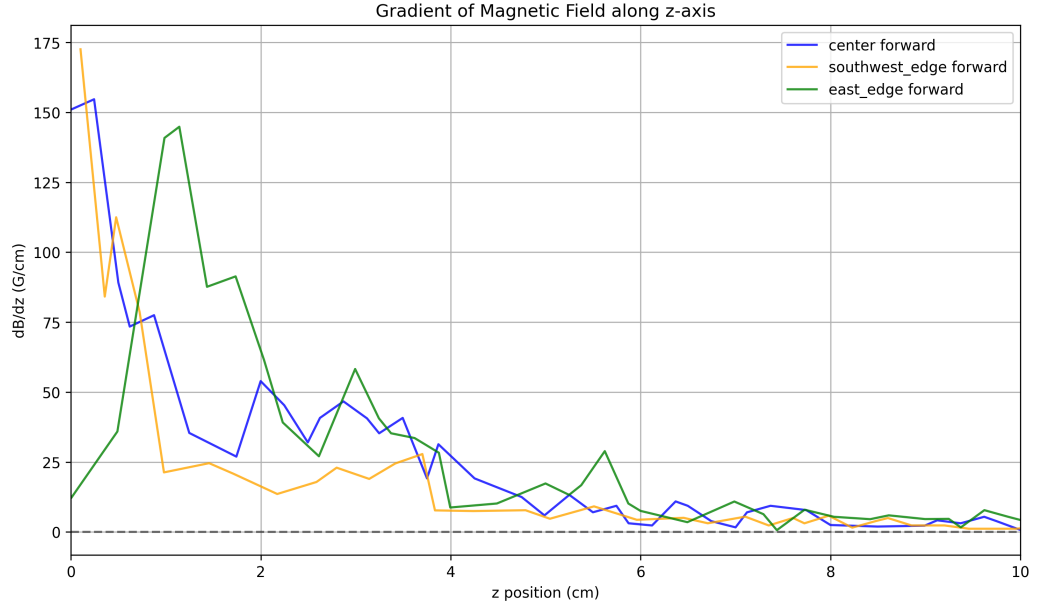
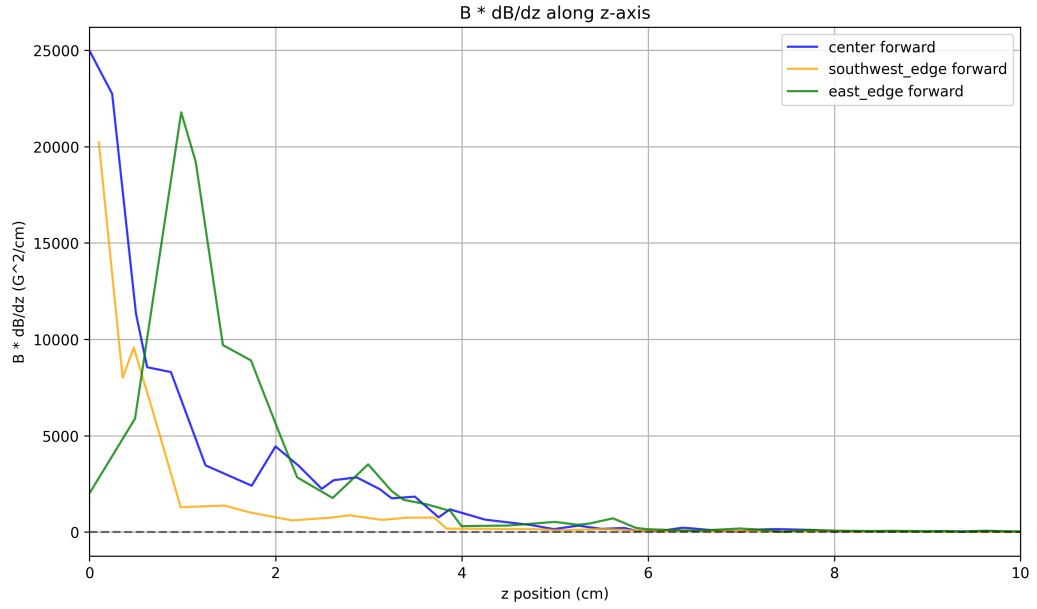


Figure 4.2: Three-dimensional reconstruction of the measured magnetic field, illustrating spatial variation across the mapping region.

As the measurement of magnetic field was done by varying current and not distance the magnetic field plots looks different of that showed by Figures 4.1 and 4.3. Figure 4.4 shows three plots, which corresponds to the three operations needed to make the field side of Equation 3.6. As highlighted, in this instance the x -axis, position, refers to the position of the pendulum's end with respect of its gravitational equilibrium point, and each point was taken with a different current applied to the solenoid while the pole remained stationary.



(a) ∇B along the measurement path.



(b) $B\nabla B$ product along the path.

Figure 4.3: Computed magnetic field gradients and the $B\nabla B$ product used for translational force characterization.

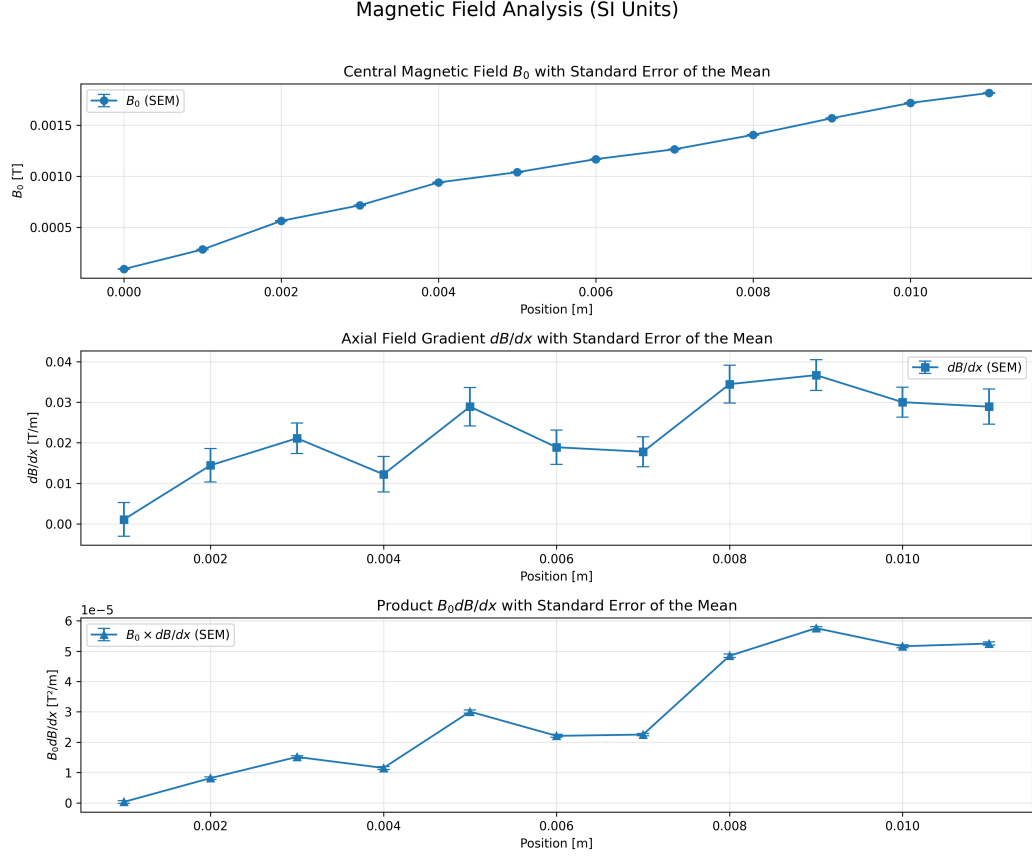


Figure 4.4: Central magnetic field B_0 , axial field gradient dB/dx , and their product $B_0 dB/dx$ as functions of position along the magnet axis. All quantities are shown in SI units (Tesla, T/m, and T^2/m , respectively) with error bars representing the standard error of the mean (SEM).

4.1.2 Mapping for 3T Siemens MagnetomTM MRI scanner

4.2 Translational

4.2.1 Translational forces from a Laboratory scale magnet (up to 180 Gauss)

Figure 4.5 shows a plot of the observed unitless displacement, $[\tan(\alpha)]$ against the magnetic field properties ($B \nabla B$) in T^2/m , which were determined from mapping the magnetic field. Figure 4.5a is a representation of Equation 3.6, it shows a linear fit

with a slope of 160.3186 ± 8.2631 and intercept 0.0160 ± 0.003 . Figure 4.5b, shows the residuals for the linear fit suggesting this fit is appropriate.

Following Equation 3.6, it can be quickly inferred that the magnetic susceptibility of the our sample is given by

$$\chi = m(\mu_0 \rho g) \quad (4.1)$$

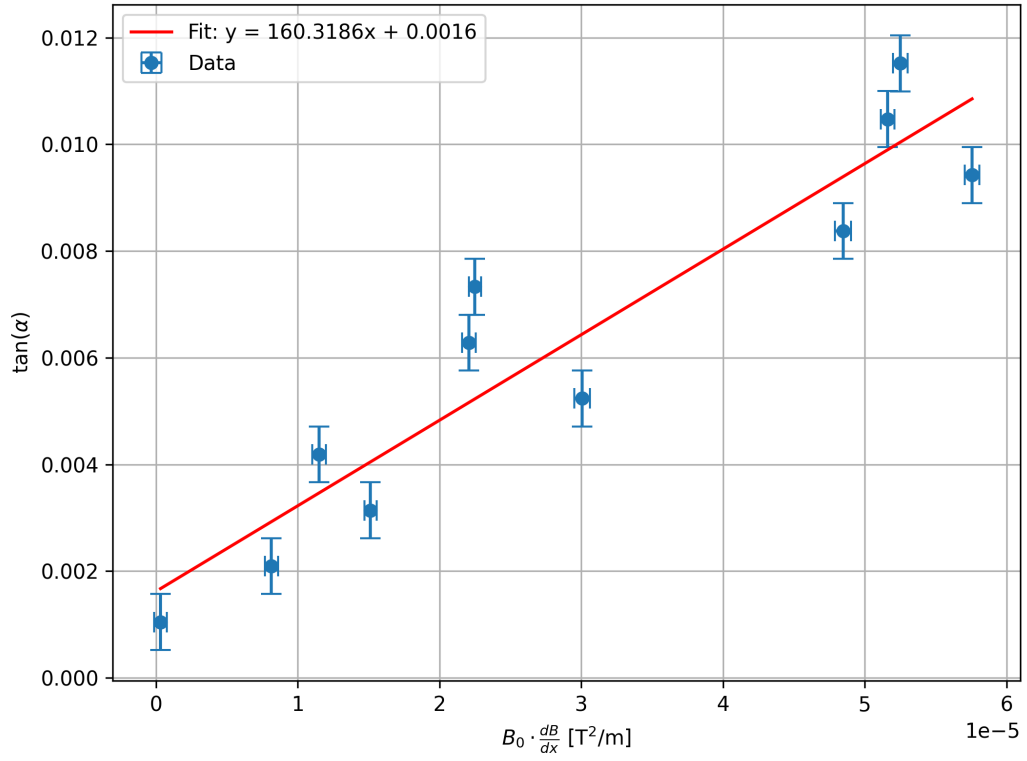
where m is the slope found with Figure's 4.5a fit, μ_0 the vacuum permittivity, ρ the material's density and g gravity. Sample characteristics are shown in Table 4.1. This table shows the measured quantities, length, diameter and mass with its calculated relevant parameters (Volume, Density $[\rho]$ and magnetic susceptibility $[\chi]$)

Table 4.1: Measured and calculated characteristics of our magnetic subject depicted in Figure 3.5

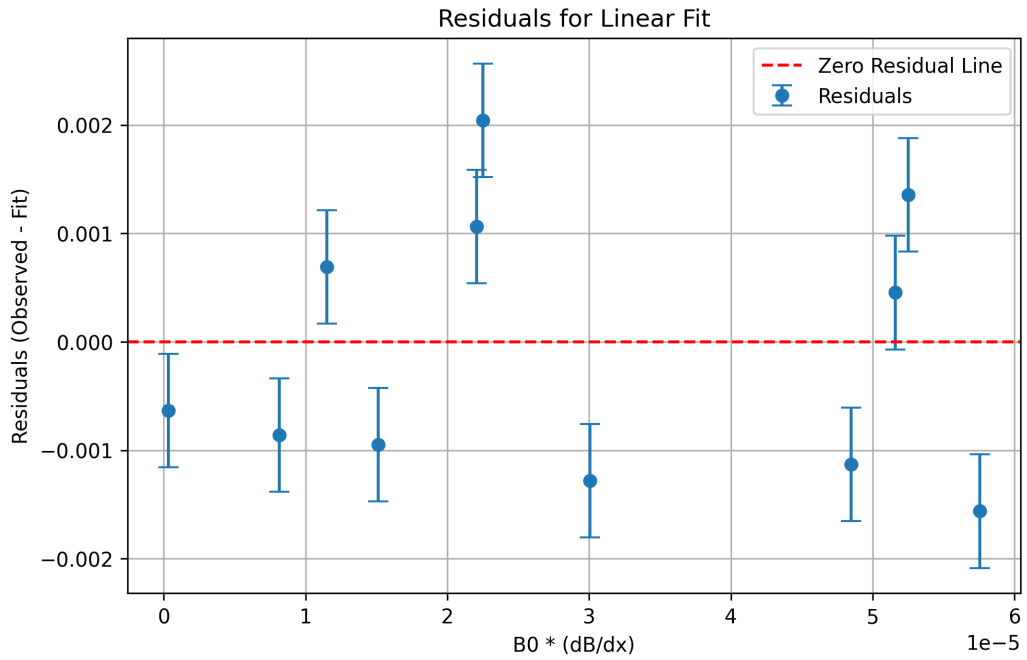
Property	Value	Unit
Length	6.00 ± 0.01	mm
Diameter	3.00 ± 0.01	mm
Mass	0.3648 ± 0.0001	g
Volume	$(4.241 \pm 0.029) \times 10^{-8}$	m ³
Density	8601 ± 59	kg/m ³
Mass magnetic susceptibility $[\frac{\chi}{\rho}]$	$(2.01 \pm 0.05) \times 10^{-3}$	m ³ /kg
Magnetic susceptibility (unitless)	17.29 ± 0.12	

Importantly, from Figure 4.5a, $\tan(\alpha)$ is in the range of 0.002 to 0.012. From Equation 2.7, and according to ASTM 2052 [7], the force ratio threshold for MRI safety indicates 1 or, equivalently, a deflection angle of 45° putting our measurements way bellow that threshold for our lab simulation. In the clinic, the magnetic field is about 1.7 hundred times bigger than what we found in our lab. With samples being paramagnetic, roughly 1 million to 10 million times smaller than what we found here.

4.2.2 Translational forces observed in a 3T Siemens MagnetomTM MRI scanner



(a) Curve fit of measured pendulum deflection vs. applied magnetic force.



(b) Residuals from linear fit of $\tan(\alpha)$ vs. $B_0 \cdot \frac{dB_0}{dz}$. Error bars indicate propagated uncertainties.

Figure 4.5: : (a) residuals from the linearized fit, and (b) fitted curve against the measured data points.

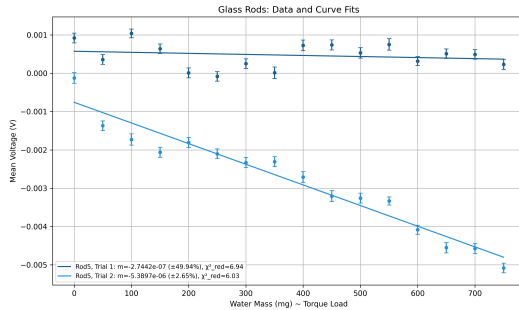
4.3 Torque

4.3.1 Sensor Calibration

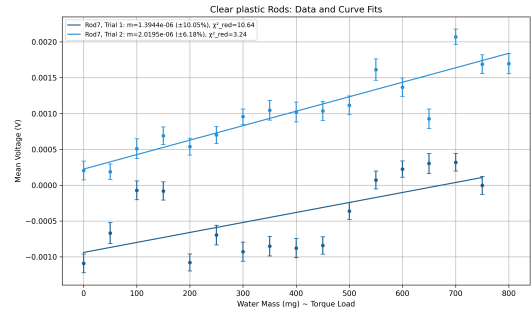
Out of Table 3.1, rods 4 and 6 (Titanium and plastic) were excluded from the calibration process. As noted, the Titanium was found not practical and the yellow plastic was found broken and needed re-fabrication, furthermore it was not possible to backtrack to its origin so uncertainty on its material makes non reproducible.

For these reasons, this section focuses on Rods 2 and 3 (aluminum), with only mentions for Rods 5 and 7 (glass and clear plastic, respectively). Aluminum based sensors proved to perform the best and allowed for repetition and documentation.

First, Figure 4.6 shows two of the trials done while calibration for glass and clear plastic. Their results shown unreliable, unpredictable changes in Voltage with weight applied. Additionally, tfor either of these two rods remained in the sub-millivolt regime, leaving the region of interest (around 0 to 200 mg) with measurements of $10^{-4}V$ too small to distinguish from noise, as explained in Section 3.3.2.



(a) Voltage readings for Rod5 (Glass) with up to 750 mg of water in increments.

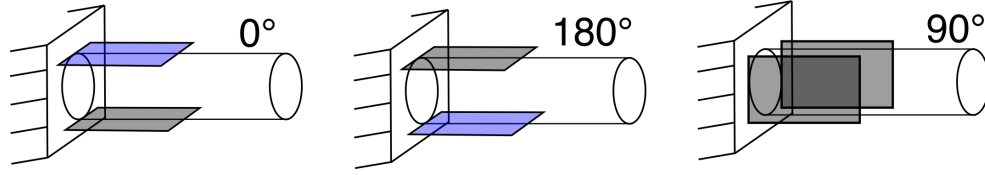


(b) Voltage readings for Rod7 (Clear Plastic) with up to 800 mg of water in increments..

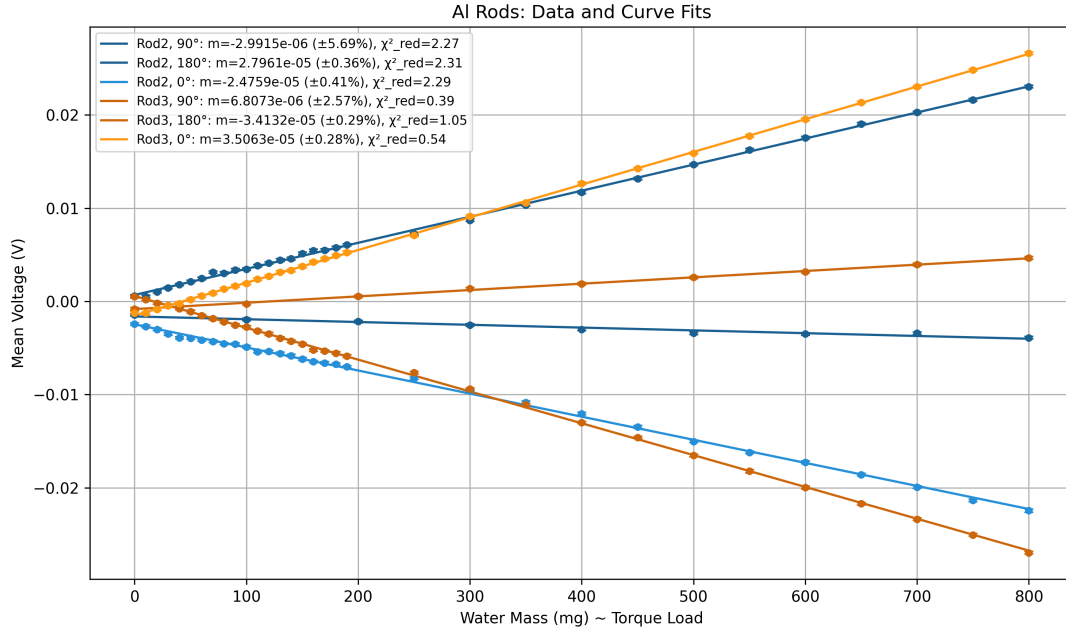
Figure 4.6: Trial examples for Rods 5 and 7. Both of them showed inconsistency across trials showing minimal to no change in voltage with weight added.

As an additional note, these two rods were the thickest of all available rods in Table 3.1. The observed behavior is expected as explained by Equation 3.12.

Next in our calibration process, the Aluminum rods, 2 and 3, showed clear linear trend along all trials and measurements. Figure 4.7 shows this linear trend for 2 identical aluminum rods. The plot shows Rod 2 and 3 color coded, blue and a alike for 2 and red for 3, to show the similarities between to identically fabricated rods. Additionally, this plot features three different sensor rotation configurations (Fig 4.7a), corresponding to three lines on each color or rod, where either 0° or 180° give maximum deformation of the strain gauges while 90° is not deforming properly.



(a) Three configurations tested, 0° and 180° are equivalent but flip voltage sign with respect of each other. 90° gives less sensibility as strain gauges does not bend in the proper axis.



(b) Calibration curves for two aluminum (AL) rods in three different orientations.

Figure 4.7: Linear fits show consistent response across orientations, confirming the reliability of the lab-built sensor.

From this result, the focus was on Rod 3 in either most sensible configuration, showing remarkable consistency across calibration trials. Focusing in the weight re-

gion of interest (0 to 200 mg), we found a similar slope than Figure 4.7 using Rod 3 in a different session a week apart. Figure 4.8 showcases this result and converts the previous units Volts per milligram of water to something more applicable to our interests (V/N).

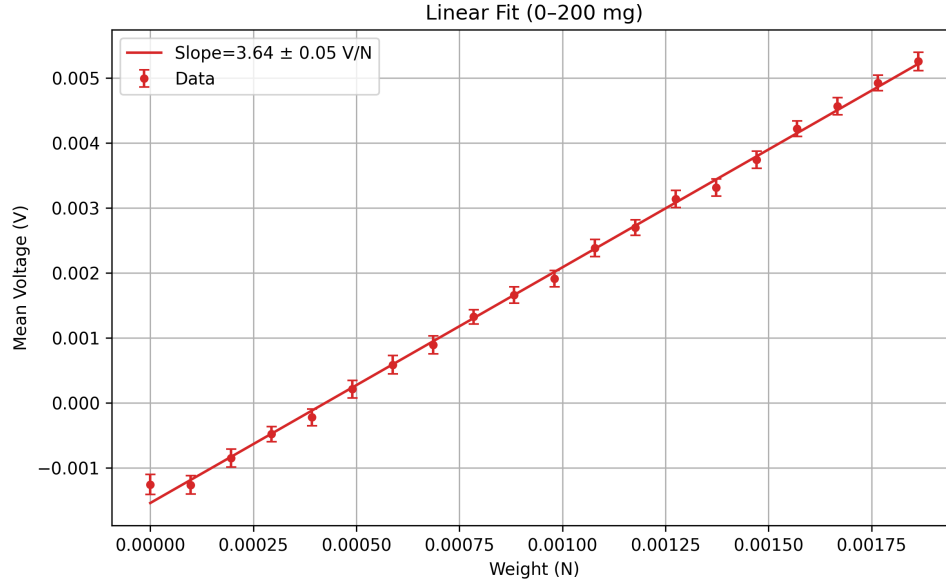


Figure 4.8: Calibration of Rod 3 (aluminum) around the region of interest, or weight of samples, for Rod 3 in the most sensitive configuration. Slope units converted from V/mg to V/N

From Fig 4.8 the smallest measurable force the instrument can be determined. Using the inverse of the slope $k = 1/m = 1/3.64 = 0.275 N/V$ we can find the smallest force detected. In this case, the smallest non-zero voltage was $\Delta V = 0.000885$ which results in $W_{min} \sim 0.243 \text{ mN}$.

Along with these measurements we also tested sensitivity along the length of sensor showing that the further out from the gauge bridge the more sensitive the sensor was. Figure 4.9 shows two lines, along the lever arm 90 and 50 mm away from the rotation axis position. Both showcase really small voltages for all forces applied at increasing lengths in the sensor. We can see that the closer to the strain gauges the force is applied the smaller the signal received, on the left side of the graph. However,

notice how the two lines deviate with distance increasing showing that at the tip of the sensor and closest to the rotation axis signal is higher for the same force.

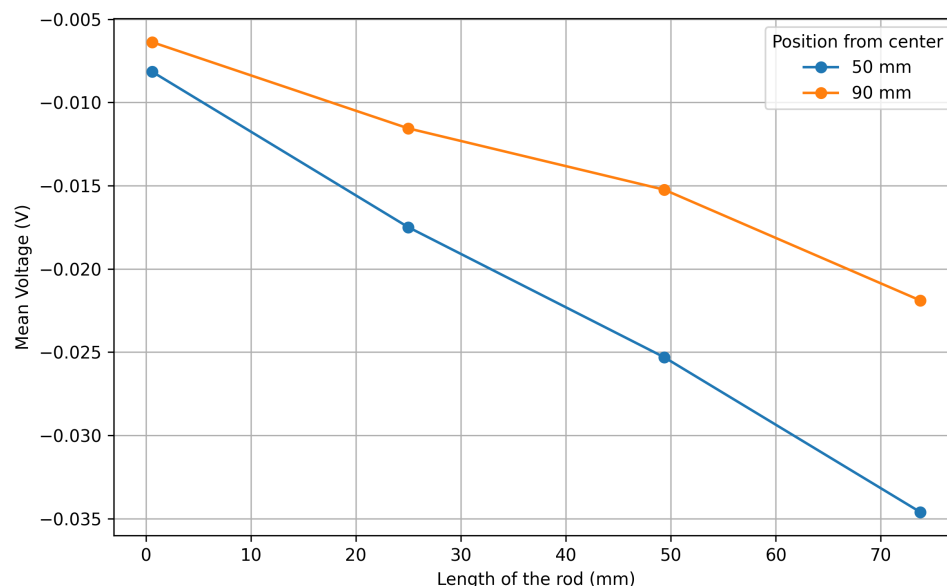
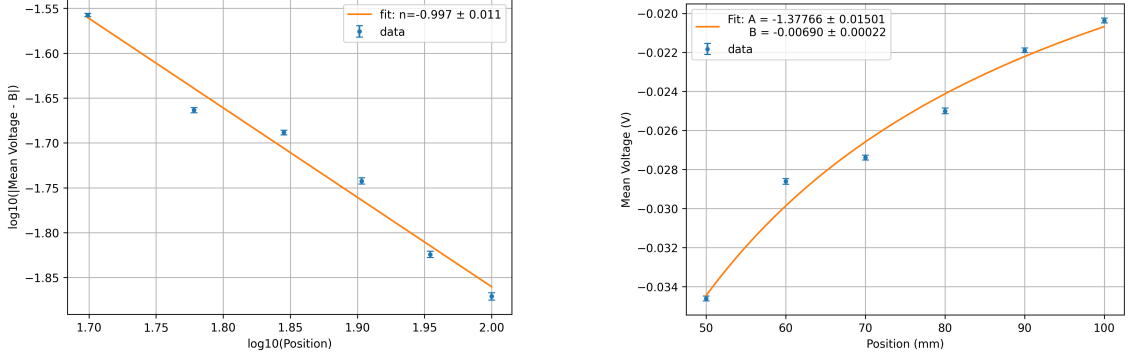


Figure 4.9: Mean voltage response of the sensor as a function of height for different distances from the center (50–100 mm). A small and relatively flat voltage is observed across all radii, while at larger heights (farther from the bridge) the measurements diverge, showing higher sensitivity near the center (50 mm). See Methods Section for details on torque transfer and distance selection.

The expected behavior of the lever arm is showcased in Figure 4.10. Similar to Figure 4.9, Figure 4.10 demonstrates the inverse relation of force and position from the rotation axis ($1/r$) in two ways. First Figure 4.10a shows a log-log plot voltage vs position with an fit n very close to -1. Additionally, a $1/r$ fit on Figure 4.10b provides the constant on top very close to -1 with almost no drift. Signifying, it is a good method to measure torque via the counter force.

Once the sensor and circuit were calibrated and the best position for measurement in the torque fixture are affixed, we moved on to the in lab measurements of torque.



(a) Log-log plot of effective measured voltage (proportional to force) versus position r , showing a power-law fit with exponent $n = -0.997 \pm 0.011$, very close to the theoretically expected value of -1 .

(b) Direct fit of the form $F(r) = A/r + B$, yielding $A = -1.378 \pm 0.015$ (very close to -1 in normalized units) and a negligible offset $B = -0.0069 \pm 0.0002$.

Figure 4.10: Demonstration of the expected inverse-distance dependence of the force (and thus torque) with respect to the distance r from the rotation axis. Both analyses confirm the $1/r$ dependence characteristic of the torque measurement via counterforce method.

4.3.2 Torque from Lab scale magnet

As described in section 3.3.5, torque measurements were performed under three different experimental configurations. When a small uniform magnetic field was applied using Helmholtz coils, the torque produced by the short (2 cm) magnetic dowel remained at or below the detection limit of the measurement system. For that reason, only the longer rod yielded reliable results. Figure ?? shows the maximum torque (N m) against the magnetic flux experimental data (in blue) exhibiting a clear parabolic dependence on B .

From the fit parameters, $k = 3.87 \pm 44.5$, $b = -5.56 \times 10^{-2}$ and $V_0 = 8.45 \times 10^{-5}$. The dominant quadratic term (kB^2) is in excellent agreement with the theoretical expectation for the torque on a magnetic dipole in a uniform field, where $\tau \propto B^2$. The linear coefficient b is very small and statistically consistent with zero within uncertainty, while the offset V_0 is negligible, confirming the absence of significant systematic bias in the measurement.

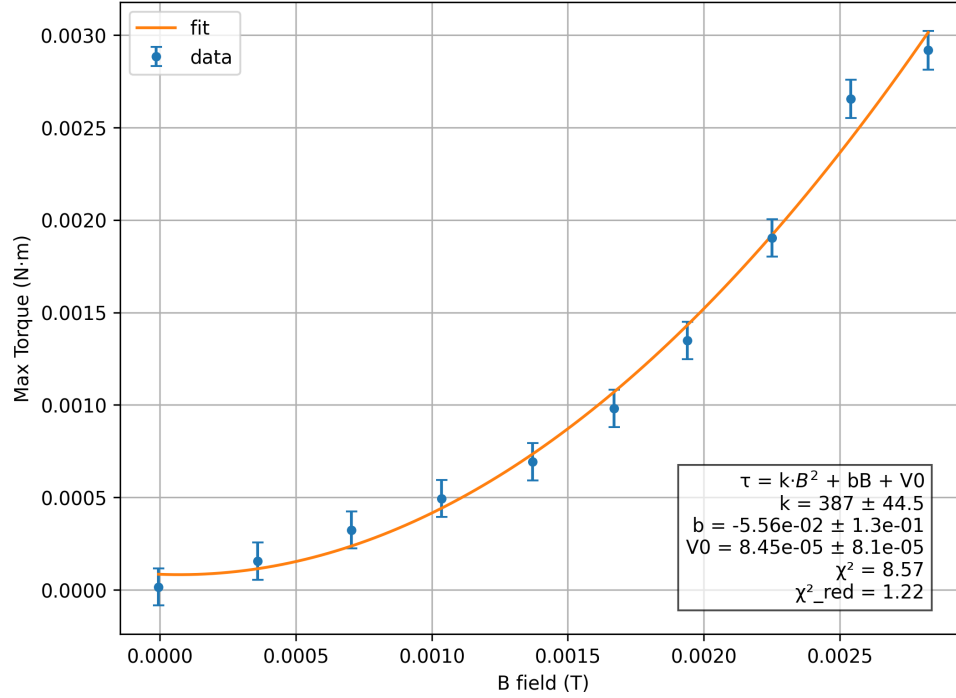


Figure 4.11: Maximum measured torque in Nm as a function of applied magnetic flux density in B for the longer rod under a Helmholtz coil configuration.

A second set of measurements was performed on the shorter (2,cm) magnetic dowel using significantly stronger magnetic fields generated by four neodymium magnets. Figure ?? shows a parabolic relationship, however the values on torque are smaller for higher magnetic fields

This time the fit yielded,

4.3.3 Torque from 3T Siemens Magnetom™ MRI scanner

Once in a clinical environment, both the circuit and the MRI-safe torque measuring fixture were functioning properly and circuit calibration was appropriate for clinical measurements.

After a single session in the MRI room no torque was perceived. Neither test material shown any perceptible or measurable torque inside the scanner.

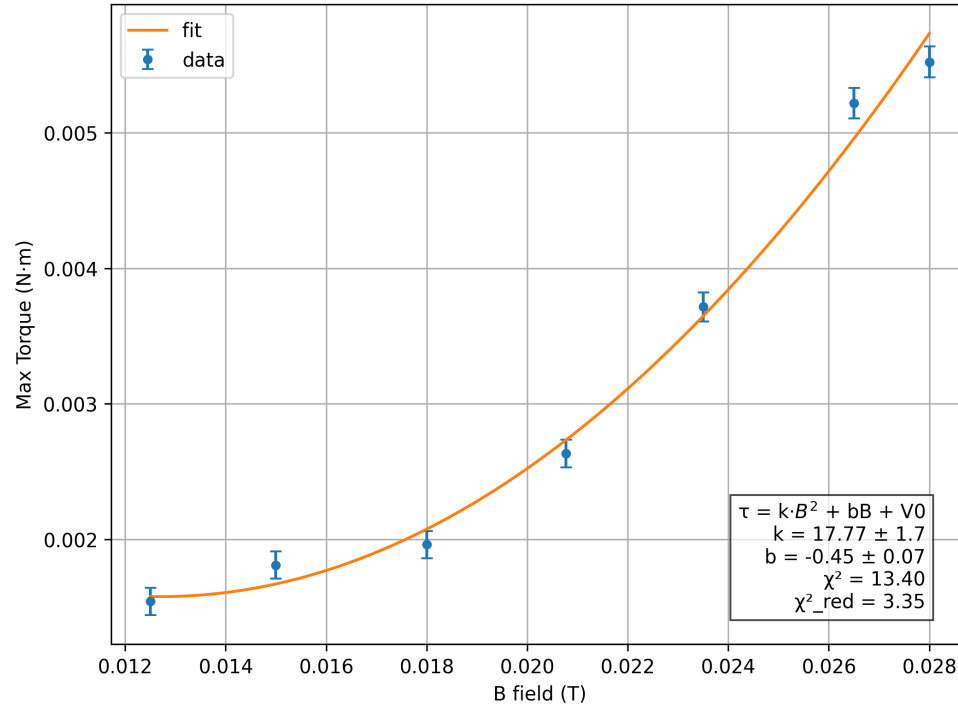


Figure 4.12: Maximum torque in Nm as a function of applied magnetic flux density in T for the shorter (2cm) magnetic rod under stronger uniform fields (Neodymium magnets). Blue points is the experimental measurements while orange curve represents a quadratic fit

4.4 Key Findings

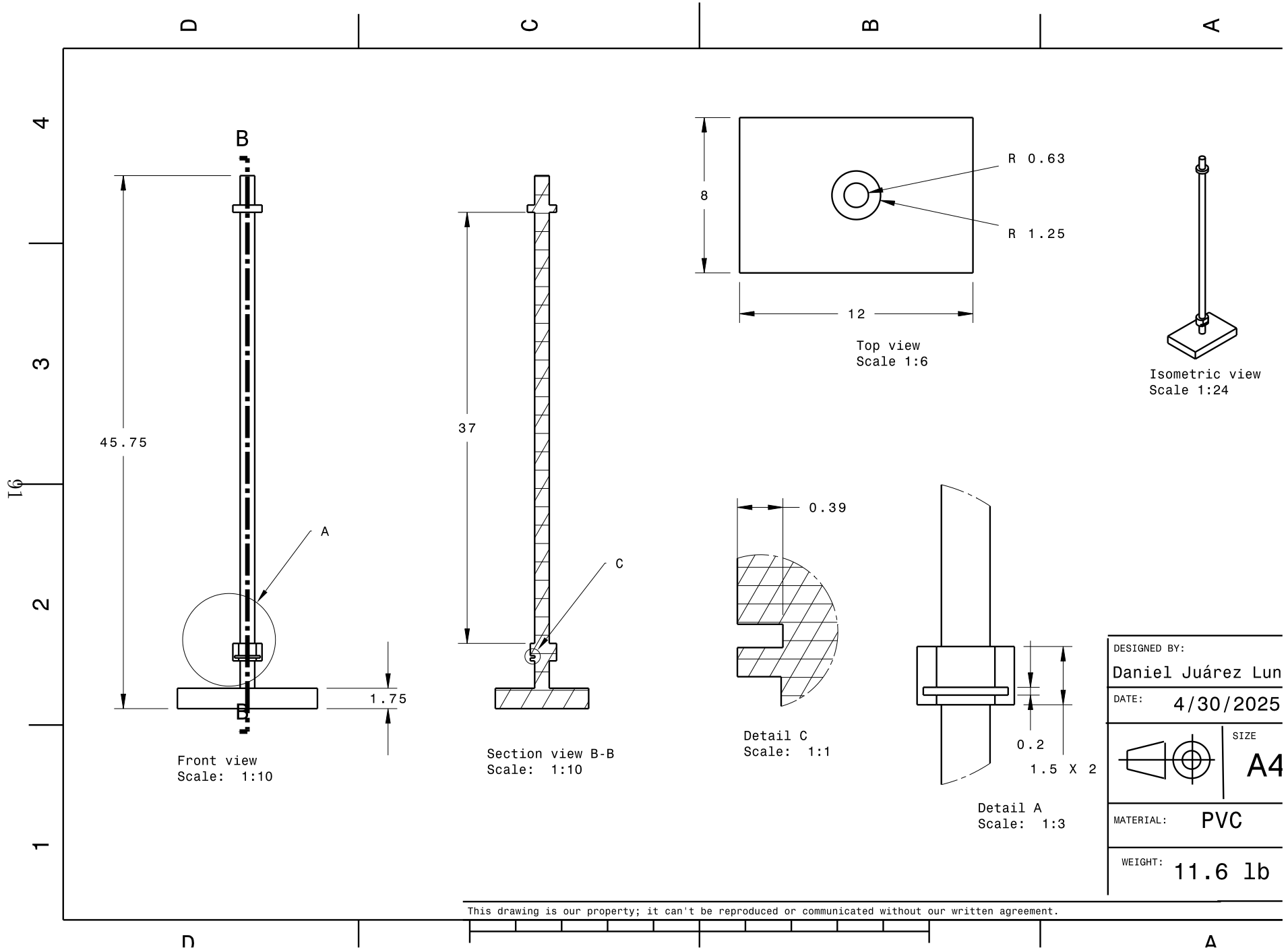
Chapter 5

Discussion and Conclusion

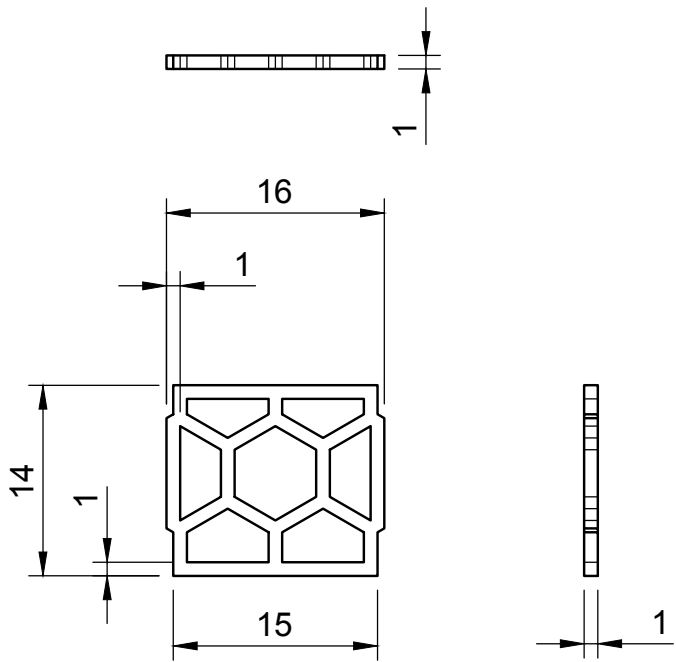
Appendix A

Engineering Drawings

Appendix A1: Pole drawing



Properties
Material: PLA
Weight:0.124 g



Dept.	Technical reference	Created by Daniel Juarez	8/5/25	Approved by
		Document type	Document status	
		Title Lead Holder Bee Comb Mesh MRI Translational Force	DWG No.	
		Rev.	Date of issue	Sheet 1/1

Appendix B

Sample calculations

B.1 Sample Calculation for Scaling at 30 G

Given that the MRI main field in the reference case is $B_0 = 3$ T, we consider a laboratory solenoid producing $B_0 = 30$ G = 3×10^{-3} T with a 5.691×10^{-3} T/m field gradient in z direction. The magnetic force acting on a weakly magnetic material is given by

$$\rho\chi = \frac{\mu_0 g \tan(\alpha)}{B_0 \frac{dB_0}{dz}}, \quad (\text{B.1})$$

where $\mu_0 = 1.256637 \times 10^{-6}$ N/A², $g = 9.8067$ m/s², ρ is the density of the laboratory material, and α is the equilibrium deflection angle of the pendulum, approximated by $\tan \alpha \approx 10/1000 = 0.01$ for a 1 cm displacement over a 1 m string.

For a scaled solenoid field of, assuming $B_0(dB_0/dz)$ scales quadratically with the magnetic field relative to the original MRI field of 3 T, we have

$$(B_0 \frac{dB_0}{dz})_{lab} = (B_0 \frac{dB_0}{dz})_{MRI} \left(\frac{B_{0,lab}}{B_{0,MRI}} \right)^2. \quad (\text{B.2})$$

Substituting the numerical values , the scaled gradient term for 30 G becomes

$$B_0 \frac{dB_0}{dz} = 1.71 \times 10^{-5} \text{ T}^2/\text{m}. \quad (\text{B.3})$$

Thus,

$$\rho\chi = \frac{(1.256637 \times 10^{-6})(9.8067)(0.01)}{1.71 \times 10^{-5}} = 7.22 \times 10^{-3}. \quad (\text{B.4})$$

Assuming $\rho = 7000 \text{ kg/m}^3$, the effective magnetic susceptibility for a ferromagnetic mimic becomes

$$\chi = \rho\chi \times \rho = (7.22 \times 10^{-3})(7000) = 5.05 \times 10^1 \approx 50. \quad (\text{B.5})$$

B.2 Smallest measurable force of translational fixture

Following section 3.2.1.1 For a specimen of $m = 0.1 \text{ g} = 1 \times 10^{-4} \text{ kg}$, the smallest measurable magnetic force is given by equation 3.2

$$F_{m,\min} = mg \frac{z}{L}, \quad (\text{Eq. 3.2})$$

where $g = 9.81 \text{ m/s}^2$, $z = 1 \times 10^{-3} \text{ m}$, and $L = 1 \text{ m}$. Substituting these values yields

$$\begin{aligned} F_{m,\min} &= (1 \times 10^{-4})(9.81) \left(\frac{1 \times 10^{-3}}{1} \right) \\ &= 9.81 \times 10^{-7} \text{ N} \approx 1.0 \times 10^{-6} \text{ N}. \end{aligned} \quad (\text{B.6})$$

Thus, for a 0.1 g mass, the smallest detectable magnetic force is approximately $1\ \mu\text{N}$. The corresponding force ratio remains

$$\frac{F_m}{F_g} = \frac{z}{L} = 10^{-3}, \quad (\text{B.7})$$

indicating that the setup can detect magnetic forces as small as one-thousandth of the object's weight.

B.3 Smallest measurable torque on fixture

Following Section 3.3.2, we evaluate the strain, voltage response, and force detection limit using the geometry reported by Chiu, *et al.* (2016) [58] For the Nitinol wire,

$$L = 69.85\ \text{mm} = 6.985 \times 10^{-2}\ \text{m},$$

$$D = 0.64\ \text{mm} = 6.40 \times 10^{-4}\ \text{m},$$

$$E = 66.2\ \text{GPa} = 6.62 \times 10^{10}\ \text{Pa}.$$

Geometric Sensitivity Factor

The geometric sensitivity term appearing in Equation 3.8 is

$$\frac{L}{D^3}. \quad (\text{B.8})$$

Substituting numerical values,

$$\begin{aligned}
D^3 &= (6.40 \times 10^{-4})^3 = 2.62 \times 10^{-10} \text{ m}^3, \\
\frac{L}{D^3} &= \frac{6.985 \times 10^{-2}}{2.62 \times 10^{-10}} = 2.66 \times 10^8 \text{ m}^{-2}.
\end{aligned} \tag{B.9}$$

This agrees with the order-of-magnitude sensitivity reported in Section 3.3.2.

Strain at the Empirical Detection Limit

Using Equation 3.8, the surface strain produced by the minimum detectable force $F = 0.49 \text{ mN} = 4.9 \times 10^{-4} \text{ N}$, as reported by Chiu, *et al.* (2016) [58] is

$$\epsilon = \frac{32FL}{\pi ED^3}. \tag{B.10}$$

Substituting,

$$\begin{aligned}
\epsilon &= \frac{32(4.9 \times 10^{-4})(6.985 \times 10^{-2})}{\pi(6.62 \times 10^{10})(2.62 \times 10^{-10})} \\
&= \frac{1.09 \times 10^{-3}}{54.5} \\
&= 2.0 \times 10^{-5}.
\end{aligned} \tag{B.11}$$

Thus, the minimum detectable force corresponds to a strain of approximately

$$\epsilon \approx 20 \mu\epsilon. \tag{B.12}$$

Expected Bridge Output

Assuming a half-bridge configuration (Equation 3.11) with

$$GF = 2.1,$$

$$V_{exc} = 5 \text{ V},$$

the predicted output voltage is

$$\begin{aligned} V_{out} &\approx \frac{V_{exc}}{2} GF \epsilon \\ &= \frac{5}{2} (2.1) (2.0 \times 10^{-5}) \\ &= 5.25 (2.0 \times 10^{-5}) \\ &= 1.05 \times 10^{-4} \text{ V}. \end{aligned} \tag{B.13}$$

Therefore,

$$V_{out} \approx 0.105 \text{ mV}. \tag{B.14}$$

Additional amplification is therefore required to reach the experimentally observed $\sim 5 \text{ mV}$ signal level.

Force Detection Limit from Noise Floor

Using Equation 3.13, the minimum detectable force for a given minimum strain ϵ_{min} is

$$F_{min} = \frac{\pi E D^3}{32 L} \epsilon_{min}. \tag{B.15}$$

If the system resolves strains on the order of $10 \mu\epsilon = 1.0 \times 10^{-5}$, then

$$\begin{aligned}
F_{min} &= \frac{\pi(6.62 \times 10^{10})(2.62 \times 10^{-10})}{32(6.985 \times 10^{-2})}(1.0 \times 10^{-5}) \\
&= 2.44 \times 10^{-4} \text{ N} \\
&= 0.24 \text{ mN}.
\end{aligned} \tag{B.16}$$

This confirms that sub-milliNewton resolution is feasible with the given geometry, and micro-Newton sensitivity becomes achievable with reduced noise, increased L/D^3 , and signal averaging.

B.4 Kirchhoff analysis of Circuit

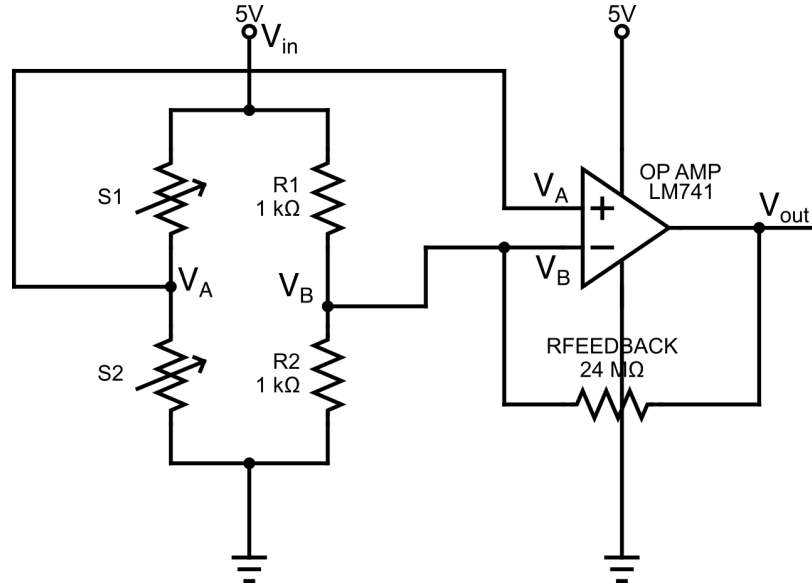


Figure B.1: Schematic of the Wheatstone bridge with strain gauges $S1$ and $S2$, fixed resistors $R = R1 = R2$, and operational amplifier with feedback resistor R_f . This configuration corresponds to the transfer function in Equation (3.14).

Using Kirchhoff's Current Law (KCL), in Figure B.1 for node V_A which corresponds to the positive leg of the op amp, or non inverting voltage, we have

$$\begin{aligned}
\frac{V_A - V_{in}}{S1} + \frac{V_A}{S2} &= 0 \\
\frac{S2(V_A - V_{in}) + S1V_A}{S1S2} &= 0 \\
V_A(S2 + S1) - S2V_{in} &= 0 \\
V_A &= \frac{S2V_{in}}{S2 + S1}.
\end{aligned}$$

Similarly, for the node V_B corresponding to the inverting input of the op-amp

$$\begin{aligned}
\frac{V_B - V_{in}}{R} + \frac{V_B}{R} + \frac{V_B - V_{out}}{R_f} &= 0 \\
\frac{R_f(2V_B - V_{in}) + R(V_B - V_{out})}{R\cancel{R_f}} &= 0 \\
V_B(2R_f + R) - R_fV_{in} - RV_{out} &= 0 \\
V_B &= \frac{R_fV_{in} + RV_{out}}{2R_f + R}.
\end{aligned}$$

At this point, we make a standard ideal op-amp assumption, the op-amp has infinite open-loop gain, meaning any finite output voltage corresponds to a vanishingly small input voltage difference. That is, for the difference amplifier

$$V_{out} = A_{ol}(V_A - V_B), \quad A_{ol} \rightarrow \infty, \quad (\text{B.17})$$

which implies $V_A \approx V_B$. Substituting $V_A = V_B$ into the previous equations allows us to solve for the closed-loop transfer function

$$\begin{aligned}
\frac{S2V_{in}}{S2 + S1} &= \frac{R_f V_{in} + R V_{out}}{2R_f + R} \\
\frac{V_{out}}{V_{in}} &= \frac{R_f(S2 - S1) + S2R}{R(S2 + S1)} \\
\frac{V_{out}}{V_{in}} &= \frac{R_f}{R} \frac{S2 - S1}{S2 + S1} + \frac{S2}{S2 + S1}
\end{aligned} \tag{3.14}$$

In summary, the output voltage is composed of two contributions: a differential term amplified by the ratio R_f/R , which captures the imbalance between the strain gauges $S1$ and $S2$, and a baseline voltage division term $S2/(S2 + S1)$ that represents the inherent bridge offset. This expression clearly shows how the Wheatstone bridge imbalance is amplified by the op-amp to produce the measurable output.

B.5 Explaining the behavior of Coarse-Fine Tuning with Potentiometers in Circuit

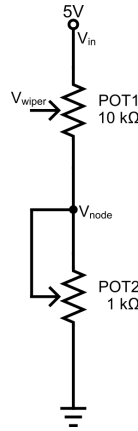


Figure B.2: Schematic of the coarse-fine potentiometer configuration used for the voltage-divider derivations in Figure 3.13.

B.5.1 1. Derivation of the Node Voltage

Starting with KVL around the loop

$$V_{\text{in}} - IR_1 - IR_2 = 0, \quad (\text{B.18})$$

Factor out the current

$$V_{\text{in}} = I(R_1 + R_2), \quad (\text{B.19})$$

Solve for the series current

$$I = \frac{V_{\text{in}}}{R_1 + R_2}, \quad (\text{B.20})$$

The node voltage is the drop across the lower resistor R_2

$$V_{\text{node}} = I R_2 = \left(\frac{V_{\text{in}}}{R_1 + R_2} \right) R_2. \quad (\text{B.21})$$

Final voltage-divider form

$$\boxed{V_{\text{node}} = V_{\text{in}} \frac{R_2}{R_1 + R_2}}. \quad (\text{3.15})$$

2. Numerical Evaluation for Fine Pot Positions

For these evaluations

$$V_{\text{in}} = 5.0 \text{ V}, \quad R_1 = 10 \text{ k}\Omega, \quad R_2 \in [0, 1000] \text{ }\Omega. \quad (\text{B.22})$$

2.1 Halfway Fine Adjustment $R_2 = 500 \text{ }\Omega$

$$V_{\text{node}} = 5.0 \frac{500}{10000 + 500} = 5.0 \left(\frac{500}{10500} \right) = 5.0(0.047619) = 0.2381 \text{ V}. \quad (\text{B.23})$$

2.2 Tabulated Values for Fine Pot Settings

$$V_{\text{node}} = 5 \frac{R_2}{10000 + R_2}. \quad (\text{B.24})$$

Fine Pot Setting	R_2 (Ohms)	V_{node} (V)
Max	1000	0.4545
Mid	500	0.2381
Min	0	0

3. Coarse Adjustment with Fine Fixed at Midpoint

Now fix the fine pot at

$$R_2 = 500 \, \Omega. \quad (\text{B.25})$$

The 10 k coarse pot wiper divides R_1 into two portions. Let the coarse wiper position be $\alpha \in [0, 1]$, where

$$\alpha = 0 : \text{wiper at top}, \quad \alpha = 1 : \text{wiper at bottom}. \quad (\text{B.26})$$

The wiper voltage is a weighted average between the supply (5 V) and the node voltage computed with the fixed fine setting

$$V_{\text{node, fine}=500} = 0.2381 \, \text{V}. \quad (\text{B.27})$$

Thus the coarse-wiper voltage is

$$V_{\text{wiper}} = (1 - \alpha)(5.0) + \alpha(0.2381). \quad (\text{B.28})$$

3.1 Coarse-Wiper Table (Fine Fixed at Half)

Coarse Setting	α	V_{wiper} (V)
Top (Min Coarse)	0	5.000
Midpoint	0.5	2.619
Bottom (Max Coarse)	1	0.238

4. Interpretation

At $\alpha = 0$ (coarse at minimum), the wiper sits at the full input voltage

$$V_{\text{wiper}} = 5 \text{ V}, \quad (\text{B.29})$$

which is independent of the fine adjustment. At $\alpha = 1$ (coarse at maximum), the wiper voltage collapses into the range controlled solely by the fine pot

$$V_{\text{wiper}} = V_{\text{node, fine}}. \quad (\text{B.30})$$

Thus the coarse pot sets the overall span, and the fine pot determines the precise lower-region tuning.

Appendix C

Calibration Logs

Additional experimental relationships and fitting diagnostics are presented to support the main analysis.

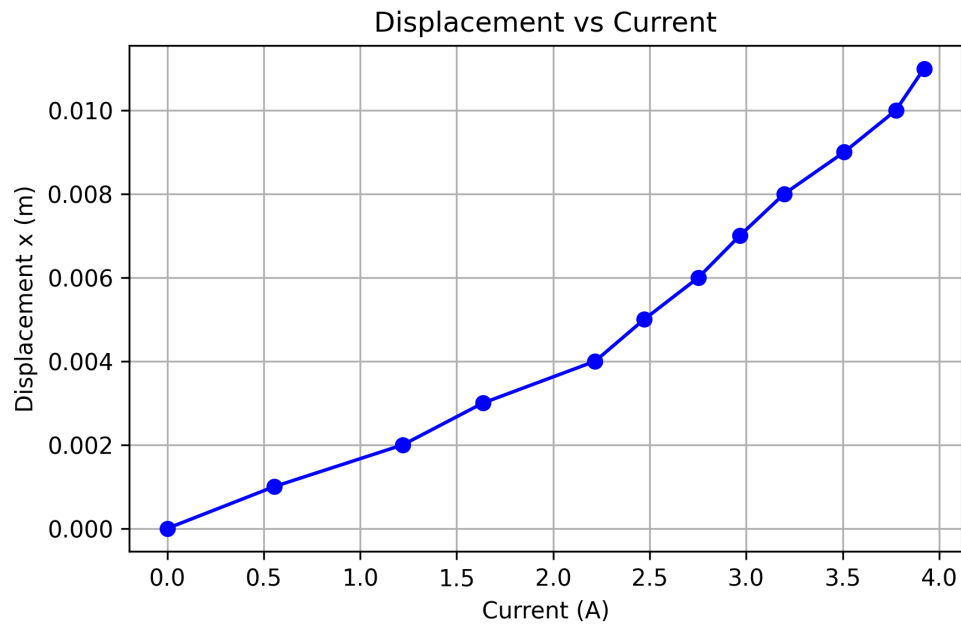


Figure B.1: Displacement as a function of current through the solenoid.

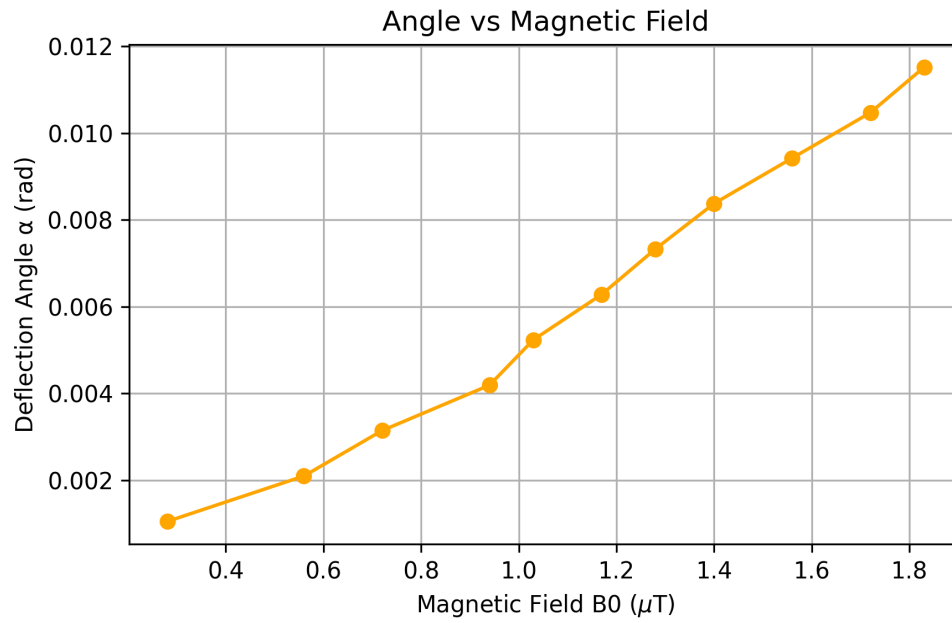


Figure B.2: Measured angle α versus magnetic field intensity B_0 .

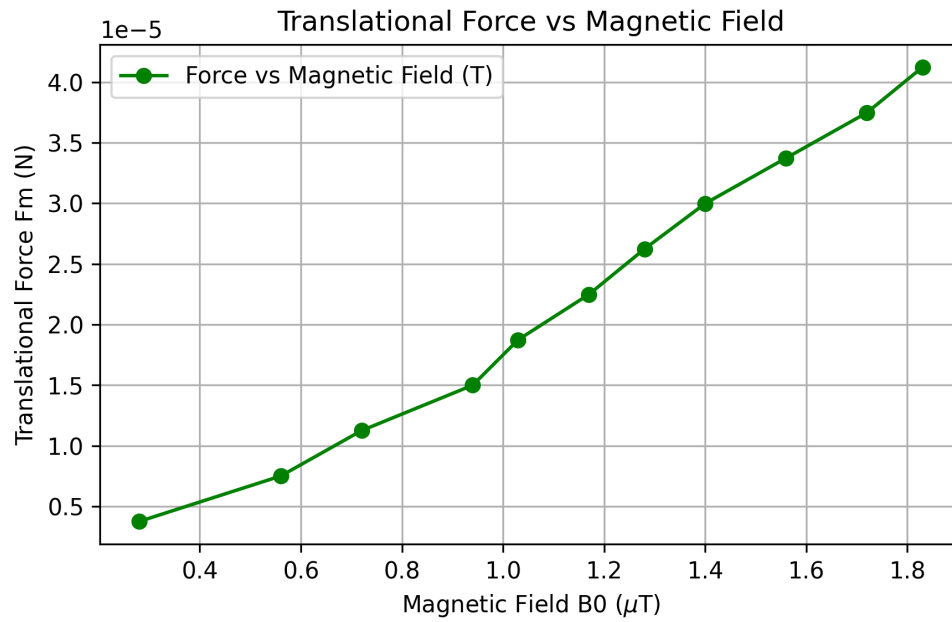


Figure B.3: Computed translational force F_m as a function of B_0 .

Appendix D

Codes and data samples

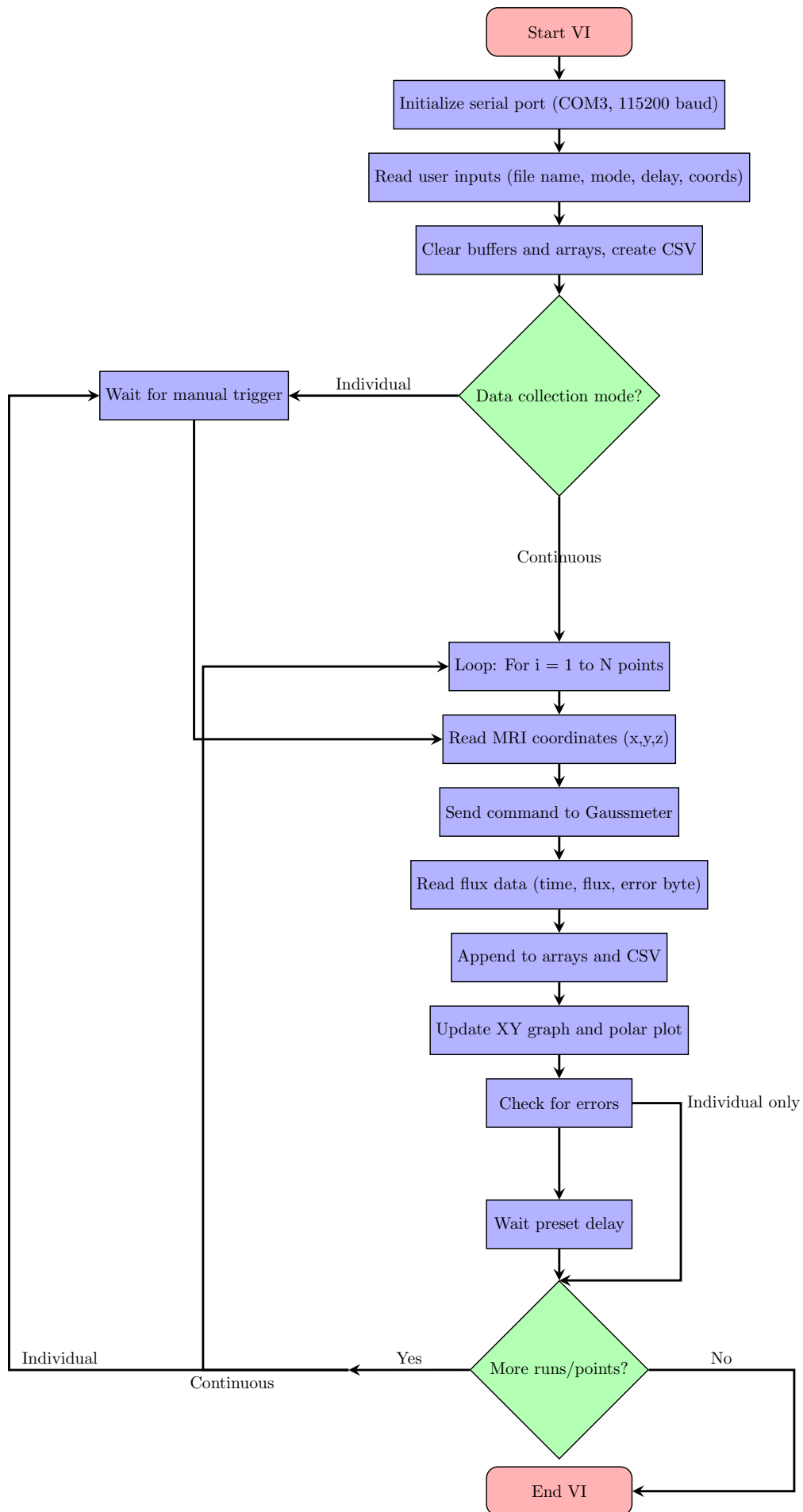
D.1 Gaussmeter LabView Protocol

Figure D.1. Upon startup, the VI initializes serial communication with the Gaussmeter, loads user-defined parameters (such as file name, mode of acquisition, delay time, and probe coordinates), and clears buffers in preparation for logging. The user may select either a continuous or an individual sampling mode. In continuous mode, the VI iterates through a defined number of data points with a fixed delay between samples; in individual mode, each data capture is triggered manually, allowing for precise positioning or timing. For both modes, each cycle involves reading the spatial coordinates, sending a measurement command, recording the returned magnetic flux value, and updating a graphical display as well as the output file.

While the flowchart reflects the core logic, it does not fully convey the customization added to the VI interface. For example, the front panel includes input fields for patient table (couch) parameters such as couch speed and displacement points. As mentioned above, phantom is discretized with radial and angular indices: angular positions span 12 sectors clockwise from the 12 o'clock position (0 to 11 or A to L), and radial depths range from the center (0) to the outer edge (6). The probe's spatial

location is represented in polar coordinates, and the system plots magnetic flux as a function of position rather than time, allowing for intuitive visualization of the MRI fringe field across the phantom surface. The VI is operated in the following way.

Flowchart of the Gaussmeter VI data acquisition algorithm



D.2 Sample Data for Stray Magnetic Field Mapping Along the Patient Couch (Backward Pass at (0,0))

Table D.1 presents a representative subset of the raw dataset from the first backward (retracting) couch translation pass along the longitudinal (z) axis at the central grid coordinate (0,0). Data were logged continuously using the Hall-effect gaussmeter probe. Flux values are raw instrument readings in gauss. The err_byte column is the gaussmeter’s reported error code.

Uncertainties: position ± 1 mm (couch display and alignment tolerance), time ± 0.01 s, flux $\pm 0.2\%$ of reading or $\approx \pm 50$ G near full scale.

z _{pos} [mm]	x _{pos} [mm]	y _{pos} [mm]	time [s]	flux [G]	err_byte
0	0	0	0.25	26823	64.039
0	0	0	0.50	26823	64.289
0	0	0	1.00	26822	64.788
0	0	0	2.00	26824	65.788
0	0	0	2.75	26672	66.539
0	0	0	3.50	24245	67.290
0	0	0	4.00	22911	67.790
0	0	0	5.00	20287	68.791
0	0	0	6.00	16493	69.789
0	0	0	7.00	14331	70.788
0	0	0	10.00	8534	73.796
0	0	0	12.00	5886	75.803
0	0	0	15.00	3563	78.807
0	0	0	17.00	2529	80.804
0	0	0	20.00	1607	83.805
0	0	0	23.00	1053	86.800
0	0	0	25.00	794	88.798
0	0	0	27.00	640	90.800
0	0	0	29.00	570	92.801
0	0	0	30.00	570	93.801

Table D.1: Subset of measurements from the first retracting translation along the z-axis at isocenter (grid 00). Flux decreases as the probe moves away from the bore entrance. Full dataset (120 rows) and all 15 CSV files were imported into Python with pandas for cleaning, alignment of forward/backward pairs, and numerical computation of the longitudinal gradient $\partial B/\partial z$.

D.3 Sample Data for Translational Force Experiment

Table D.2 presents a representative subset of the displacement–current dataset used for field–gradient and angle calculations for experiment en section 3.2.2 Uncertainties: position ± 0.05 mm, current ± 0.001 A.

N	x [mm]	I [A]	$B_{-0.5}$	B_0	$B_{+0.5}$	dB/dx	θ	$B_0 dB/dx$
1	0	0.000	–	–	–	–	–	–
2	1	0.554	2.86	2.81	2.84	0.1111	0.001047	0.3123
3	2	1.221	5.56	5.66	5.70	1.4444	0.002094	8.1691
4	3	1.636	7.03	7.22	7.24	2.1111	0.003141	15.2469
5	4	2.217	9.32	9.44	9.44	1.2222	0.004188	11.5432
6	5	2.471	10.27	10.40	10.56	2.8889	0.005235	30.0444
7	6	2.752	11.61	11.66	11.80	1.8889	0.006282	22.0160
8	7	2.966	12.58	12.63	12.76	1.7778	0.007329	22.4593
9	8	3.196	13.92	14.02	14.27	3.4444	0.008377	48.2988
10	9	3.507	15.52	15.69	15.89	3.6667	0.009424	57.5259
11	10	3.775	17.01	17.28	17.31	3.0000	0.010471	51.8333
12	11	3.921	18.03	18.17	18.32	2.8889	0.011518	52.4815

Table D.2: Representative dataset of displacement, current, magnetic field samples at ± 0.5 mm and equilibrium, computed gradients, and corresponding angular measurements.

Appendix E

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Glossary

AIMD active implantable medical devices. 8, 9, 10, 11, 16

CDR Creighton Digital Repository. 111

CIED cardiac implantable electronic device. 1, 2, 3, 4, 5, 25

CRT cardiac resynchronization therapy. 25

CSV Comma Separated Values. 47

DAC data acquisition card. 72

DAQ Data Acquisition. 70

ECG electrocardiogram. 2

FEG Frequency Encoding Gradient. 24

HRS Heart Rhythm Society. 1

ICD implantable cardioverter-defibrillators. 9, 25

KCL Kirchhoff's Current Law. 98

MRI Magnetic Resonance Imaging. viii, xi, 1, 2, 3, 4, 5, 10, 11, 12, 13, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 26, 30, 31, 33, 39, 40, 41, 42, 43, 44, 48, 50, 51, 52, 53, 57, 74, 79, 80, 87

PEG Phase Encoding Gradient. 24

PLA Polylactic Acid. 52

PNS peripheral nerve stimulation. 31

PVC Polyvinyl Chloride. 12, 42, 48, 59

RF radiofrequency. 2, 3, 10, 11, 16, 17, 23, 30, 31, 32

SAR Specific Absorption Rate. 32

SSG Slice Select Gradient. 24

VI LabVIEW Virtual Instrument. 43, 53, 72, 106, 107

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